Scientific Computing

Mathematical Foundations for Machine Learning

Course Conspect

2026

Course Structure

Block I: Probability and Statistics

Lecture	Topic
1	Probability models, conditioning, and random variables
2	Distributions, expectation, and variance
3	Joint distributions, dependence, and random vectors
4	Samples, concentration, and limit theorems
5	Statistical estimation: method of moments, maximum likelihood, and MAP
6	Sampling distributions, confidence intervals, and bootstrap
7	Entropy, cross-entropy, KL divergence, Fisher information, and the Cramér–Rao bound

Block II: Linear Algebra, Calculus, and Optimization

Lecture	Topic
8	Geometry of data, projections, and least squares
9	Eigenvalues, positive-definite matrices, singular value decomposition, and PCA
10	Multivariable calculus, matrix calculus, and automatic differentiation
11	Optimization for machine learning

Block III: Numerical Computing and Autodiff

Lecture	Topic
12	NumPy, visualization, and numerical linear algebra
13	PyTorch, autograd, and the integrated capstone

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Chapter 1

Probability Models, Conditioning, and Random Variables

Probability starts with a model: a precise description of what may happen, which statements about the outcome are meaningful, and how probabilities are assigned. Conditioning changes the reference population, while a random variable turns outcomes into numerical values whose distribution can be studied.

1.1 Probability spaces and events

Definition 1.1: Probability space

A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Here Ω is the *sample space*, whose elements are outcomes; $\mathcal{F}$ is a collection of subsets of Ω, called events; and $\mathbb{P}$ assigns a probability to every event.

The event collection $\mathcal{F}$ is closed under complements and countable unions:

$$A \in \mathcal{F} \Longrightarrow A^c \in \mathcal{F}, \qquad A_1, A_2, \ldots \in \mathcal{F} \Longrightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

It also contains Ω. In a finite model one normally takes $\mathcal{F} = 2^{\Omega}$, the set of all subsets of Ω.

For events $A, B \subseteq \Omega$:

- $A \cap B$ means that both events occur;
- $A \cup B$ means that at least one occurs;
- $A^c = \Omega \setminus A$ means that A does not occur;
- $A \setminus B = A \cap B^c$ means that A occurs but B does not.

Definition 1.2: Probability axioms

For events in $\mathcal{F}$, a probability measure satisfies:

$$\begin{aligned} &\mathbb{P}(A) \geq 0, \\ &\mathbb{P}(\Omega) = 1, \\ &\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad \text{for pairwise disjoint } A_i. \end{aligned}$$

Several basic rules follow from these axioms:

$$\mathbb{P}(\varnothing) = 0, \tag{1.1}$$
$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \tag{1.2}$$
$$A \subseteq B \Longrightarrow \mathbb{P}(A) \le \mathbb{P}(B), \tag{1.3}$$
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \tag{1.4}$$

The last identity is the inclusion–exclusion rule. Adding $\mathbb{P}(A)$ and $\mathbb{P}(B)$ counts the overlap twice, so it must be subtracted once.

Example 1.3 (Two fair coin tosses)**.** Toss a fair coin twice and record the ordered result. Then

$$\Omega = \{HH, HT, TH, TT\}, \qquad \mathbb{P}(\{\omega\}) = \frac{1}{4} \quad \text{for every } \omega \in \Omega.$$

Let

$$A = \{\text{first toss is } H\} = \{HH, HT\}, \qquad B = \{\text{second toss is } H\} = \{HH, TH\}.$$

Then

$$A \cap B = \{HH\}, \qquad A \cup B = \{HH, HT, TH\}, \qquad A^c = \{TH, TT\}.$$

Inclusion–exclusion gives

$$\mathbb{P}(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}.$$

For a finite sample space with equally likely outcomes,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

This is a counting shortcut, not an additional axiom: it is valid only after equal likelihood has been justified.

1.2 Conditional probability and total probability

Definition 1.4: Conditional probability

For $\mathbb{P}(B) > 0$, the probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \tag{1.5}$$

Conditioning restricts the reference set to B and renormalizes the remaining probabilities. The event after the vertical bar determines the denominator. In general, $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ are different quantities.

Rearranging (1.5) gives the multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A). \tag{1.6}$$

Example 1.5 (At least one head)**.** In the two-toss model, let

$$E = \{\text{at least one head}\} = \{HH, HT, TH\}.$$

Given E, the probability that the second toss is heads is

$$\mathbb{P}(B \mid E) = \frac{\mathbb{P}(B \cap E)}{\mathbb{P}(E)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

The unconditional probability is $\mathbb{P}(B) = 1/2$. Conditioning removes the outcome TT, in which B fails.

Theorem 1.6: Law of total probability

Let $B_1, \dots, B_m$ be a partition of Ω: the events are pairwise disjoint and their union is Ω. If $\mathbb{P}(B_i) > 0$, then

$$\mathbb{P}(A) = \sum_{i=1}^{m} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i). \tag{1.7}$$

Indeed, the events $A \cap B_i$ are disjoint and their union is A. Add their probabilities and then apply the multiplication rule. For a binary partition B, B^c,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c).$$

1.3 Independence and mutual exclusivity

Definition 1.7: Independence

Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \tag{1.8}$$

If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Independence means that learning whether one event occurred does not change the probability of the other. In the coin example,

$$\mathbb{P}(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2},$$

so the events “first toss is heads” and “second toss is heads” are independent.

Definition 1.8: Mutual exclusivity

Events A and B are mutually exclusive, or disjoint, if

$$A \cap B = \varnothing.$$

They cannot occur together.

Proposition 1.9

Two mutually exclusive events with positive probability are not independent.

Proof. Mutual exclusivity gives $\mathbb{P}(A \cap B) = 0$, while positive probabilities give $\mathbb{P}(A)\mathbb{P}(B) > 0$. Thus (1.8) cannot hold. □

	Mutual exclusivity	**Independence**
Can both occur?	No	Usually yes
Defining condition	$A \cap B = \varnothing$	$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$
Information meaning	One rules out the other	One does not change the other

1.4 Bayes' theorem and base rates

Theorem 1.10: Bayes' theorem

Let $B_1, \dots, B_m$ be a partition of Ω. For $\mathbb{P}(A) > 0$,

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{m} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}. \tag{1.9}$$

The numerator is the probability of the route B_j followed by A. The denominator adds every mutually exclusive route that can produce A.

Example 1.11 (Rare-defect classifier)**.** Let D mean that an item is defective and $+$ mean that a classifier produces a positive flag. Assume

$$\mathbb{P}(D) = 0.01, \qquad \mathbb{P}(+ \mid D) = 0.99, \qquad \mathbb{P}(+ \mid D^c) = 0.05.$$

The probability of a positive result is

$$\mathbb{P}(+) = 0.99(0.01) + 0.05(0.99) = 0.0594.$$

Therefore

$$\mathbb{P}(D \mid +) = \frac{0.99(0.01)}{0.0594} = \frac{1}{6} \approx 0.167.$$

The same calculation can be expressed with expected counts in a population of 10,000 items:

Actual state	**Positive**	**Negative**	**Total**
Defect D	99	1	100
No defect D^c	495	9,405	9,900
Total	594	9,406	10,000

Only 99 of the 594 positive items are defective. The sensitivity $\mathbb{P}(+ \mid D) = 0.99$ is not the posterior $\mathbb{P}(D \mid +)$. The low base rate and the false positives among the much larger non-defective group dominate the denominator.

For a Bayesian classifier with classes $C_1, \dots, C_K$ and observed features x,

$$\mathbb{P}(C_k \mid x) = \frac{p(x \mid C_k)\mathbb{P}(C_k)}{\sum_{j=1}^{K} p(x \mid C_j)\mathbb{P}(C_j)}. \tag{1.10}$$

The factors have distinct roles:

- $\mathbb{P}(C_k)$ is the prior or class base rate;

- $p(x \mid C_k)$ is the class-conditional likelihood;
- $\mathbb{P}(C_k \mid x)$ is the posterior after observing x.

The maximum a posteriori rule is

$$\widehat{C}_{\mathrm{MAP}}(x) = \arg\max_k p(x \mid C_k)\mathbb{P}(C_k).$$

1.5 Random variables

Definition 1.12: Random variable

A real-valued random variable is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

such that $\{\omega : X(\omega) \leq x\}$ is an event for every $x \in \mathbb{R}$. In finite examples this event condition is automatic.

The sampled outcome is random; the mapping X is fixed.

Example 1.13 (Number of heads)**.** For two coin tosses, let X be the number of heads. Then

Outcome ω	HH	HT	TH	TT
$X(\omega)$	2	1	1	0

The support $\{0, 1, 2\}$ is the set of possible values of X; it is not the original sample space.

A *discrete* random variable takes values in a finite or countable set. An *absolutely continuous* random variable assigns probabilities to intervals through a density. Mixed random variables, with both point masses and a continuous part, also exist.

1.6 PMFs, PDFs, and CDFs

1.6.1 Probability mass functions

Definition 1.14: PMF

For a discrete random variable X, its probability mass function is

$$p_X(x) = \mathbb{P}(X = x), \qquad p_X(x) \geq 0, \qquad \sum_x p_X(x) = 1.$$

For the number of heads in two fair tosses,

$$p_X(0) = \frac{1}{4}, \qquad p_X(1) = \frac{1}{2}, \qquad p_X(2) = \frac{1}{4}.$$

The probability of an event is obtained by summing the masses of its values; for example, $\mathbb{P}(X \geq 1) = p_X(1) + p_X(2) = 3/4$.

1.6.2 Cumulative distribution functions

Definition 1.15: CDF

For every real-valued random variable X, the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \le x).$$

Every CDF is nondecreasing and right-continuous, with

$$\lim_{x \to -\infty} F_X(x) = 0, \qquad \lim_{x \to \infty} F_X(x) = 1.$$

It determines interval probabilities through

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a). \tag{1.11}$$

For the number of heads,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \le x < 1, \\ \frac{3}{4}, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

For a discrete variable, the jump at x equals its point mass:

$$p_X(x) = F_X(x) - F_X(x^-),$$

where $F_X(x^-)$ denotes the left limit.

1.6.3 Probability density functions

Definition 1.16: PDF

An absolutely continuous random variable Y has a probability density function f_Y satisfying

$$f_Y(y) \ge 0, \qquad \int_{-\infty}^{\infty} f_Y(y)\, dy = 1,$$

and

$$\mathbb{P}(a < Y \le b) = \int_a^b f_Y(y)\, dy. \tag{1.12}$$

Warning. A density value is not a point probability. For an absolutely continuous random variable, $\mathbb{P}(Y = y) = 0$ for every single point, and $f_Y(y)$ may be larger than one. Probability is the area under the density.

The CDF and density are related by

$$F_Y(y) = \int_{-\infty}^{y} f_Y(t)\, dt, \qquad F_Y'(y) = f_Y(y)$$

at points where the derivative exists.

Example 1.17 (Uniform distribution on the unit interval)**.** Let

$$f_Y(y) = \begin{cases} 1, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y, & 0 \le y \le 1, \\ 1, & y > 1. \end{cases}$$

Thus

$$\mathbb{P}(0.2 < Y \le 0.7) = \int_{0.2}^{0.7} 1 \, dy = F_Y(0.7) - F_Y(0.2) = 0.5.$$

Chapter 2

Distributions, Expectation, and Variance

2.1 Exponential waiting times

Imagine watching a help desk's inbox. Start a stopwatch now and stop it when the next message arrives. The number on the stopwatch is T, the waiting time in minutes. On one occasion we might wait 0.3 minutes; on another, 7 minutes. We want to answer questions such as: what is the probability that we are still waiting after five minutes?

2.1.1 A model of arrivals without a schedule

First build a simple model using one-second intervals. Suppose that in each interval a message arrives with probability $1/300$, independently of what happened in earlier intervals. Most seconds contain no message. Occasionally two successful seconds occur close together; occasionally there is a long gap.

In this model, waiting more than five minutes means getting no message in any of 300 consecutive seconds. By independence,

$$\mathbb{P}(T > 5) = \left(1 - \frac{1}{300}\right)^{300} \approx 0.3673.$$

A long wait is therefore quite plausible. The model does not place an arrival at regular five-minute intervals.

Now shorten the time intervals. Write λ for the arrival rate, measured in messages per minute. A small interval of length Δ minutes has arrival probability $\lambda\Delta$. In our example, $\lambda = 0.2$, because $0.2 \times (1/60) = 1/300$. The rate describes how frequently messages arrive on average; it is not itself a probability.

Divide a time span of t minutes into n equal intervals. In the discrete approximation, each has arrival probability $\lambda t/n$. The probability that all of them are empty is

$$\left(1 - \frac{\lambda t}{n}\right)^{n}.$$

Letting the intervals become arbitrarily short gives

$$\mathbb{P}(T > t) = \lim_{n\to\infty} \left(1 - \frac{\lambda t}{n}\right)^{n} = e^{-\lambda t}, \qquad t \geq 0. \tag{2.1}$$

This is the probability that *no message has arrived by time t*. The exponential function appears because we multiply the no-arrival probabilities of many short intervals. The resulting continuous waiting time has an *exponential distribution*.

The assumptions describe an inbox with no timetable and no change in activity during the period being studied. They would be unsuitable for notifications sent every hour or an inbox that becomes much busier at lunchtime.

2.1.2 From the waiting probability to the density

A message has either arrived by time t, or it has not. Thus the CDF is

$$F_T(t) = \mathbb{P}(T \le t) = 1 - e^{-\lambda t}, \qquad t \ge 0.$$

Differentiating this CDF, using the relation from Chapter 1, gives the density of the waiting time.

Definition 2.1: Exponential distribution

For $\lambda > 0$, the notation $T \sim \text{Exp}(\lambda)$ means

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \ge 0, \\ 0, & t < 0. \end{cases}$$

The parameter λ is the arrival rate.

Increasing λ makes short waits more likely. For example, at rate 0.2 per minute the probability of waiting more than five minutes is $e^{-1} \approx 0.3679$. At rate 0.5, that probability drops to $e^{-2.5} \approx 0.0821$. Figure 2.1 shows this comparison directly.

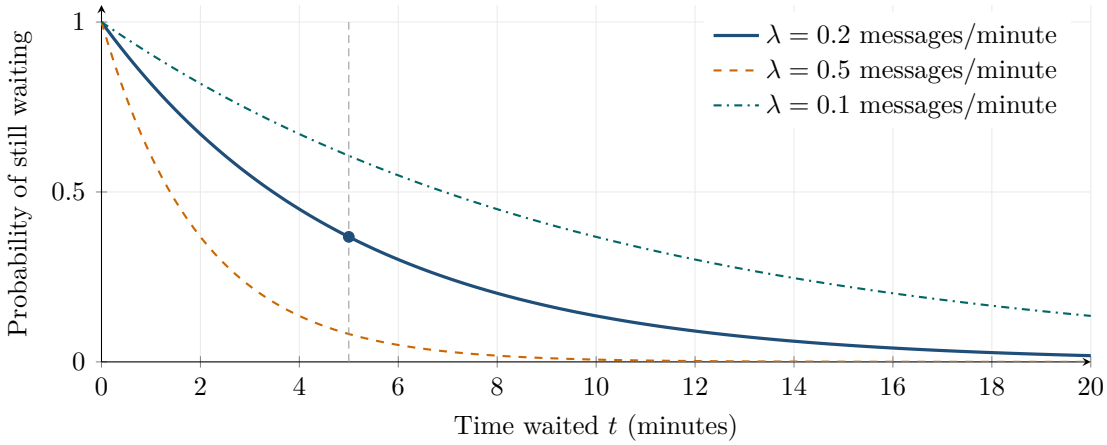


Figure 2.1: The probability that the inbox is still empty after t minutes. Every curve starts at one: at time zero we have not yet waited. A busier inbox has a larger rate and a faster decrease in the probability of still waiting. The dashed vertical line marks five minutes.

For an interval question, subtract two waiting probabilities. With $\lambda = 0.2$,

$$\mathbb{P}(2 < T \le 5) = \mathbb{P}(T > 2) - \mathbb{P}(T > 5) = e^{-0.4} - e^{-1} \approx 0.3024.$$

We keep the cases that are still waiting at minute two and remove those that are still waiting at minute five.

A different practical question is how long we must wait to have seen a message in 95% of repetitions. Solve

$$1 - e^{-0.2t} = 0.95 \quad \Longrightarrow \quad t = -\frac{\log(0.05)}{0.2} \approx 14.98 \text{ minutes}.$$

This cutoff is the 95-th percentile of the waiting time. More generally, the q-quantile is $t_q = -\log(1-q)/\lambda$, for $0 < q < 1$.

2.1.3 Does a long wait make an arrival overdue?

Suppose we have already waited ten minutes with no message. In the model, the next short interval has the same arrival chance as it did at the start. Earlier empty intervals do not change that chance. Therefore the probability of waiting another five minutes is still $e^{-0.2 \cdot 5} \approx 0.3679$.

We can verify this using conditional probability. For $s, t \geq 0$,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

The right-hand side is $\mathbb{P}(T > t)$. The distribution of the remaining wait is the same as the distribution of the original wait. This property is called *memorylessness*.

A scheduled bus behaves differently: if it comes every ten minutes, waiting nine minutes tells us something about the remaining wait. The exponential model is appropriate only when this kind of schedule, ageing, or change in activity is absent.

2.2 Expectation and functions of a random variable

The probability of a long wait answers one question. The average amount of waiting answers another. If we repeat the inbox experiment independently many times and average the recorded times, what value should that average be close to?

Start with a simpler example. Suppose a task takes one minute with probability $3/4$ and nine minutes with probability $1/4$. In many repetitions, about three quarters of the tasks take one minute and one quarter take nine minutes. Their average is close to

$$\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 9 = 3 \text{ minutes}.$$

This probability-weighted average is the *expectation*. It can be three minutes even though no individual task takes three minutes.

2.2.1 Averaging with probabilities

For a discrete random variable, weight each possible value by its probability and add. For a continuous variable, a small interval near x has probability approximately $f_X(x)\Delta x$, so its contribution to the average is approximately $x f_X(x)\Delta x$. Summing these contributions over increasingly small intervals gives an integral.

Definition 2.2: Expectation

The expectation of X is

$$\mathbb{E}[X] = \begin{cases} \sum_x x\, p_X(x), & X \text{ discrete}, \\ \int_{-\infty}^{\infty} x f_X(x)\, dx, & X \text{ has a density}. \end{cases}$$

For a finite expectation we require $\mathbb{E}[|X|] < \infty$: the sum or integral of absolute contributions must converge.

This condition matters when very large values, although rare, contribute so much that no finite average exists. All the distributions in the main part of this chapter have finite means. The Student extension provides an example where the condition can fail.

For exponential waiting times, the definition gives

$$\mathbb{E}[T] = \int_0^{\infty} t\lambda e^{-\lambda t}\, dt.$$

To evaluate it, integrate by parts with $u = t$ and $dv = \lambda e^{-\lambda t}\, dt$:

$$\mathbb{E}[T] = [-te^{-\lambda t}]_0^{\infty} + \int_0^{\infty} e^{-\lambda t}\, dt = \frac{1}{\lambda}.$$

At rate 0.2 messages per minute, the mean wait is five minutes. This is an average over repeated waits, not an appointment for the next arrival. We already found that more than a third of waits exceed five minutes.

2.2.2 First transform the observation, then average

Sometimes we want to average something other than the waiting time itself. For example, assign a response score

$$g(t) = e^{-at}, \qquad a > 0.$$

An immediate response scores one; longer waits receive smaller scores. The parameter a controls how quickly the score decreases with time.

To find the average score, calculate the score for each possible wait and then weight it by the probability of that wait:

$$\mathbb{E}[g(T)] = \int_0^{\infty} g(t) f_T(t)\, dt.$$

For this particular score,

$$\mathbb{E}[e^{-aT}] = \int_0^{\infty} e^{-at}\lambda e^{-\lambda t}\, dt = \frac{\lambda}{\lambda + a}.$$

Result 2.3: Expectation of a function

This calculation illustrates the general rule

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x)p_X(x), & X \text{ discrete}, \\ \int_{-\infty}^{\infty} g(x)f_X(x)\,dx, & X \text{ has a density}. \end{cases} \tag{2.2}$$

We assume $\mathbb{E}[|g(X)|] < \infty$. The rule lets us use the distribution of X directly, without first deriving the distribution of $g(X)$.

Scoring the average wait is a different operation. For $a = \lambda = 1$,

$$\mathbb{E}[e^{-T}] = \frac{1}{2}, \qquad e^{-\mathbb{E}[T]} = e^{-1} \approx 0.3679.$$

The nonlinear transformation makes the order of the two operations matter.

Another useful choice is $g(t) = t^2$. Large waits receive much more weight after squaring. Again integrating by parts,

$$\mathbb{E}[T^2] = \int_0^{\infty} t^2 \lambda e^{-\lambda t}\,dt = \frac{2}{\lambda}\mathbb{E}[T] = \frac{2}{\lambda^2}.$$

We will use this second moment to measure spread.

2.2.3 Why expectations add

Suppose completing a job requires a wait X and a processing time Y. On each repetition, the total is $X+Y$. Averaging the totals gives the average waiting time plus the average processing time. This remains true even if long waits tend to accompany long processing times.

Theorem 2.4: Linearity of expectation

Let $X_1, \dots, X_n$ have finite expectations. For constants a_i and b,

$$\mathbb{E}\left[b + \sum_{i=1}^{n} a_i X_i\right] = b + \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

Independence is not required.

For a finite sample space, distributing the sum makes the reason explicit:

$$\sum_{\omega \in \Omega} (X(\omega) + Y(\omega))\mathbb{P}(\{\omega\}) = \mathbb{E}[X] + \mathbb{E}[Y].$$

The integral version works in the same way.

For instance, if a job has a fixed two-minute setup followed by $T \sim \text{Exp}(0.2)$, its mean duration is $2 + \mathbb{E}[T] = 7$ minutes. Expressing T in seconds instead multiplies its mean by 60.

Linearity also makes counts easy to analyze. For an event A, the indicator $\mathbf{1}_A$ is one if A occurs and zero otherwise. Its expectation is

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A).$$

Consequently, the expected number of events occurring among $A_1, \dots, A_n$ is

$$\mathbb{E}\left[\sum_{i=1}^{n} \mathbf{1}_{A_i}\right] = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

No independence assumption is needed for this mean. Dependence will matter when we ask about the count's spread.

2.3 Variance and standard deviation

A constant five-minute wait and an exponential wait with mean five minutes have the same expectation. Their behavior is very different: the first is perfectly predictable, while the second sometimes ends almost immediately and sometimes takes much longer. We need a measure of this variation around the mean.

If $\mu = \mathbb{E}[X]$, the deviation of an observation is $X - \mu$. Averaging the signed deviations gives zero: $\mathbb{E}[X - \mu] = 0$. Positive and negative deviations cancel. Squaring them prevents that cancellation and gives more weight to large departures from the mean.

Definition 2.5: Variance and standard deviation

For X with a finite second moment,

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mu)^2], \qquad \mu = \mathbb{E}[X], \qquad \operatorname{sd}(X) = \sqrt{\operatorname{Var}(X)}.$$

Variance is the average squared deviation. If waiting time is in minutes, variance is in minutes squared. Taking the square root returns to minutes; this is why standard deviation is useful when describing the size of typical fluctuations. It is not a bound on individual observations.

Expanding the square gives a convenient way to calculate variance:

$$\begin{aligned} \operatorname{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \end{aligned} \tag{2.3}$$

For exponential waiting times, we have already computed both terms:

$$\operatorname{Var}(T) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}, \qquad \operatorname{sd}(T) = \frac{1}{\lambda}.$$

The inbox with rate 0.2 has both mean and standard deviation equal to five minutes. A constant five-minute wait instead has variance zero.

2.3.1 Changing the origin or the units

Adding a fixed delay b moves every observation and its mean by the same amount. The deviations from the mean stay unchanged. Multiplying times by a multiplies every deviation by a, so their squares are multiplied by a^2.

Result 2.6: Changing the origin and scale

For constants a, b and a variable with finite second moment,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \qquad \operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

For example, converting minutes to seconds multiplies the mean and standard deviation by 60, and the variance by 60^2. Variance is not linear.

2.3.2 Adding two variable delays

For a job with waiting time X and processing time Y, let $U = X - \mathbb{E}[X]$ and $V = Y - \mathbb{E}[Y]$. The deviation of the total duration is $U + V$. Squaring gives

$$(U + V)^2 = U^2 + V^2 + 2UV.$$

The first two terms describe the separate spreads. The last term tells us whether the two delays fluctuate together. If both are above their means, or both are below, their product UV is positive. If one is above its mean while the other is below, the product is negative.

Definition 2.7: Covariance

For variables with finite second moments, the average product of their centered deviations is the *covariance*:

$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Result 2.8: Variance of a sum

Taking expectations in the squared-sum identity gives

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y). \tag{2.4}$$

For independent variables, expectations of products factor. Thus $\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] = 0$, and the covariance term vanishes. For independent variables with finite second moments, this extends to

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i).$$

Example 2.9 (Three different ways to combine fluctuations)**.** Let X, Y each have variance one. If they are independent, the variance of $X + Y$ is two. If $Y = X$, the fluctuations reinforce each other and $\operatorname{Var}(X + Y) = \operatorname{Var}(2X) = 4$. If $Y = -X$, they cancel exactly and the sum has variance zero.

Zero covariance is enough for variances to add, even without full independence. We will study that distinction further when considering joint distributions. For the sums used below, independence will be an explicit part of the model.

2.4 Poisson counts

Return to the inbox, but change the question. Instead of timing the next message, watch for ten minutes and count how many messages arrive. A waiting time can be 2.4 minutes; a message count must be an integer such as zero, one, or two. The same arrivals produce two different kinds of random variables.

Write $N(t)$ for the number of messages arriving between time zero and time t. Figure 2.2 shows one possible record. At minute six we have counted two messages; at minute ten we have counted three.

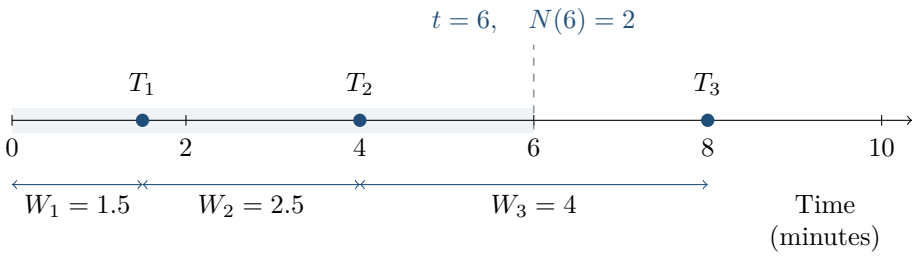


Figure 2.2: One possible sequence of arrivals, at 1.5, 4, and 8 minutes. The gaps W_1, W_2, W_3 are waiting times; the arrival times T_k are cumulative sums of those gaps. By the marked time $t = 6$, two messages have arrived, so $N(6) = 2$.

2.4.1 Counting successful short intervals

We can use the same short-interval model as before. Divide t minutes into n tiny intervals, each independently containing a message with probability $p_n = \lambda t/n$. To get exactly k messages, exactly k intervals must contain a message and the others must be empty.

For a particular choice of those k intervals, independence gives probability $p_n^k(1-p_n)^{n-k}$. The number of ways to choose the intervals is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

For example, if there are three intervals and exactly one message, it can be in the first, second, or third interval: three possibilities. Here $n! = 1 \cdot 2 \cdots n$, with $0! = 1$. Adding the equally probable possibilities gives

$$\binom{n}{k}\left(\frac{\lambda t}{n}\right)^k\left(1-\frac{\lambda t}{n}\right)^{n-k}.$$

Put $m = \lambda t$. For fixed k, as n increases,

$$\frac{\binom{n}{k}}{n^k} \longrightarrow \frac{1}{k!}, \qquad \left(1-\frac{m}{n}\right)^{n-k} \longrightarrow e^{-m}.$$

Thus the continuous-time count has probability $e^{-m}m^k/k!$ of equalling k. This is the *Poisson distribution*.

Definition 2.10: Poisson distribution

For $m > 0$, a count $N \sim \text{Poisson}(m)$ has PMF

$$\mathbb{P}(N = k) = e^{-m}\frac{m^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

For our inbox observed for t minutes, $N(t) \sim \text{Poisson}(\lambda t)$.

The rate λ describes arrivals per minute; $m = \lambda t$ describes the expected number over the whole observation window. At rate 0.2 per minute, a ten-minute window has $m = 2$, while a thirty-minute window has $m = 6$. Neither count is guaranteed to equal its mean.

Example 2.11 (No messages or several messages)**.** For the ten-minute count with $m = 2$,

$$\mathbb{P}(N(10) = 0) = e^{-2} \approx 0.1353,$$

and

$$\mathbb{P}(N(10) \geq 2) = 1 - \mathbb{P}(N(10) = 0) - \mathbb{P}(N(10) = 1) = 1 - e^{-2}(1 + 2) \approx 0.5940.$$

No messages in ten minutes is exactly the event that the first waiting time exceeds ten minutes. The Poisson and Exponential calculations agree: $\mathbb{P}(N(10) = 0) = \mathbb{P}(T > 10) = e^{-2}$.

2.4.2 Mean and variance of the count

To check normalization, sum the PMF and use the exponential series:

$$\sum_{k=0}^{\infty} e^{-m} \frac{m^k}{k!} = e^{-m} e^{m} = 1.$$

For the mean, the factor k cancels one factor in $k!$. Reindexing the remaining sum with $j = k - 1$ gives

$$\mathbb{E}[N] = \sum_{k=1}^{\infty} k e^{-m} \frac{m^k}{k!} = m \sum_{j=0}^{\infty} e^{-m} \frac{m^j}{j!} = m.$$

For the variance we need $\mathbb{E}[N^2]$. It is easier to first calculate $\mathbb{E}[N(N-1)]$, since $k(k-1)$ cancels two factors in $k!$:

$$\mathbb{E}[N(N-1)] = m^2 \sum_{j=0}^{\infty} e^{-m} \frac{m^j}{j!} = m^2.$$

Now use $N^2 = N(N-1) + N$:

$$\mathbb{E}[N^2] = m^2 + m, \qquad \mathrm{Var}(N) = m.$$

For ten minutes, the mean and variance are both two, and the standard deviation is $\sqrt{2}$ messages.

The equality of mean and variance will later help us recognize when a single Poisson model misses part of the variation in a population.

2.4.3 Combining independent arrival streams

Suppose two independent inboxes receive messages at rates λ_1 and λ_2. Counting all their messages over the same interval should add the expected counts. In fact, the entire distribution remains Poisson:

$$N_1 + N_2 \sim \mathrm{Poisson}(m_1 + m_2), \qquad m_i = \lambda_i t.$$

To see why, split a total of k messages into j from the first inbox and $k - j$ from the second. Independence lets us multiply their probabilities; summing over j and using the binomial theorem gives

$$\mathbb{P}(N_1 + N_2 = k) = \frac{e^{-(m_1+m_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} m_1^j m_2^{k-j} = \frac{e^{-(m_1+m_2)}(m_1 + m_2)^k}{k!}.$$

In the short-interval model, counts in disjoint time windows are also independent. If arrivals instead trigger further arrivals or the rate changes with the time of day, these assumptions need to be reconsidered.

2.5 Gamma waiting times and the event-count connection

Suppose the help desk collects three messages before starting a batch. Now we need the time until the third message, rather than the time until the next one or the number received by a fixed deadline.

Let W_1 be the wait from time zero to the first message, W_2 the gap from the first to the second, and so on. In our arrival model these gaps are independent and each follows $\text{Exp}(\lambda)$. The k-th arrival time is

$$T_k = W_1 + \cdots + W_k.$$

For example, the gaps in Figure 2.2 are 1.5, 2.5, and 4 minutes, so $T_3 = 8$ minutes.

Before deriving a new density, we can already find the mean and variance:

$$\mathbb{E}[T_k] = \frac{k}{\lambda}, \qquad \text{Var}(T_k) = \frac{k}{\lambda^2}.$$

Expectation adds by linearity; variance adds because the gaps are independent. At rate 0.2, waiting for three messages takes 15 minutes on average, with variance 75 minutes squared.

2.5.1 The distribution of a sum of waiting times

Consider two messages first. A total wait near five minutes could consist of a first gap near one minute and a second near four, or a first near two and a second near three, and so on. To find the density of the total, we must account for every possible split.

If the first gap is u, the second must be near $t - u$. Independence gives a product of densities for these two contributions. Integrating over $0 < u < t$ adds all the possible splits:

$$\begin{aligned} f_{T_2}(t) &= \int_0^t f_{W_1}(u) f_{W_2}(t-u)\, du \\ &= \int_0^t \lambda e^{-\lambda u} \lambda e^{-\lambda(t-u)}\, du = \lambda^2 t e^{-\lambda t}, \qquad t > 0. \end{aligned}$$

This way of finding the density of a sum is called *convolution*.

Repeating the same step for k independent gaps gives

$$f_{T_k}(t) = \frac{\lambda^k}{(k-1)!} t^{k-1} e^{-\lambda t}, \qquad t > 0.$$

For $k = 1$, this is the exponential density. For $k > 1$, the density starts near zero: fitting several arrivals into a very short time is unlikely. As t increases, enough time becomes available for all k arrivals. Eventually the density decreases because exceptionally long total waits are also unlikely.

2.5.2 From an integer number of arrivals to a shape parameter

The density above gives a useful family of positive, often skewed distributions. We can extend its shape beyond integer values of k. The only obstacle is the factorial $(k-1)!$. Its continuous extension is the gamma function:

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u}\, du, \qquad \alpha > 0.$$

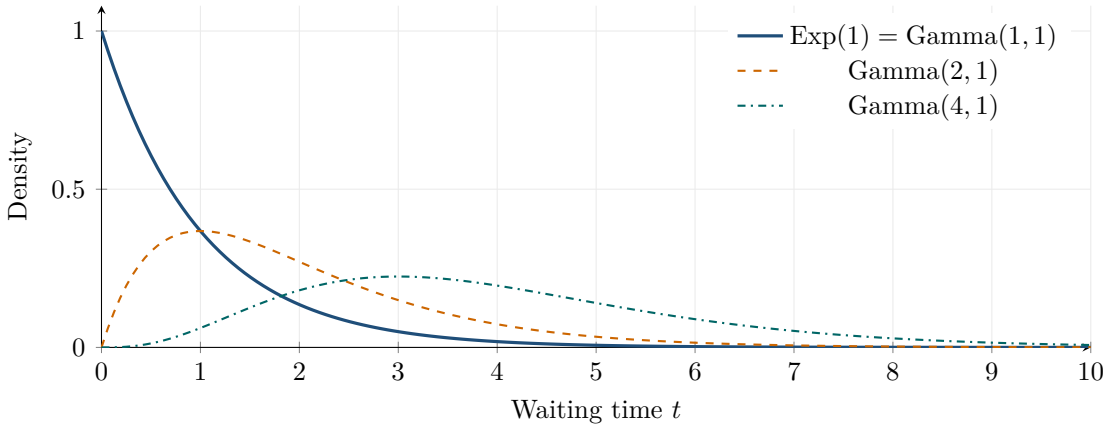


Figure 2.3: Time until the first, second, or fourth arrival, at rate one. One arrival can occur almost immediately. Several arrivals require a sum of waiting times, moving the typical total wait to the right. The means are $1, 2, 4$; the standard deviations are $1, \sqrt{2}, 2$.

Integration by parts gives $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$. Since $\Gamma(1) = 1$, this implies $\Gamma(k) = (k-1)!$ for positive integers k.

Replacing k by a positive real number α and the factorial by $\Gamma(\alpha)$ gives the *Gamma distribution*. We write its rate as β so that λ can continue to denote the particular inbox rate.

Definition 2.12: Gamma distribution

For shape $\alpha > 0$ and rate $\beta > 0$, $X \sim \text{Gamma}(\alpha, \beta)$ has density

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \qquad x > 0,$$

and zero density for $x < 0$.

The substitution $u = \beta x$ shows that its integral is one. The case $\alpha = 1$ is Exponential; integer $\alpha = k$ describes waiting for k arrivals at rate β. For noninteger shape, the density still makes sense, but there is no literal fractional-number-of-arrivals interpretation.

We can calculate the moments for every positive shape by inserting x or x^2 into the density integral:

$$\mathbb{E}[X] = \frac{\Gamma(\alpha+1)}{\beta\Gamma(\alpha)} = \frac{\alpha}{\beta},$$
$$\mathbb{E}[X^2] = \frac{\Gamma(\alpha+2)}{\beta^2\Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\beta^2}.$$

Subtracting the squared mean gives

$$\text{Var}(X) = \frac{\alpha}{\beta^2}.$$

Increasing the rate β shortens the waits. Increasing the shape α at fixed rate increases the mean, while the spread relative to that mean decreases: $\text{sd}(X)/\mathbb{E}[X] = 1/\sqrt{\alpha}$.

Throughout these notes the second Gamma parameter is a rate. Another common convention uses a scale $\theta = 1/\beta$, measured in the same units as X. In that convention the mean and variance are $\alpha\theta$ and $\alpha\theta^2$. Checking which parameterization is used prevents a reciprocal error.

2.5.3 A deadline question can be answered by counting

To have received the third message by minute ten means to have counted at least three messages by minute ten. These are the same event, expressed using different random variables:

$$\{T_k \le t\} = \{N(t) \ge k\}, \qquad k \ge 1. \tag{2.5}$$

Therefore, instead of integrating a Gamma density, we can add Poisson probabilities:

$$\mathbb{P}(T_k \le t) = 1 - \mathbb{P}(N(t) < k) = 1 - e^{-\lambda t} \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!}.$$

For rate 0.2, a ten-minute window has expected count two, so

$$\mathbb{P}(T_3 \le 10) = 1 - e^{-2}(1 + 2 + 2^2/2) \approx 0.3233.$$

Receiving exactly k messages also has a waiting-time description: the k-th has arrived, but the $(k+1)$-th has not. For $k \ge 1$, $\mathbb{P}(N(t) = k) = \mathbb{P}(T_k \le t) - \mathbb{P}(T_{k+1} \le t)$. For $k = 0$, it is simply $\mathbb{P}(T_1 > t)$.

2.6 Bernoulli and Binomial distributions

The inbox model asked whether a message arrived during each small time interval. Many other observations are also binary: a link was clicked or it was not, a transmission succeeded or failed, an item passed an inspection or failed it. We now describe these trials without reference to time.

2.6.1 One binary outcome

Code a click as one and no click as zero.

Definition 2.13: Bernoulli distribution

If the click probability is p, then

$$\mathbb{P}(B = 1) = p, \qquad \mathbb{P}(B = 0) = 1 - p, \qquad 0 \le p \le 1.$$

This is a *Bernoulli distribution*, written $B \sim \text{Bernoulli}(p)$.

The same construction applies to the indicator of any event.

For a binary variable, squaring changes neither possible value: $0^2 = 0$, $1^2 = 1$. Thus

$$\mathbb{E}[B] = p, \qquad \mathbb{E}[B^2] = p, \qquad \text{Var}(B) = p(1-p).$$

At $p = 1/2$, neither outcome is favored and the variance is largest. At $p = 0$ or $p = 1$, every observation is the same and variance is zero.

2.6.2 A fixed number of independent trials

Show a link to n people. Assume their clicks are independent and each has the same probability p. If B_i indicates the i-th click, then the total number of clicks is

$$S_n = B_1 + \cdots + B_n.$$

Definition 2.14: Binomial distribution

The number of successes in n independent Bernoulli trials with the same probability p follows a *Binomial distribution*, $S_n \sim \text{Binomial}(n, p)$, with

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

The displayed expression is read directly for $0 < p < 1$; at $p = 0$ or $p = 1$ the count is fixed.

The counting argument is the one used for the short arrival intervals: choose the k successful trials, multiply their success and failure probabilities, and add over all choices.

We do not need to sum this PMF to obtain its moments. Since the count is a sum of independent Bernoulli variables,

$$\mathbb{E}[S_n] = np, \qquad \text{Var}(S_n) = np(1-p).$$

For $n \geq 1$, the observed click proportion is $\widehat{p} = S_n/n$. The scaling rules for mean and variance give

$$\mathbb{E}[\widehat{p}] = p, \qquad \text{Var}(\widehat{p}) = \frac{p(1-p)}{n}.$$

With more independent observations, the proportion fluctuates less. For example, multiplying n by four halves its standard deviation.

Example 2.15 (A small click probability)**.** For 100 independent impressions with $p = 0.03$, the count has mean 3 and variance 2.91. The chance of no clicks is

$$\mathbb{P}(S_{100} = 0) = (1 - 0.03)^{100} \approx 0.0476.$$

The common-p assumption treats all impressions alike. We will revisit it when different users have different propensities to click.

2.6.3 Why rare Binomial successes look Poisson

The short-interval model contained many independent trials, each with a tiny success probability. A Binomial count behaves similarly whenever n is large and p is small, with $np = m$ held fixed. For any fixed k,

$$\binom{n}{k} \left(\frac{m}{n}\right)^k \left(1 - \frac{m}{n}\right)^{n-k} \longrightarrow e^{-m} \frac{m^k}{k!}.$$

This is the Poisson limit derived earlier. Its moments agree with the Binomial moments in the limit:

$$np = m, \qquad np(1-p) \longrightarrow m.$$

For the click example, the approximation is Poisson(3), which gives no-click probability $e^{-3} \approx 0.0498$, close to the exact 0.0476.

2.7 Geometric waiting times

Now show the link repeatedly until the first click, instead of fixing the number of impressions in advance. Let G be the number of impressions needed, including the successful one. For example, the sequence

$$\text{no click}, \quad \text{no click}, \quad \text{click}$$

gives $G = 3$.

Definition 2.16: Geometric distribution

Assume independent trials with the same success probability $0 < p \leq 1$. For $0 < p < 1$, the event $G = k$ requires $k - 1$ failures followed by one success:

$$\mathbb{P}(G = k) = (1 - p)^{k-1} p, \qquad k = 1, 2, \ldots.$$

This is the *Geometric distribution*, $G \sim \text{Geometric}(p)$. If $p = 1$, the first trial always succeeds and $G = 1$.

2.7.1 Waiting longer means accumulating failures

The event $G > k$ means that all of the first k trials failed. Hence

$$\mathbb{P}(G > k) = (1 - p)^k, \qquad k = 0, 1, \ldots.$$

With $p = 0.1$, the chance of still waiting for a success after twenty trials is $0.9^{20} \approx 0.1216$.

This model has the same memoryless logic as exponential waiting. Once the first m trials have failed, the next trials still have the original success probability. Formally, for $0 < p < 1$,

$$\mathbb{P}(G > m + k \mid G > m) = \frac{(1 - p)^{m+k}}{(1 - p)^m} = (1 - p)^k.$$

An example is a scrolling session in which a user independently stops after each video with probability p. The number of videos watched is Geometric. If fatigue makes stopping more likely after each video, the constant-p assumption no longer holds.

2.7.2 Mean and variance

To compute the mean, weight each possible stopping trial by its probability. Write $q = 1 - p$. The geometric series

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1 - q}, \qquad |q| < 1,$$

can be differentiated to obtain the factor k needed in the mean:

$$\sum_{k=1}^{\infty} k q^{k-1} = \frac{1}{(1 - q)^2}.$$

Multiplying by p gives $\mathbb{E}[G] = 1/p$. A second differentiation gives

$$\mathbb{E}[G(G - 1)] = p \sum_{k=2}^{\infty} k(k - 1) q^{k-1} = \frac{2(1 - p)}{p^2}.$$

As in the Poisson calculation, use $G^2 = G(G-1) + G$ to get

$$\mathrm{Var}(G) = \frac{1-p}{p^2}.$$

At $p = 0.1$, the mean is ten trials and the variance is ninety.

Some sources count failures before success instead. That variable is $H = G - 1$, so its mean is $(1-p)/p$, its variance is unchanged, and it can equal zero. In these notes, Geometric always counts the successful trial too.

2.7.3 Turning a trial count into a time

Return to the inbox approximation, where one trial corresponds to a short interval of duration Δ minutes. A count of G intervals represents a waiting time of ΔG minutes. Set $p = \lambda\Delta$, just as in the first section.

By time t, we have completed $\lfloor t/\Delta \rfloor$ intervals; the floor symbol means the integer number of whole intervals. No arrival by that time has probability

$$\mathbb{P}(\Delta G > t) = (1 - \lambda\Delta)^{\lfloor t/\Delta \rfloor} \longrightarrow e^{-\lambda t} \qquad (\Delta \downarrow 0).$$

The Geometric waiting count therefore becomes an Exponential waiting time as the time intervals shrink. The moments show the same connection:

$$\mathbb{E}[\Delta G] = \frac{1}{\lambda}, \qquad \mathrm{Var}(\Delta G) = \frac{1-\lambda\Delta}{\lambda^2} \longrightarrow \frac{1}{\lambda^2}.$$

The time-unit conversion is what connects the two distributions: Geometric counts trials; Exponential measures elapsed time.

2.8 The Gaussian distribution

The waiting times studied so far are nonnegative and usually skewed: a wait can be much longer than average, but cannot extend below zero. Measurement errors pose a different problem. A sensor measuring a fixed quantity may sometimes read too high and sometimes too low. Several small disturbances can add together, with positive and negative contributions partly cancelling.

A useful model for such errors has a central peak and decreases symmetrically on either side. This is the *Gaussian*, also called the *normal*, distribution. It is often a useful approximation when many small independent disturbances contribute and no one disturbance dominates. The connection between sums and Gaussian approximations will be developed further in the chapter on limit theorems.

2.8.1 Center and width

Let μ describe the center of the errors and $\sigma > 0$ their scale. The center can be nonzero if the sensor is systematically biased.

Definition 2.17: Gaussian distribution

For $\mu \in \mathbb{R}$ and $\sigma > 0$, the density is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], \qquad x \in \mathbb{R}.$$

We write $X \sim \mathcal{N}(\mu, \sigma^2)$.

The expression $(x-\mu)/\sigma$ measures how far x lies from the center relative to the width of the curve. Squaring it treats positive and negative deviations equally. The negative exponential makes values far from the center less likely. Increasing σ widens the curve; the factor $1/\sigma$ reduces its height to preserve total area one.

The simplest member is $Z \sim \mathcal{N}(0,1)$, with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

The constant $\sqrt{2\pi}$ is the value of $\int_{-\infty}^{\infty} e^{-z^2/2}\,dz$, so this density integrates to one. Every other Gaussian can be obtained by shifting and scaling:

$$X = \mu + \sigma Z.$$

2.8.2 Why the parameters are the mean and standard deviation

Symmetry of ϕ means that the contributions of z and $-z$ to the mean cancel, giving $\mathbb{E}[Z] = 0$. For the second moment, differentiating the density gives $\phi'(z) = -z\phi(z)$. This turns the moment integral into an integration-by-parts calculation:

$$\begin{aligned}\mathbb{E}[Z^2] &= \int_{-\infty}^{\infty} z^2\phi(z)\,dz = -\int_{-\infty}^{\infty} z\phi'(z)\,dz \\ &= [-z\phi(z)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z)\,dz = 1.\end{aligned}$$

Thus $\mathrm{Var}(Z) = 1$. Applying the shift and scale rules to $X = \mu + \sigma Z$ gives

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2, \qquad \mathrm{sd}(X) = \sigma.$$

This justifies the parameters' interpretation. In $\mathcal{N}(\mu, \sigma^2)$, the second argument is the variance.

2.8.3 Expressing an error in standard-deviation units

For a sensor with mean error zero, an error of 0.4 units may be large or small relative to its usual variation. If the standard deviation is 0.2, that error is two standard deviations above the mean; if it is 2, it is only 0.2 standard deviations above the mean.

The transformation

$$Z = \frac{X-\mu}{\sigma}$$

subtracts the center and expresses the deviation in standard-deviation units. This is *standardization*. For a Gaussian X, the result always follows $\mathcal{N}(0,1)$. We can therefore answer interval questions for any Gaussian using a single CDF:

$$\Phi(z) = \mathbb{P}(Z \le z) = \int_{-\infty}^{z} \phi(u)\,du.$$

The endpoints of the original interval are transformed in the same way:

$$\mathbb{P}(a < X \le b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

The integral defining Φ is evaluated numerically. Symmetry gives $\Phi(-z) = 1 - \Phi(z)$.

Example 2.18 (How often does a sensor make a large error?)**.** Suppose the error is $E \sim \mathcal{N}(0, 0.2^2)$. An absolute error larger than 0.4 is an error more than two standard deviations from zero. The two symmetric tails give

$$\mathbb{P}(|E| > 0.4) = \mathbb{P}(|Z| > 2) = 2[1 - \Phi(2)] \approx 0.0455.$$

Under this model, about 4.55% of repeated readings have such an error. Likewise,

$$\mathbb{P}(|Z| \le 1) \approx 0.6827, \qquad \mathbb{P}(|Z| \le 1.96) \approx 0.95.$$

The central 95% interval for the sensor error is therefore

$$[-1.96(0.2), 1.96(0.2)] = [-0.392, 0.392].$$

2.8.4 Adding errors and connecting back to earlier distributions

Changing units or adding a fixed bias preserves the Gaussian shape:

$$aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2), \qquad a \neq 0.$$

For $a = 0$, the result is simply the constant b.

Independent Gaussian errors also add to another Gaussian. If $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ are independent, then

$$X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

The convolution method used for Gamma waiting times gives this density. The means and variances agree with the addition rules we already know.

For example, two independent centered errors with standard deviations 0.2 and 0.3 produce a total error with variance $0.2^2 + 0.3^2 = 0.13$, and standard deviation $\sqrt{0.13} \approx 0.361$. Standard deviations do not simply add.

A sum can also be approximately Gaussian even when its individual terms are not Gaussian. We have already met two such sums: Binomial counts add Bernoulli outcomes, and integer-shape Gamma waits add exponential gaps. For sufficiently large parameters, useful approximations are

$$\begin{aligned} \text{Binomial}(n, p) &\approx \mathcal{N}(np, np(1-p)), \\ \text{Gamma}(k, \lambda) &\approx \mathcal{N}(k/\lambda, k/\lambda^2), \\ \text{Poisson}(m) &\approx \mathcal{N}(m, m). \end{aligned}$$

Each Gaussian uses the mean and variance of the distribution being approximated. For the Binomial, both the expected successes np and failures $n(1-p)$ should be large; for Gamma and Poisson, the approximations improve as k and m grow.

The Gaussian allows negative values, while counts and waiting times do not. Thus approximating a very small count or a strongly skewed waiting time by a Gaussian can assign substantial probability to impossible outcomes. The original distributions remain the appropriate starting point in those cases.

2.9 Mixtures and heterogeneity

Suppose two sensors measure the same reference quantity. One tends to read two units too low; the other tends to read two units too high. Within either sensor, the random error has standard deviation one.

Choose a sensor with equal probability for each reading. If we combine all readings without keeping track of which sensor produced them, what distribution do the errors follow? Averaging the two sensor biases gives zero, but that does not mean either sensor usually makes an error near zero. Most errors are still near -2 or 2.

2.9.1 Choose a source, then draw an observation

Let Z record which sensor is selected:

$$\mathbb{P}(Z = 1) = \mathbb{P}(Z = 2) = \frac{1}{2}.$$

Given the chosen sensor, the error distribution is

$$X \mid Z = 1 \sim \mathcal{N}(-2, 1), \qquad X \mid Z = 2 \sim \mathcal{N}(2, 1).$$

The notation $X \mid Z = k$ means that we are considering only readings from sensor k.

Write the two Gaussian densities as f_1 and f_2. For any interval, half the readings come from each sensor, so the law of total probability gives half of each sensor's interval probability. The density is therefore

$$f_X(x) = \frac{1}{2} f_1(x) + \frac{1}{2} f_2(x).$$

This is a *mixture distribution*. Its two peaks reflect the two sources of readings.

Definition 2.19: Finite mixture

The same construction works with K sources chosen with probabilities $\pi_1, \ldots, \pi_K$:

$$f_X(x) = \sum_{k=1}^{K} \pi_k f_k(x), \qquad \pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1. \tag{2.6}$$

The f_k are called the mixture's *components*. For discrete observations, use component PMFs in the same formula.

If the source label is not recorded, it is an unobserved, or *latent*, variable.

2.9.2 Where the pooled variance comes from

In the two-sensor example the overall mean is

$$\mathbb{E}[X] = \tfrac{1}{2}(-2) + \tfrac{1}{2}(2) = 0.$$

Within either sensor, the second moment equals its variance plus the square of its mean:

$$\mathbb{E}[X^2 \mid Z = 1] = 1 + (-2)^2 = 5, \qquad \mathbb{E}[X^2 \mid Z = 2] = 1 + 2^2 = 5.$$

Averaging these gives $\mathbb{E}[X^2] = 5$. Hence the pooled variance is $5 - 0^2 = 5$, even though each sensor separately has variance one.

There are two sources of spread: random fluctuation within a sensor, and the difference between the two sensors' centers. To separate them in general, let component k have mean μ_k and variance σ_k^2. Then

$$\mu = \mathbb{E}[X] = \sum_k \pi_k \mu_k, \qquad \mathbb{E}[X^2] = \sum_k \pi_k (\sigma_k^2 + \mu_k^2).$$

Result 2.20: Within-source and between-source variance

For a finite mixture with finite second moments, subtracting μ^2 and regrouping the terms gives

$$\mathrm{Var}(X) = \underbrace{\sum_k \pi_k \sigma_k^2}_{\text{variation within sources}} + \underbrace{\sum_k \pi_k (\mu_k - \mu)^2}_{\text{variation between sources}} . \tag{2.7}$$

For the sensors, these contributions are 1 and 4, respectively.

The inner averages can be written as conditional expectations. The quantity $\mathbb{E}[X \mid Z]$ equals μ_k when source k is selected; it varies because the selected source varies. Similarly, $\mathrm{Var}(X \mid Z)$ is the variance inside the selected source. The outer expectation averages over the choice of source. With this notation, the two identities become

Theorem 2.21: Total expectation and total variance

For the finite mixture above, with finite second moments,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Z]], \tag{2.8}$$
$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Z)] + \mathrm{Var}(\mathbb{E}[X \mid Z]). \tag{2.9}$$

2.9.3 The same mean and variance can hide different shapes

A single Gaussian $\mathcal{N}(0, 5)$ has the same mean and variance as the two-sensor mixture. Its most likely region is near zero, whereas the mixture has two peaks near the sensor biases (Figure 2.4).

For example, in sensor 1 the event $-1 \le X \le 1$ corresponds to a standardized error between one and three. Sensor 2 gives the same probability by symmetry. Therefore the pooled probability is

$$\mathbb{P}(|X| \le 1) = \Phi(3) - \Phi(1) \approx 0.1573.$$

The Gaussian $\mathcal{N}(0, 5)$ instead gives $2\Phi(1/\sqrt{5}) - 1 \approx 0.3453$. Knowing the mean and variance alone does not determine interval probabilities.

Also distinguish selecting one sensor from averaging two readings. If we take one independent reading from *each* sensor and average them, their opposite biases cancel:

$$\frac{X_1 + X_2}{2} \sim \mathcal{N}(0, 1/2).$$

A mixture chooses which source supplies an observation. An average combines observations from both sources. These are different experiments.

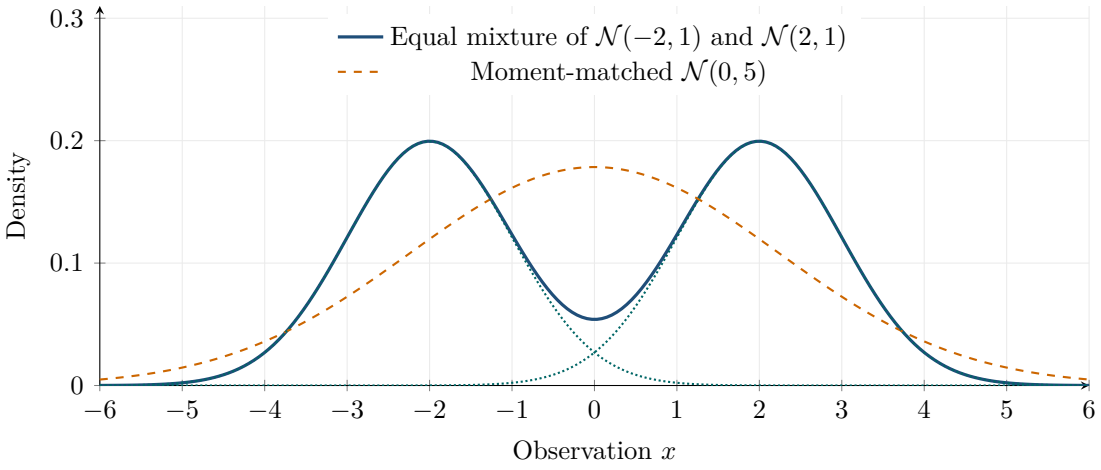


Figure 2.4: Pooling two biased sensors produces the blue mixture density. The dotted curves are the weighted sensor densities, which add to the blue curve. A single Gaussian with the same mean and variance puts substantially more probability near zero.

2.9.4 Different users have different message rates

Two sensors give a small number of groups. In a user population, the differences can vary continuously. Some users receive very few messages; others receive many. This variation between members of a population is called *heterogeneity*.

Observe each user for one day. Let Λ be the user's expected number of messages in that day. A particular user's value, denoted λ, is fixed during the observation; the uppercase variable Λ varies when we choose a different user. Assume that, for a user with value λ,

$$N \mid \Lambda = \lambda \sim \text{Poisson}(\lambda).$$

Thus even for a fixed user the realized count N fluctuates from day to day. Across users there is additional variation in the mean count.

The within-user mean and variance both equal Λ. Averaging over users, the laws of total expectation and variance give

$$\mathbb{E}[N] = \mathbb{E}[\Lambda], \qquad \text{Var}(N) = \mathbb{E}[\Lambda] + \text{Var}(\Lambda).$$

The first contribution to the variance is daily Poisson variation. The second is variation in the users' average activity. If the latter is positive, the pooled variance exceeds the pooled mean. This is called *overdispersion* relative to a Poisson distribution.

Replacing every user's rate by the population average removes this second contribution. It also misses the extra zeros contributed by quiet users and the large counts contributed by busy users.

2.9.5 A Gamma distribution of Poisson means

We need a distribution for the positive quantity Λ. One possible choice is the Gamma family, whose shape and rate let us describe different amounts of variation among users:

$$\Lambda \sim \text{Gamma}(\alpha, \beta).$$

The Gamma moments immediately give

$$\mathbb{E}[N] = \frac{\alpha}{\beta}, \qquad \operatorname{Var}(N) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

For $\alpha = 2, \beta = 1/2$, the mean user count is four and the variance between user means is eight. The pooled count therefore has mean four and variance $4 + 8 = 12$.

We can also find the full count distribution. Let $g(\lambda)$ be the Gamma density of user means. To find the probability of k messages, average the Poisson probability over all possible user means:

$$\mathbb{P}(N = k) = \int_0^\infty \mathbb{P}(N = k \mid \Lambda = \lambda)\, g(\lambda)\, d\lambda.$$

This is the continuous version of the weighted sum for the sensors: possible user means replace the discrete source labels, and the integral replaces the sum. The count remains discrete; it is the mixing parameter that varies continuously.

Substituting the Poisson PMF and Gamma density gives

$$\begin{aligned}\mathbb{P}(N = k) &= \int_0^\infty \frac{e^{-\lambda}\lambda^k}{k!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}\, d\lambda \\ &= \frac{\Gamma(k+\alpha)}{\Gamma(\alpha)k!} \left(\frac{\beta}{\beta+1}\right)^\alpha \left(\frac{1}{\beta+1}\right)^k, \qquad k = 0, 1, \ldots.\end{aligned}$$

The second line follows by combining the powers and exponentials, then using the gamma integral with shape $k+\alpha$ and rate $\beta+1$. The resulting distribution is called *negative binomial*. Here its meaning is supplied by the construction: a Poisson count whose mean differs from user to user according to a Gamma distribution.

For our numerical example,

$$\mathbb{P}(N = 0) = \left(\frac{1/2}{1 + 1/2}\right)^2 = \frac{1}{9} \approx 0.1111.$$

A single Poisson(4) distribution instead gives $e^{-4} \approx 0.0183$. The mean counts agree, but the predicted fraction of users receiving no messages is very different.

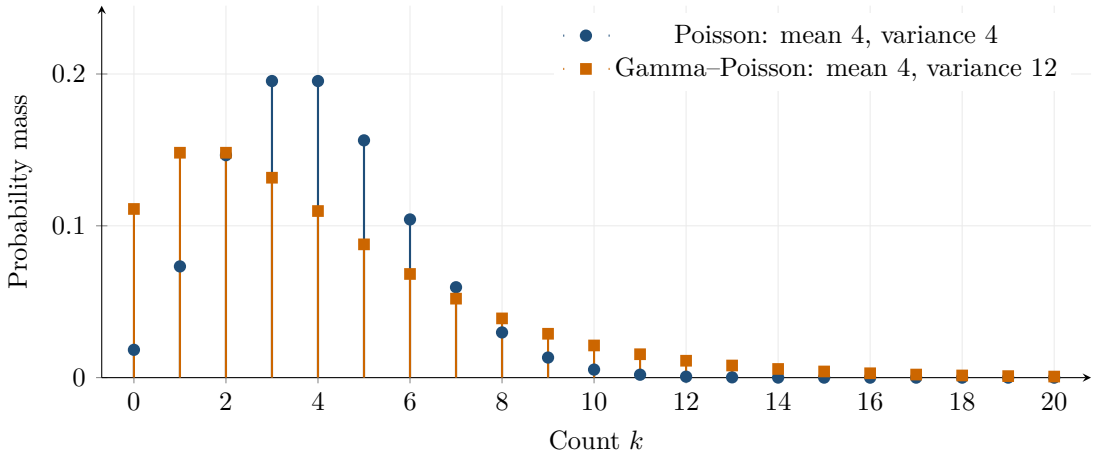


Figure 2.5: Both populations average four messages per user per day. With a common Poisson mean of four, the variance is four. With Gamma-distributed user means, the variance is twelve, with more users receiving zero messages or unusually many messages.

The conditional-moment rules used here extend the finite-mixture identities by replacing weighted sums with integrals. They require finite moments, but they do not require us to evaluate the full mixture PMF before calculating its mean and variance.

2.9.6 Ten responses from one user or one response from ten users?

Heterogeneity also matters for binary observations, but the way we group observations changes the result. Suppose half the users click with probability 0.1, and half with probability 0.9. A randomly chosen user's response is a click with probability

$$\tfrac{1}{2}(0.1) + \tfrac{1}{2}(0.9) = 0.5.$$

That overall probability does not tell us how much the individual user probabilities differ.

Now compare two experiments. In the first, choose one user and show them ten posts, with independent responses once their click probability is fixed. The chosen user is likely to click about once or about nine times, depending on which group they belong to. In the second, choose ten users independently and record one response from each. Each response is independently a click with probability 0.5, so the count is Binomial$(10, 0.5)$.

The means are both five. The variances are different. Write P for the first experiment's shared user probability and S for the ten-response count. Within either user group, the variance is $10(0.1)(0.9) = 0.9$. The group means are one and nine, whose variance around five is sixteen. Therefore

$$\operatorname{Var}(S) = 0.9 + 16 = 16.9,$$

whereas the second experiment has variance $10(0.5)(0.5) = 2.5$.

More generally, let $q = \mathbb{E}[P]$ and suppose $S \mid P \sim \text{Binomial}(n, P)$. Then

$$\begin{aligned} \mathbb{E}[S] &= nq, \\ \operatorname{Var}(S) &= n\mathbb{E}[P(1-P)] + n^2 \operatorname{Var}(P) \\ &= nq(1-q) + n(n-1)\operatorname{Var}(P). \end{aligned}$$

The final step uses $\mathbb{E}[P(1-P)] = q(1-q) - \operatorname{Var}(P)$, obtained by expanding $\mathbb{E}[P^2] = \operatorname{Var}(P) + q^2$. The extra variance comes from sharing the same random user probability across trials.

If a fresh P_i is independently drawn for each trial instead, and the responses are independently generated given those probabilities, the responses are independent Bernoulli variables with parameter q. Their sum is Binomial(n, q). Which quantities are shared across observations is part of specifying the experiment.

2.10 Extension: Gaussian scale mixtures and Student's t

Return to measurement errors. So far the two sensors differed in their average errors. Another possibility is that a sensor is usually precise but occasionally operates under much noisier conditions, without changing its average error.

For example, suppose 95% of readings have error $\mathcal{N}(\mu, \sigma^2)$, and the remaining 5% have standard deviation 5σ:

$$X \sim 0.95\,\mathcal{N}(\mu, \sigma^2) + 0.05\,\mathcal{N}(\mu, 25\sigma^2).$$

This is called a *contaminated Gaussian* model. The noisier readings make large errors more common. The mean stays μ, while the mixture variance is

$$\operatorname{Var}(X) = 0.95\sigma^2 + 0.05(25\sigma^2) = 2.2\sigma^2.$$

Instead of two noise levels, we can allow a continuous range. Describe the noise using its inverse variance τ, often called *precision*. A small τ means a large variance and a noisy

reading; a large τ means a narrow error distribution. For a standardized error, use

$$Z \mid \tau \sim \mathcal{N}(0, 1/\tau).$$

Model the positive precision by

$$\tau \sim \text{Gamma}(\nu/2, \nu/2), \qquad \nu > 0.$$

Using equal shape and rate fixes $\mathbb{E}[\tau] = 1$; the Gamma variance formula gives $\text{Var}(\tau) = 2/\nu$. Thus ν controls how much the noise level varies: large ν keeps precision close to one, while small ν permits more variation.

Average the conditional Gaussian density over these precision values, just as we averaged Poisson probabilities over user means. Let g_τ denote the Gamma density and $f_{Z|\tau}(z \mid u)$ the Gaussian density with variance $1/u$. Then

$$\begin{aligned} f_Z(z) &= \int_0^\infty f_{Z|\tau}(z \mid u)\, g_\tau(u)\, du \\ &= \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{z^2}{\nu}\right)^{-(\nu+1)/2}. \end{aligned}$$

The product inside the integral is proportional, as a function of u, to $u^{(\nu-1)/2} e^{-(\nu+z^2)u/2}$. This is another gamma integral, now with shape $(\nu+1)/2$ and rate $(\nu+z^2)/2$. The resulting density defines *Student's t-distribution*, written $Z \sim t_\nu$. In this construction, ν controls variation in precision and therefore how often extreme errors occur.

Occasionally drawing a small precision produces an unusually large Gaussian error. After averaging over precisions, extreme errors remain much more likely than under a standard Gaussian. This is the meaning of *heavy tails* here: for large $|z|$, the density decreases like $|z|^{-(\nu+1)}$, rather than like $e^{-z^2/2}$.

For $\nu > 1$, the mean exists and is zero. For $\nu > 2$, total variance and the Gamma integral give

$$\text{Var}(Z) = \mathbb{E}[1/\tau] = \frac{\nu}{\nu - 2}.$$

When $1 < \nu \leq 2$, rare large observations make the second-moment integral diverge, so the variance is infinite. When $0 < \nu \leq 1$, even the absolute first moment diverges and the mean is undefined.

For an error with center μ and scale $s > 0$, write $X = \mu + sZ$. For $\nu > 2$,

$$\mathbb{E}[X] = \mu, \qquad \text{Var}(X) = s^2 \frac{\nu}{\nu - 2}.$$

The scale s is therefore not its standard deviation. As ν increases, precision concentrates near one and the standard Student distribution approaches $\mathcal{N}(0, 1)$.

2.11 Distribution reference

The table collects the means and variances derived above. The support lists the values the variable can take, distinguishing counts from waiting times and signed errors.

Distribution	**Support**	$\mathbb{E}[X]$	$\mathbf{Var}(X)$
$\mathrm{Exp}(\lambda)$	$[0, \infty)$	$1/\lambda$	$1/\lambda^2$
$\mathrm{Poisson}(m)$	$0, 1, \ldots$	m	m
$\mathrm{Gamma}(\alpha, \beta)$	$[0, \infty)$	α/β	α/β^2
$\mathrm{Bernoulli}(p)$	$0, 1$	p	$p(1-p)$
$\mathrm{Binomial}(n, p)$	$0, \ldots, n$	np	$np(1-p)$
$\mathrm{Geometric}(p)$	$1, 2, \ldots$	$1/p$	$(1-p)/p^2$
$\mathcal{N}(\mu, \sigma^2)$	$\mathbb{R}$	μ	σ^2
t_ν	$\mathbb{R}$	0 $(\nu > 1)$	$\nu/(\nu-2)$ $(\nu > 2)$

Here $\lambda, \alpha, \beta, m, \nu > 0$, $\sigma > 0$, and n is a nonnegative integer. The Bernoulli and Binomial parameters allow $0 \le p \le 1$; the Geometric requires $0 < p \le 1$. Gamma uses shape and rate. Geometric counts trials including the first success. Student moments exist only under the conditions shown.

Connection	**Required mechanism or condition**
Exponential to Gamma	Sum independent waiting times with a common rate.
Exponential to Poisson	Count arrivals generated by independent exponential interarrival times.
Bernoulli to Binomial	Sum a fixed number of independent trials with the same success probability.
Binomial to Poisson	Take $n \to \infty$, $p \to 0$, with $np = m$.
Geometric to Exponential	Rescale trial duration to Δ, set $p = \lambda\Delta$, and let $\Delta \downarrow 0$.
Poisson to negative binomial	Mix the Poisson rate with a Gamma distribution.
Gaussian to Student	Mix the Gaussian precision with $\mathrm{Gamma}(\nu/2, \nu/2)$.

Chapter 3

Joint Distributions, Dependence, and Random Vectors

3.1 Joint and marginal distributions

In the shared-user experiment of Section 2.9, record the first two responses separately: $X, Y \in \{0, 1\}$, rather than only their total.

Definition 3.1: Joint probability mass function

The *joint probability mass function*

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$$

records the probability of each pair.

With the same equally likely user probabilities 0.1 and 0.9, it is the table below. For example, the two-click cell is $p_{X,Y}(1, 1) = \frac{1}{2}(0.1)^2 + \frac{1}{2}(0.9)^2 = 0.41$.

	$Y = 0$	$Y = 1$	Row total
$X = 0$	0.41	0.09	0.50
$X = 1$	0.09	0.41	0.50
Column total	0.50	0.50	1

3.1.1 Marginalizing a table

If we only care about the first response, either value of the second is acceptable. We therefore add across a row.

Result 3.2: Marginal PMFs

Sum the joint PMF over the other variable:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y). \tag{3.1}$$

These are the *marginal* PMFs, written in the margins of the table.

Marginalizing means ignoring a variable by summing over its possible values, not setting it to zero.

If the two responses instead came from independently selected users, every interior cell would be 0.25. The row and column totals would still be 0.5. Thus the same two marginal distributions can belong to very different joint distributions.

3.1.2 Conditional PMFs and factorization

Applying conditional probability (1.5) to the table gives

$$\mathbb{P}(Y = 1 \mid X = 1) = \frac{0.41}{0.50} = 0.82, \qquad \mathbb{P}(Y = 1 \mid X = 0) = \frac{0.09}{0.50} = 0.18.$$

Definition 3.3: Conditional PMF

For a fixed x, the *conditional PMF* is the selected row divided by its total. In notation, for $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x, y)}{p_X(x)}, \qquad p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y \mid x). \tag{3.2}$$

Result 3.4: Independence and factorization

The independence condition (1.8), applied to all pairs of values, becomes

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y. \tag{3.3}$$

The table fails this test: $0.41 \neq 0.5 \cdot 0.5$. Once the user's probability P is fixed, however, the conditional joint PMF factors. This is *conditional independence*, written $X \perp\!\!\!\perp Y \mid P$. The shared-user effect from Chapter 2 is therefore a distinction between conditional and unconditional factorization, not a causal effect of one click on the other.

3.2 Joint densities: probability over a region

For a pair with a joint density, replace the probability table by a density over the plane. Near (x, y), a small rectangle of side lengths $\Delta x, \Delta y$ has probability approximately $f_{X,Y}(x, y)\Delta x\Delta y$. Adding small rectangles becomes a double integral.

Definition 3.5: Joint density

For a pair with a joint density and a region R,

$$\mathbb{P}((X, Y) \in R) = \iint_R f_{X,Y}(x, y)\, dx\, dy.$$

The density is nonnegative, and its integral over the whole plane is one.

3.2.1 Two arrival times, put in order

Suppose two people each independently choose an arrival time uniformly within a one-hour window. Let U, V be their times in hours after the window begins. Their unordered pair is uniform on the unit square: $f_{U,V}(u, v) = 1$ for $0 < u, v < 1$.

Now record the earlier and later times,

$$T_1 = \min(U, V), \qquad T_2 = \max(U, V).$$

For $t_1 < t_2$, the ordered pair near (t_1, t_2) can arise from (U, V) near either (t_1, t_2) or (t_2, t_1). These two equally sized regions contribute equally. Hence

$$f_{T_1,T_2}(t_1, t_2) = \begin{cases} 2, & 0 < t_1 < t_2 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The allowed triangle has area $1/2$; multiplying by the density two gives total probability one. Sorting has introduced dependence: the later time cannot be smaller than the earlier time.

For instance, both arrivals occur in the first half-hour exactly when $T_2 \le 1/2$. The relevant triangle has area $\frac{1}{2}(\frac{1}{2})^2 = 1/8$, so

$$\mathbb{P}(T_2 \le 1/2) = 2 \cdot \frac{1}{8} = \frac{1}{4}.$$

This agrees with the probability that both independent original times fall in the first half of the window.

3.2.2 Marginalizing integrates along a slice

Fix an earlier time t. The later time can range from t to one, so the marginal density of the earlier time is

$$f_{T_1}(t) = \int_t^1 2\, ds = 2(1 - t), \qquad 0 < t < 1.$$

There are more possible later arrivals when the first arrival is early. Conversely, if the later arrival is at s, the earlier one can be anywhere between zero and s:

$$f_{T_2}(s) = \int_0^s 2\, dt = 2s, \qquad 0 < s < 1.$$

Result 3.6: Marginal densities

The general marginalization rules are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx. \tag{3.4}$$

Although the bounds are written over the whole line, the integrand is zero outside the joint support. The triangular support is what produced the variable integration limits in the example.

3.2.3 A conditional density is a normalized slice

If the first arrival is at time t, how much longer do we wait for the second? Dividing the joint density by the earlier-time marginal gives

$$f_{T_2|T_1}(s \mid t) = \frac{2}{2(1 - t)} = \frac{1}{1 - t}, \qquad t < s < 1.$$

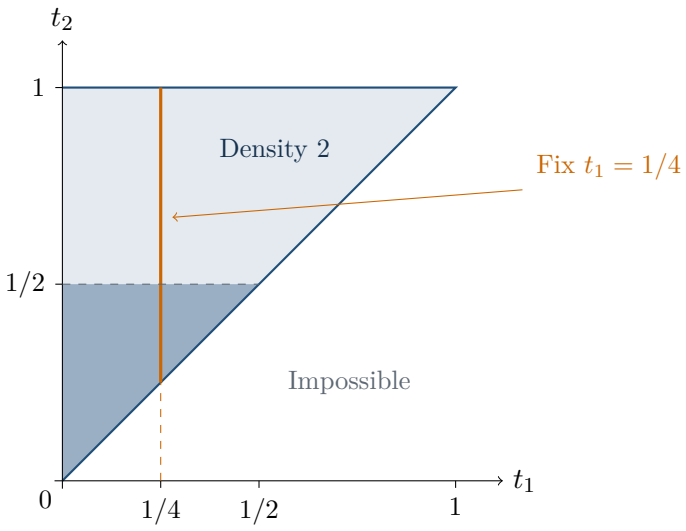


Figure 3.1: Every possible ordered pair lies above the diagonal $t_2 = t_1$. The dark triangle represents two arrivals before half an hour. At a fixed $t_1 = 1/4$, the later time ranges along the marked vertical segment from $1/4$ to one. Marginalizing adds density along such a segment; conditioning normalizes the segment's density.

Thus, conditional on the earlier time, the later time is uniform over the remaining part of the window. At $t = 1/4$, for example,

$$\mathbb{P}(T_2 > 3/4 \mid T_1 = 1/4) = \int_{3/4}^{1} \frac{1}{1 - 1/4}\, ds = \frac{1}{3}.$$

Definition 3.7: Conditional density

Wherever $f_X(x) > 0$, the conditional-density formula is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}. \tag{3.5}$$

In a smooth density model, conditioning on a shrinking interval around x gives this normalized slice. The denominator is a marginal density, not a point probability. Conditional densities are specified up to changes on sets of conditioning values with probability zero.

For a pair with a joint density, independence is equivalent to $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ almost everywhere. The ordered times fail this test: both marginals are positive inside $(0, 1)$, but their joint density is zero whenever $t_1 > t_2$.

3.2.4 Functions of the pair

Applying the expectation rule (2.2) to the joint distribution gives

$$\mathbb{E}[g(X, Y)] = \begin{cases} \displaystyle\sum_x \sum_y g(x, y) p_{X,Y}(x, y), & \text{discrete pair,} \\ \displaystyle\iint_{\mathbb{R}^2} g(x, y) f_{X,Y}(x, y)\, dx\, dy, & \text{pair with a density.} \end{cases} \tag{3.6}$$

We require $\mathbb{E}[|g(X,Y)|] < \infty$ for a finite expectation. For example, the average gap between the ordered arrivals is

$$\begin{aligned}\mathbb{E}[T_2 - T_1] &= \int_0^1 \int_t^1 (s-t)\, 2\, ds\, dt \\ &= \int_0^1 (1-t)^2\, dt = \frac{1}{3} \text{ hour.}\end{aligned}$$

3.3 Correlation and nonlinear dependence

For variables with finite second moments, expanding the covariance definition from Section 2.3 gives a convenient calculation from a joint distribution:

$$\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \tag{3.7}$$

For the click table, XY is one only in the $(1,1)$ cell, so $\mathbb{E}[XY] = 0.41$ and

$$\operatorname{Cov}(X,Y) = 0.41 - (0.5)(0.5) = 0.16.$$

3.3.1 Correlation removes the units

Converting one measurement from minutes to seconds multiplies its covariance with another by 60, although the relationship has not changed. To remove this dependence on units, use *correlation*.

Definition 3.8: Correlation

For means μ_X, μ_Y and finite, positive standard deviations σ_X, σ_Y, it is

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y} = \mathbb{E}\left[\frac{X-\mu_X}{\sigma_X}\frac{Y-\mu_Y}{\sigma_Y}\right]. \tag{3.8}$$

This is the average product of the standardized deviations. Changing minutes to seconds no longer affects the answer. Reversing the sign of one variable reverses the sign of the correlation. For our clicks, $\sigma_X = \sigma_Y = 0.5$, hence $\rho_{X,Y} = 0.64$.

To see why correlation lies between -1 and one, put $U = (X-\mu_X)/\sigma_X$ and $V = (Y-\mu_Y)/\sigma_Y$. Both have mean zero and variance one, so

$$0 \le \mathbb{E}[(U-V)^2] = 2 - 2\rho_{X,Y}, \qquad 0 \le \mathbb{E}[(U+V)^2] = 2 + 2\rho_{X,Y}.$$

At $\rho = 1$, the first square has expectation zero, so $U = V$ with probability one. At $\rho = -1$, $U = -V$ with probability one. Thus perfect correlation means an exact affine relationship, with positive or negative slope. Correlation is undefined if either variable has variance zero: a constant has no fluctuations to standardize.

3.3.2 Zero covariance can hide complete dependence

Let X be uniform on $[-1,1]$, and let $Y = X^2$. Then

$$\mathbb{E}[X] = 0, \qquad \mathbb{E}[Y] = \mathbb{E}[X^2] = \frac{1}{3}, \qquad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

The odd powers average to zero by symmetry. Thus $\mathrm{Cov}(X, Y) = 0$, and the correlation is zero since both variances are positive. Yet knowing X tells us Y exactly.

An event calculation also rules out independence. Here $Y \le 1/4$ is exactly the event $|X| \le 1/2$, so

$$\mathbb{P}(Y \le 1/4 \mid |X| \le 1/2) = 1, \qquad \mathbb{P}(Y \le 1/4) = \frac{1}{2}.$$

Correlation detects alignment of signed deviations, not every possible relationship. An upward-then-downward or U-shaped relationship can have zero correlation while remaining highly informative.

3.4 Conditional means as prediction rules

The conditional moments from Section 2.9 also determine which prediction rule has the smallest mean squared error. Write $m(x) = \mathbb{E}[Y \mid X = x]$ and $v(x) = \mathrm{Var}(Y \mid X = x)$, and assume Y has a finite second moment. For a prediction $a(x)$, expand $Y - a(x) = (Y - m(x)) + (m(x) - a(x))$ inside the group $X = x$:

$$\mathbb{E}[(Y - a(x))^2 \mid X = x] = v(x) + (m(x) - a(x))^2.$$

The cross term has conditional mean zero. Applying total expectation (2.8), for a rule with finite mean squared error,

Result 3.9: The conditional mean minimizes squared error

With $m(X) = \mathbb{E}[Y \mid X]$ and $v(X) = \mathrm{Var}(Y \mid X)$,

$$\mathbb{E}[(Y - a(X))^2] = \mathbb{E}[v(X)] + \mathbb{E}[(m(X) - a(X))^2].$$

The second term is nonnegative and vanishes for $a(X) = m(X)$. Thus the conditional mean gives the smallest expected squared prediction error among rules using the same information.

The first term is uncertainty left in Y after observing X. The second is the extra error from predicting something other than the conditional mean. In the click table, $m(0) = 0.18$ and $m(1) = 0.82$, so its error is $0.18(0.82) = 0.1476$, compared with 0.25 for the constant prediction 0.5. This is a property of the probability model; estimating such a rule from a dataset is a separate statistical problem.

3.5 Random vectors and covariance matrices

A pair is the simplest example of several measurements made together. For a single object we might record height, width, temperature, and two sensor readings. Collect d measurements into a column vector

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_d \end{pmatrix} \in \mathbb{R}^d.$$

This is a *random vector*: all coordinates refer to the same randomly selected object or experimental repetition. A lowercase $\mathbf{x}$ denotes one realized vector. The joint distribution specifies probabilities for regions in $\mathbb{R}^d$.

The mean vector averages each coordinate:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix}.$$

To record how every pair of coordinates fluctuates together, arrange their covariances into a matrix.

Definition 3.10: Covariance matrix

For a random vector with finite second moments,

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top], \qquad \Sigma_{ij} = \operatorname{Cov}(X_i, X_j). \tag{3.9}$$

The superscript $\top$ denotes transpose: it turns the column of deviations into a row. Their product is a $d \times d$ matrix whose (i, j) entry is the product of the two deviations.

The diagonal entries are variances; the off-diagonal entries are covariances. For the two-click vector,

$$\boldsymbol{\mu} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 0.25 & 0.16 \\ 0.16 & 0.25 \end{pmatrix}.$$

The matrix is symmetric because $\operatorname{Cov}(X_i, X_j) = \operatorname{Cov}(X_j, X_i)$. If the coordinates have positive variances, standardizing each one gives the correlation matrix, with entries $R_{ij} = \Sigma_{ij}/\sqrt{\Sigma_{ii}\Sigma_{jj}}$ and ones on its diagonal. For the clicks its off-diagonal entries are 0.64.

3.5.1 A covariance matrix cannot imply a negative variance

Result 3.11: Variance of a linear combination

For a random vector with covariance matrix Σ, take a weighted sum $S = a_1X_1 + \cdots + a_dX_d = \mathbf{a}^\top\mathbf{X}$. Expanding its squared centered value gives

$$\operatorname{Var}(S) = \sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j \operatorname{Cov}(X_i, X_j) = \mathbf{a}^\top\Sigma\mathbf{a} \geq 0. \tag{3.10}$$

A symmetric matrix satisfying this inequality for every real vector $\mathbf{a}$ is called *positive semidefinite*. Every covariance matrix must have this property. It is *positive definite* if the inequality is strict for every nonzero $\mathbf{a}$, meaning that no nontrivial weighted combination has zero variance.

For example, the symmetric matrix

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

cannot be a covariance matrix. It would imply $\operatorname{Var}(X_1 - X_2) = 1 + 1 - 2(2) = -2$. Positive diagonal entries alone do not make a covariance matrix valid. In two dimensions, a symmetric matrix

$$\begin{pmatrix} v_1 & c \\ c & v_2 \end{pmatrix}$$

is positive semidefinite exactly when $v_1 \geq 0$, $v_2 \geq 0$, and $c^2 \leq v_1v_2$. For positive variances, the last condition is the correlation bound $|\rho| \leq 1$.

A diagonal covariance matrix means that distinct coordinates are uncorrelated. It does not generally mean that they are independent: (X, X^2), with X uniform on $[-1, 1]$, remains a counterexample.

3.6 Linear transformations of random vectors

Two sensors measuring the same quantity can be summarized by their average and their difference. Converting units, subtracting a known bias, or combining features into a score are other examples of linear combinations. We want rules for the means and covariances of the results.

Let A be a fixed $k \times d$ matrix and $\mathbf{b} \in \mathbb{R}^k$ a fixed vector, and set

$$\mathbf{W} = A\mathbf{X} + \mathbf{b}.$$

Row i of A supplies the weights for the i-th new variable.

Result 3.12: Mean under an affine transformation

Linearity of expectation gives

$$\mathbb{E}[\mathbf{W}] = A\boldsymbol{\mu} + \mathbf{b}. \tag{3.11}$$

Subtracting this mean yields $\mathbf{W} - \mathbb{E}[\mathbf{W}] = A(\mathbf{X} - \boldsymbol{\mu})$. Hence

Result 3.13: Covariance under an affine transformation

$$\begin{aligned} \mathrm{Cov}(\mathbf{W}) &= \mathbb{E}[A(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top A^\top] \\ &= A\Sigma A^\top. \end{aligned} \tag{3.12}$$

Here $\mathrm{Cov}(\mathbf{W})$ denotes the covariance matrix of a vector. These rules need finite second moments, not independence or Gaussian distributions.

Example 3.14 (Average error and disagreement between sensors)**.** Let X_1, X_2 be centered sensor errors with variances one and covariance 0.8. Define

$$M = \frac{X_1 + X_2}{2}, \qquad D = X_1 - X_2, \qquad A = \begin{pmatrix} 1/2 & 1/2 \\ 1 & -1 \end{pmatrix}.$$

Both new means are zero, and matrix multiplication gives

$$\mathrm{Cov}\begin{pmatrix} M \\ D \end{pmatrix} = A \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} A^\top = \begin{pmatrix} 0.9 & 0 \\ 0 & 0.4 \end{pmatrix}.$$

Positively correlated errors often move together, so averaging them removes little of the shared error. Subtracting them cancels that shared variation. With independent unit-variance errors, the two variances would instead be 0.5 and 2.

The zero covariance between M and D follows from the equal original variances. It does not by itself establish independence. The Gaussian model below makes them independent.

3.7 The multivariate Gaussian

3.7.1 Construct correlated readings from independent noise

Let Z_1, Z_2 be independent standard Gaussian variables. Consider the sensor-error model

$$X_1 = Z_1, \qquad X_2 = 0.8Z_1 + 0.6Z_2.$$

Both sensors contain the disturbance Z_1; the second also has an independent disturbance Z_2. The known rules for independent Gaussian sums give

$$\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0, \qquad \mathrm{Var}(X_1) = 1, \qquad \mathrm{Var}(X_2) = 0.8^2 + 0.6^2 = 1,$$

and

$$\mathrm{Cov}(X_1, X_2) = 0.8\,\mathrm{Var}(Z_1) + 0.6\,\mathrm{Cov}(Z_1, Z_2) = 0.8.$$

Thus both marginal distributions are standard Gaussian, but their joint distribution is correlated. It is a *bivariate Gaussian.*

The construction extends to any number of measurements.

Definition 3.15: Multivariate Gaussian

Let $\mathbf{Z}$ have m independent standard Gaussian coordinates, and let A be a fixed $d \times m$ matrix. A vector of the form

$$\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}$$

has a multivariate Gaussian distribution, written

$$\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma), \qquad \Sigma = AA^\top.$$

Its mean vector is $\boldsymbol{\mu}$, and its covariance matrix is Σ.

The independent noises supply the randomness; the rows of A specify which noises each measurement shares. Equivalently, a vector is jointly Gaussian if every real linear combination of its coordinates is a univariate Gaussian, allowing a constant when the variance is zero. For the construction above this follows because each combination is itself a sum of independent Gaussian noises. In particular,

$$\mathbf{a}^\top \mathbf{X} \sim \mathcal{N}(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \Sigma \mathbf{a}). \tag{3.13}$$

3.7.2 From circular to elliptical density contours

Independence gives the joint density of two standard noises:

$$f_{Z_1,Z_2}(z_1, z_2) = \phi(z_1)\phi(z_2) = \frac{1}{2\pi} \exp\left[-\frac{z_1^2 + z_2^2}{2}\right].$$

It has the same value wherever $z_1^2 + z_2^2$ is constant, so its equal-density contours are circles centered at zero. These contours describe density values, not a claim that all observations lie inside a particular circle.

For a correlation $-1 < \rho < 1$, replace the numerical sensor model by

$$X_1 = Z_1, \qquad X_2 = \rho Z_1 + \sqrt{1 - \rho^2}\, Z_2.$$

The inverse relations are $z_1 = x_1$ and $z_2 = (x_2 - \rho x_1)/\sqrt{1-\rho^2}$. The transformation scales areas by $\sqrt{1-\rho^2}$, so preserving probabilities requires dividing the density by this factor. Substitution then gives

$$f_{X_1,X_2}(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\right]. \tag{3.14}$$

The cross term changes which pairs are likely. With positive ρ, equal-sign deviations are favored; with negative ρ, opposite-sign deviations are favored. Both marginals are still $\mathcal{N}(0,1)$.

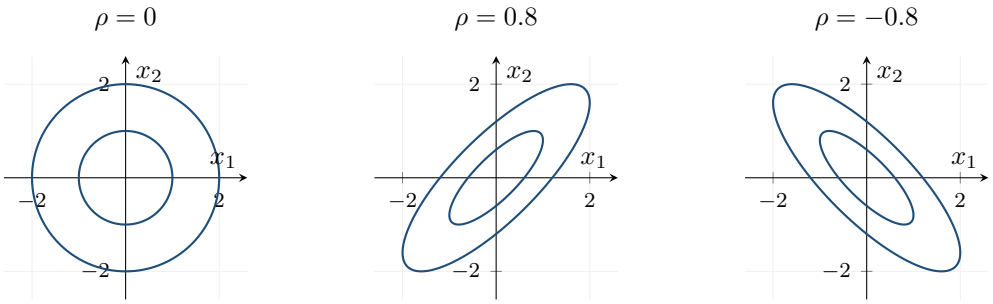


Figure 3.2: Equal-density contours for three bivariate Gaussian models. In every panel both individual features have mean zero and variance one. Correlation changes the joint shape, which separate histograms of the features cannot reveal. Each curve joins points with the same density; the outer curve has lower density than the inner one.

The ellipse directions can be understood without a matrix decomposition. For $\rho = 0.8$, use the rotated coordinates

$$U = \frac{X_1 + X_2}{\sqrt{2}}, \qquad V = \frac{X_1 - X_2}{\sqrt{2}}.$$

Their variances are $1 + \rho = 1.8$ and $1 - \rho = 0.2$, respectively, and their covariance is zero. In these coordinates the expression in the exponent becomes

$$q = \frac{u^2}{1.8} + \frac{v^2}{0.2}.$$

For fixed $q = c > 0$, the ellipse extends by $\sqrt{1.8c}$ along the equal-sign direction and $\sqrt{0.2c}$ along the opposite-sign direction. The first is three times longer than the second. Shared sensor fluctuations stretch the distribution along the diagonal.

3.7.3 The density in matrix notation

Result 3.16: Multivariate Gaussian density

For a positive-definite covariance matrix, the general Gaussian density is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}\sqrt{\det \Sigma}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right]. \tag{3.15}$$

Here Σ^{-1} is the inverse matrix, and $\det \Sigma$ is its determinant. These are the matrix versions of dividing by variance and correcting the density for a change of scale. For an invertible square A, the transformation $\mathbf{x} = \boldsymbol{\mu} + A\mathbf{z}$ scales volumes by $|\det A| = \sqrt{\det \Sigma}$, while

$$\sum_i z_i^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$

This explains both parts of the density formula from the independent standard-noise density.

In two dimensions,

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \qquad \det\Sigma = \sigma_1^2\sigma_2^2(1-\rho^2).$$

For positive σ_1, σ_2 and $|\rho| < 1$, putting $u = (x_1 - \mu_1)/\sigma_1$, $v = (x_2 - \mu_2)/\sigma_2$ gives

$$(\mathbf{x}-\boldsymbol{\mu})^\top\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}) = \frac{u^2 - 2\rho uv + v^2}{1-\rho^2}.$$

For a diagonal covariance matrix this reduces to a sum of squared standardized deviations, and the density factors into the univariate Gaussian densities.

The quadratic expression is called the *squared Mahalanobis distance*. It measures how unusual a displacement is relative to the model's spread and dependence. In the unit-variance model with $\rho = 0.8$, the points $(1, 1)$ and $(1, -1)$ have the same ordinary distance from the origin, but

$$q(1,1) = \frac{2-1.6}{0.36} = \frac{10}{9}, \qquad q(1,-1) = \frac{2+1.6}{0.36} = 10.$$

Disagreement between the sensors is much less typical than a shared positive error. Euclidean distance alone misses that distinction.

3.7.4 Marginals, transformations, and the independence exception

Selecting some coordinates is a linear transformation. Their marginal distribution is Gaussian with the corresponding entries of the mean vector and the corresponding rows and columns of the covariance matrix. In particular, $X_i \sim \mathcal{N}(\mu_i, \Sigma_{ii})$.

Result 3.17: Affine transformations preserve Gaussianity

For $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, fixed B and $\mathbf{b}$,

$$B\mathbf{X} + \mathbf{b} \sim \mathcal{N}(B\boldsymbol{\mu} + \mathbf{b}, B\Sigma B^\top). \tag{3.16}$$

The result is Gaussian because substituting $\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}$ gives another linear transformation of independent standard Gaussian noises. The moment rules in Section 3.6 now describe the *whole distribution*, not just two summaries of it.

Result 3.18: The jointly Gaussian independence criterion

For jointly Gaussian variables, zero covariance *does* imply independence.

In the nondegenerate case, a diagonal covariance matrix makes the joint density a product of marginal densities. Therefore the average and difference in the sensor example are independent under the joint Gaussian assumption, as well as uncorrelated.

It is not enough for each marginal to be Gaussian. To see why, take $X \sim \mathcal{N}(0, 1)$ and an independent random sign S, equal to -1 or one with probability $1/2$. Set $Y = SX$. Symmetry makes $Y \sim \mathcal{N}(0, 1)$, and

$$\mathrm{Cov}(X, Y) = \mathbb{E}[SX^2] = \mathbb{E}[S]\mathbb{E}[X^2] = 0.$$

Yet $|Y| = |X|$ always, so they are dependent. The pair is not jointly Gaussian: $X + Y$ is exactly zero with probability $1/2$, but is not a constant. A nondegenerate Gaussian has no such point mass.

Finally, a Gaussian covariance matrix can be singular. If $X_2 = X_1 \sim \mathcal{N}(0,1)$, then

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

All pairs lie on a line. This is a valid Gaussian random vector, but it has no density with respect to two-dimensional area. Formula (3.15) requires a positive-definite matrix; it cannot be used when the determinant is zero.

3.8 What one Gaussian measurement tells us about another

Return to $X_1 = Z_1$, $X_2 = 0.8Z_1 + 0.6Z_2$. Before seeing the first sensor, the second error has variance one. After observing $X_1 = x$, the shared noise Z_1 is known and only the independent noise Z_2 remains:

$$X_2 \mid X_1 = x \sim \mathcal{N}(0.8x, 0.36).$$

If the first sensor reads two units too high, the conditional mean error of the second is 1.6, with standard deviation 0.6. Marginalizing over the first sensor and conditioning on its reading are different operations: the marginal distribution is still $\mathcal{N}(0,1)$.

For the general bivariate Gaussian with positive marginal variances and $|\rho| < 1$, write

$$X = \mu_X + \sigma_X Z_1, \qquad Y = \mu_Y + \sigma_Y(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2).$$

Observing $X = x$ fixes $Z_1 = (x - \mu_X)/\sigma_X$. Therefore

Result 3.19: Conditional bivariate Gaussian distribution

For a jointly Gaussian pair with $\sigma_X, \sigma_Y > 0$ and $|\rho| < 1$,

$$Y \mid X = x \sim \mathcal{N}\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1-\rho^2)\right). \tag{3.17}$$

Equivalently, the conditional mean's slope is $\operatorname{Cov}(X,Y)/\operatorname{Var}(X)$. Positive correlation shifts the conditional mean in the direction of the observation; negative correlation shifts it in the opposite direction. The conditional variance is constant in x for this Gaussian model. General conditional distributions need not have a linear mean or a constant variance.

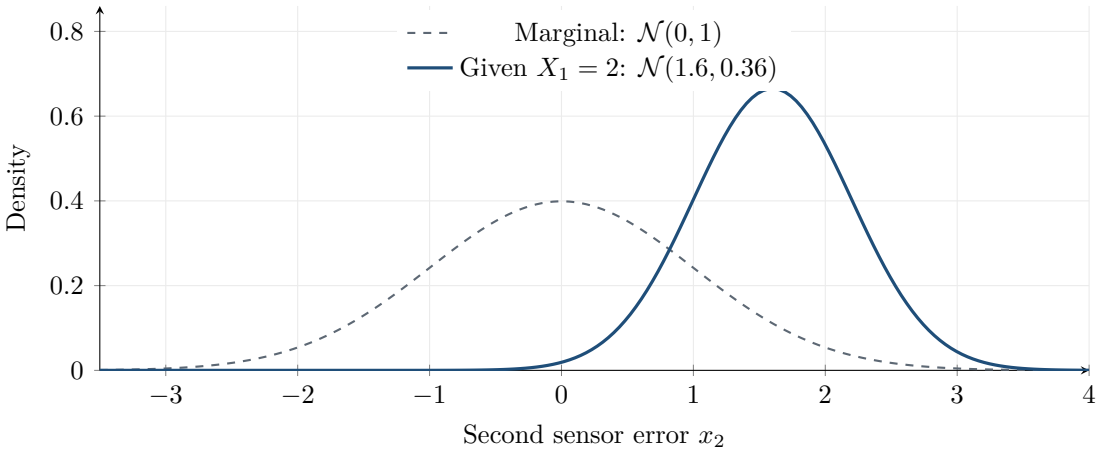


Figure 3.3: Before observing the first sensor, the second error has distribution $\mathcal{N}(0,1)$. After observing $X_1 = 2$, it has distribution $\mathcal{N}(1.6, 0.36)$. Knowledge of the shared disturbance shifts the mean and reduces the remaining spread.

As a check using total variance (2.9), the numerical model gives $1 = 0.36 + 0.64$: residual variance plus variance of the conditional mean. If $\rho = 0$, observing X changes neither the mean nor the variance, consistent with Gaussian independence. As $|\rho|$ approaches one, the remaining variance approaches zero. At $|\rho| = 1$, the relationship is exact and the conditional distribution is a point mass.

3.9 Gaussian class-conditional models

Let C denote the class label, taking values $1, \dots, K$. Use Gaussian class-conditional densities in (1.10):

$$\mathbb{P}(C = k) = \pi_k, \qquad \mathbf{X} \mid C = k \sim \mathcal{N}_d(\boldsymbol{\mu}_k, \Sigma_k).$$

For positive-definite Σ_k, let f_k be the corresponding Gaussian density. The new modeling choice is the covariance structure *within each class*.

3.9.1 The relationship between features can identify the class

Consider two equally likely operating modes of a device. In mode $C = +$, its two sensor errors tend to move together; in mode $C = -$, they tend to move in opposite directions. Use

$$\boldsymbol{\mu}_+ = \boldsymbol{\mu}_- = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \Sigma_+ = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}, \qquad \Sigma_- = \begin{pmatrix} 1 & -0.8 \\ -0.8 & 1 \end{pmatrix}.$$

Each feature alone has distribution $\mathcal{N}(0, 1)$ in both classes. Its value alone therefore leaves the class probabilities at $1/2$. The distinguishing information is entirely in the pair.

The determinants are both 0.36, so the density normalization constants cancel in the likelihood ratio. Subtracting the exponents in (3.14) gives

$$\log \frac{f_+(x_1, x_2)}{f_-(x_1, x_2)} = \frac{2(0.8)}{1 - 0.8^2} x_1 x_2 = \frac{40}{9} x_1 x_2.$$

With equal prior probabilities, mode $+$ is more probable when $x_1 x_2 > 0$, and mode $-$ is more probable when $x_1 x_2 < 0$. At $(1, 1)$,

$$\mathbb{P}(C = + \mid \mathbf{X} = (1, 1)) = \frac{1}{1 + e^{-40/9}} \approx 0.9884.$$

At $(1, -1)$, this probability is approximately 0.0116. The individual feature values have equal marginal densities in both classes; their agreement or disagreement changes the conclusion.

If we incorrectly replace each class density by the product of its feature marginals, both models become $\phi(x_1)\phi(x_2)$. Then every pair receives posterior class probability $1/2$. Assuming conditional independence has erased all the useful information in this example.

3.9.2 What the Gaussian decision rule uses

For positive class priors, maximizing the posterior is equivalent to maximizing the log-score

$$s_k(\mathbf{x}) = \log \pi_k - \frac{1}{2} \log \det \Sigma_k - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k).$$

The omitted term $-\frac{d}{2} \log(2\pi)$ is the same for every class. The three remaining terms reflect the class's prevalence, the volume over which its density is spread, and the observation's

displacement relative to the class's covariance. A broad class cannot receive the same peak density as a narrow class with the same total probability.

If all classes share a common covariance Σ, expanding the quadratic expression shows that the term $-\frac{1}{2}\mathbf{x}^\top\Sigma^{-1}\mathbf{x}$ is also common. The class-dependent part becomes

$$\mathbf{x}^\top\Sigma^{-1}\boldsymbol{\mu}_k - \frac{1}{2}\boldsymbol{\mu}_k^\top\Sigma^{-1}\boldsymbol{\mu}_k + \log\pi_k,$$

which is linear in the observed features, plus a constant. Comparing two classes therefore gives a linear boundary. If the covariance matrices differ, quadratic terms generally remain, as they did in the two-mode example.

Chapter 4

Samples, Concentration, and Limit Theorems

4.1 An average changes from dataset to dataset

Record one hundred waiting times from the inbox of Section 2.1 and report their average. Repeat the experiment with another hundred waits. The two averages will usually differ. How far might either average be from the population mean, and how does collecting more observations change that uncertainty?

Keep the same model: independent waits $T_i \sim \text{Exp}(0.2)$, measured in minutes. Their mean and standard deviation are both five minutes, as calculated in Chapter 2. Here the new object is the *dataset* and the average computed from it, not another distribution for an individual wait.

4.1.1 A population, a sample, and a sampling distribution

Definition 4.1: Population, sample, and i.i.d. observations

The *population distribution* describes the observations we want to learn about. A *sample* is the collection $X_1, \dots, X_n$ that we observe. We call these observations *independent and identically distributed*, abbreviated i.i.d., when they are mutually independent and all follow the same distribution.

Both requirements matter: measurements from different operating conditions need not share a distribution, and repeated measurements of one object need not be independent.

Before collection, $X_1, \dots, X_n$ are random variables; after collection, $x_1, \dots, x_n$ are the recorded numbers.

Definition 4.2: Sample mean and sampling distribution

The *sample mean* is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{4.1}$$

Its distribution over repetitions of the entire sampling experiment is its *sampling distribution*. It is not the distribution of values within one dataset.

To visualize the distinction, imagine collecting many independent datasets of size one hundred, calculating one average for each, and plotting those averages.

Result 4.3: Mean and variance of the sample mean

For i.i.d. observations with finite mean μ and variance σ^2, the sum and scaling rules from Sections 2.2 and 2.3 immediately give

$$\mathbb{E}[\bar{X}_n] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}. \tag{4.2}$$

This is the general form of the click-proportion calculation in Section 2.6. We now use it to study accuracy.

4.1.2 Standard deviation versus standard error

Definition 4.4: Standard error

The standard deviation of a statistic across repeated samples is called its *standard error*. For an i.i.d. sample mean,

$$\mathrm{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}. \tag{4.3}$$

For the inbox, a single wait has standard deviation five minutes, while the average of one hundred waits has standard error $5/\sqrt{100} = 0.5$ minutes. Individual waits remain just as variable when the dataset grows. It is their average that becomes more precise.

In this particular model, the Gamma sum result in Section 2.5 supplies an exact sampling distribution:

$$\sum_{i=1}^{n} T_i \sim \mathrm{Gamma}(n, 0.2), \qquad \bar{T}_n \sim \mathrm{Gamma}(n, 0.2n). \tag{4.4}$$

The second parameter is a rate. Dividing a positive variable by n multiplies its Gamma rate by n. Figure 4.1 shows how the resulting density narrows while retaining mean five.

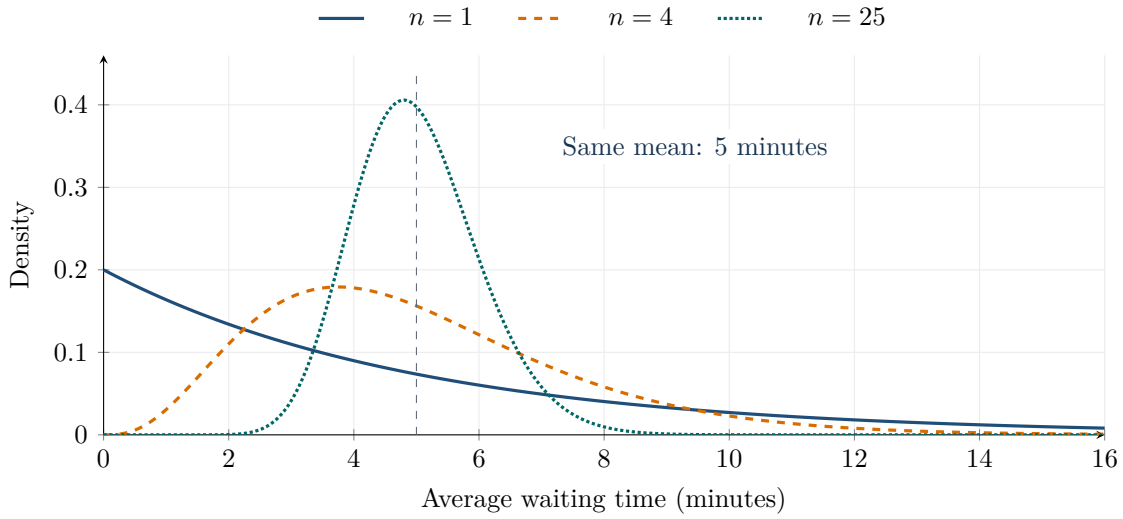


Figure 4.1: Exact sampling densities of the mean of n independent exponential waits, each with mean five minutes. For $n = 1$, the curve is the distribution of a single observation. The larger-sample curves describe averages of entire datasets, not less variable individual waits. Their standard errors are 5, 2.5, and 1 minutes.

In a real dataset, the population variance is usually unknown.

Definition 4.5: Sample variance

A common measure of the observed spread is the *sample variance*,

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2, \qquad n \geq 2.$$

The denominator $n-1$ corrects for centering the data at their own sample mean; the statistical-estimation chapter will derive that correction. Writing $S_n = \sqrt{S_n^2}$, the corresponding estimate of the mean's standard error is $S_n/\sqrt{n}$. It is an estimate, not the exact value of $\sigma/\sqrt{n}$, and its interpretation depends on the sampling assumptions.

4.2 Bounding the probability of a large error

Suppose we want the average waiting time to be within one minute of the population mean. Knowing its standard error gives a scale for the error, but does not by itself give a probability. We first seek a bound that does not require the full sampling distribution.

4.2.1 Markov: a nonnegative quantity cannot often be very large

Let $W \geq 0$ have finite expectation. On the event $W \geq a$, where $a > 0$, its contribution is at least a. Pointwise,

$$W \geq a\,\mathbf{1}_{\{W \geq a\}}.$$

Taking expectations, using the indicator rule from Section 2.2, gives $\mathbb{E}[W] \geq a\mathbb{P}(W \geq a)$. Rearranging yields

Theorem 4.6: Markov's inequality

For $W \geq 0$, $\mathbb{E}[W] < \infty$, and $a > 0$,

$$\mathbb{P}(W \geq a) \leq \frac{\mathbb{E}[W]}{a}. \tag{4.5}$$

For a nonnegative wait with mean five minutes, Markov gives $\mathbb{P}(T \geq 20) \leq 1/4$. The exponential model gives the much smaller exact value $e^{-4} \approx 0.0183$. Markov is valid for *every* nonnegative distribution with that mean, including one that puts probability $1/4$ at twenty and $3/4$ at zero. For that latter distribution the bound is attained. Without more information about the distribution, we cannot improve this bound uniformly.

Nonnegativity is essential. A signed quantity could take very large positive and negative values whose contributions cancel in its mean. To control deviations on both sides, apply Markov to their squares.

4.2.2 Chebyshev: turn a variance into an error bound

For X with mean μ and finite variance σ^2, take $W = (X-\mu)^2$ and $a = \varepsilon^2$ in Markov's inequality. The event $W \geq \varepsilon^2$ is exactly $|X - \mu| \geq \varepsilon$. Thus

Theorem 4.7: Chebyshev's inequality

For X with mean μ, finite variance σ^2, and $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}. \tag{4.6}$$

Applying this to $\bar{X}_n$, using (4.2), gives

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \min\left(1, \frac{\sigma^2}{n\varepsilon^2}\right). \tag{4.7}$$

The minimum with one simply discards a bound larger than any probability. For one hundred inbox waits and tolerance one minute,

$$\mathbb{P}(|\bar{T}_{100} - 5| \geq 1) \leq \frac{25}{100} = 0.25.$$

This is an upper bound, not a prediction that a quarter of datasets will miss the target.

Conversely, a sufficient sample size for failure probability at most $\delta \in (0, 1)$ is

$$n \geq \frac{\sigma^2}{\delta\varepsilon^2}.$$

For the inbox, $n = 500$ guarantees an error below one minute except on an event of probability at most 0.05. The guarantee uses the true variance, or a known upper bound on it. Substituting a random sample variance without further justification does not preserve the same finite-sample guarantee.

4.3 The law of large numbers

Chebyshev answers a question at a specified sample size. It also gives a limiting conclusion: for a fixed error tolerance, the probability of exceeding it tends to zero as the number of observations grows.

Theorem 4.8: Weak law of large numbers: finite-variance version

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and finite variance. Then, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0. \tag{4.8}$$

Proof. By (4.7), the probability is bounded above by $\sigma^2/(n\varepsilon^2)$, which tends to zero. □

Definition 4.9: Convergence in probability to a constant

The condition $\mathbb{P}(|Z_n - c| > \varepsilon) \to 0$ for every $\varepsilon > 0$ is called *convergence in probability* to c, written $Z_n \xrightarrow{\mathbb{P}} c$. Here $Z_n = \bar{X}_n$ and $c = \mu$.

Choose a fixed tolerance first, such as one minute or one tenth of a minute. Among repeatedly collected datasets, the fraction whose average lies outside that tolerance becomes arbitrarily small as the dataset size increases.

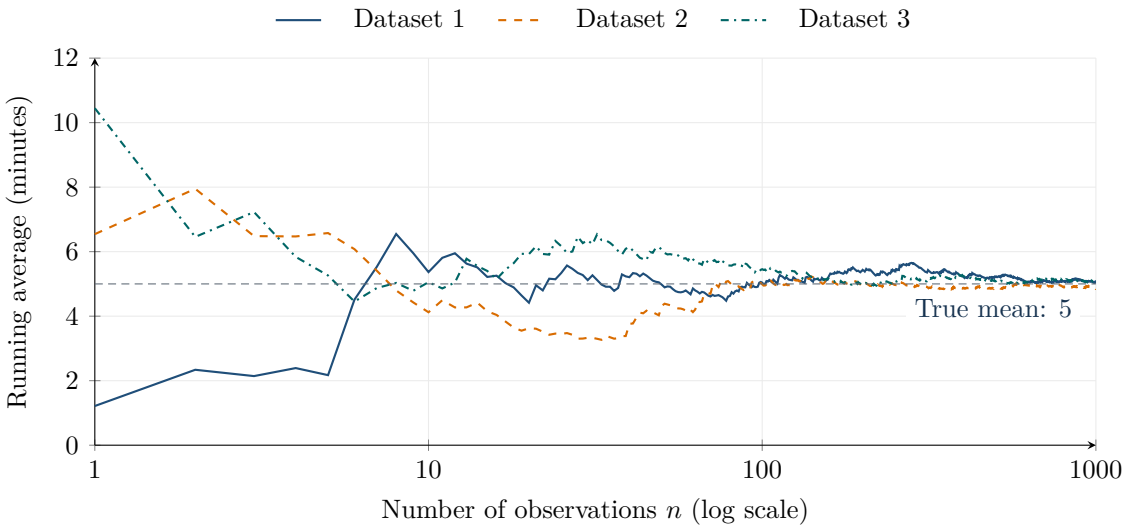


Figure 4.2: Three independent simulated sequences of exponential waits. Each curve uses the first n observations of its own sequence. The horizontal axis is logarithmic. Averages can move away from the mean after a new observation; convergence does not mean monotone improvement. These are illustrative paths, not a probability bound.

The theorem does not require the error to decrease at every step of one running experiment. A new large wait can move an otherwise accurate average farther from five. Nor does it claim that individual waits approach five: their distribution is unchanged. The running averages in Figure 4.2 fluctuate even as their overall spread shrinks.

Finite variance is sufficient for this short proof, but is not necessary for the i.i.d. law of large numbers. The more general result requires only $\mathbb{E}[|X_1|] < \infty$. That distinction will matter for the heavy-tailed examples in Section 4.5.

The distribution being sampled also matters. If only short waits are recorded, their average can settle very precisely near the mean of that selected population, not the mean of all waits. Increasing sample size reduces sampling fluctuations; it does not correct an unrepresentative sampling process.

4.4 The central limit theorem

The law of large numbers tells us that the error becomes small. It does not tell us its shape. For the inbox we have an exact Gamma answer; for a complicated observation or a model's loss, that distribution may be unavailable. The central limit theorem supplies a broadly applicable approximation under stated assumptions.

4.4.1 Keep the shrinking fluctuations visible

The error $\bar{X}_n - \mu$ has standard deviation $\sigma/\sqrt{n}$. Divide by this scale:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n}(X_i - \mu)}{\sigma\sqrt{n}}.$$

This expresses the sampling error in standard-error units. The error itself shrinks, while Z_n has mean zero and variance one for every n. Equal mean and variance do not determine a shape; the new claim concerns the distribution of Z_n.

Theorem 4.10: Classical i.i.d. central limit theorem

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and $0 < \sigma^2 = \mathrm{Var}(X_1) < \infty$. Then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1). \tag{4.9}$$

Explicitly, for every real z,

$$\mathbb{P}(Z_n \le z) \longrightarrow \Phi(z).$$

The symbol $\xrightarrow{d}$ means *convergence in distribution*: here, convergence of the CDFs to the Gaussian CDF. This is different from saying that the rescaled errors converge to zero. They retain nonzero spread. It also does not say that the original X_i become Gaussian or that the unscaled sample mean approaches a Gaussian of fixed positive variance.

Returning to the original units gives the approximation

$$\bar{X}_n \;\approx\; \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

For Gaussian observations this distribution is exact for every n, by Chapter 3. For exponential observations it is an approximation to a Gamma density. Figure 4.3 shows the rescaled densities: the narrowing has been removed, allowing us to see the change in shape.

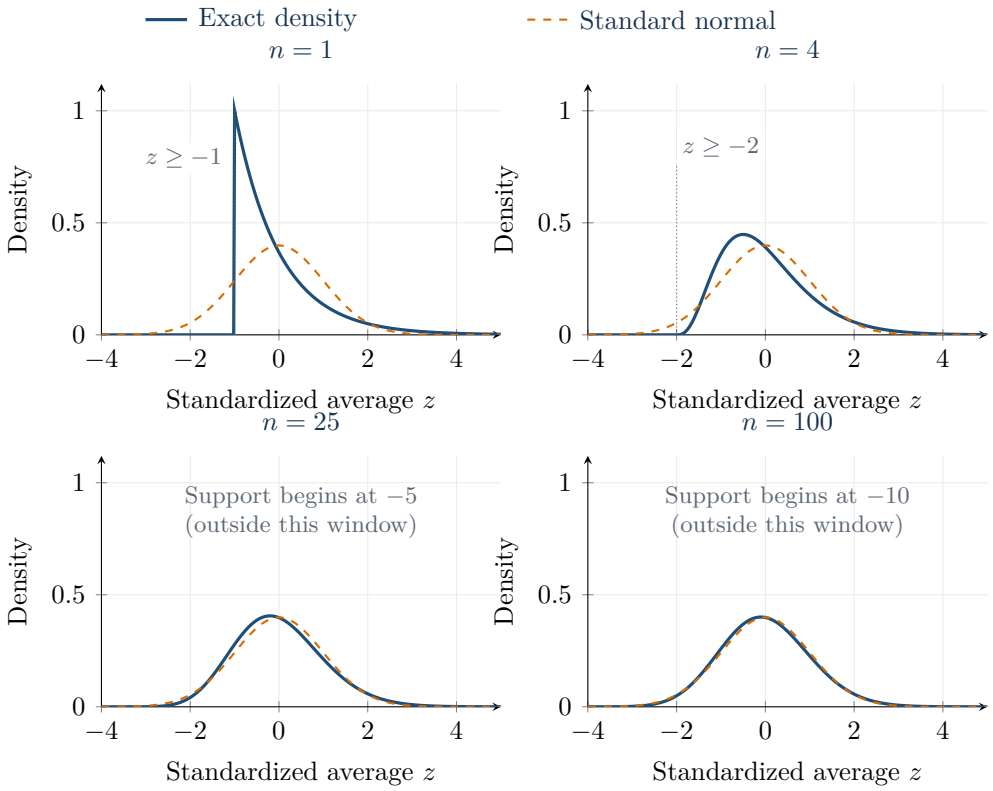


Figure 4.3: Exact densities of $Z_n = \sqrt{n}(\bar{T}_n - 5)/5$, compared with the standard Gaussian. All panels use the same axes. The original waits remain exponential; it is the standardized average whose shape becomes nearly Gaussian. The lower support endpoint is $-\sqrt{n}$, so the small-sample curves are visibly asymmetric.

4.4.2 A bound, an approximation, and an exact answer

For one hundred waits, the one-minute error tolerance is two standard errors. The Gaussian approximation gives

$$\begin{aligned}\mathbb{P}(|\bar{T}_{100} - 5| > 1) &= \mathbb{P}(|Z_{100}| > 2) \\ &\approx 2[1 - \Phi(2)] \approx 0.04550.\end{aligned}$$

The exact Gamma distribution in (4.4) lets us check it. Let G_n be the CDF of $\text{Gamma}(n, 0.2n)$. The count–waiting-time identity (2.5) gives, for $t \geq 0$,

$$G_n(t) = 1 - e^{-0.2nt} \sum_{j=0}^{n-1} \frac{(0.2nt)^j}{j!}.$$

Numerically,

$$G_{100}(4) \approx 0.017108, \qquad 1 - G_{100}(6) \approx 0.027864.$$

The two tails are unequal because the exact sampling distribution remains somewhat skewed. Their sum is about 0.04497.

Method	Value	Meaning	Information used
Chebyshev	0.25000	Upper bound	Independence and variance
CLT	0.04550	Approximation	Gaussian approximation to the average
Gamma	0.04497	Exact probability	Full exponential model

The approximation is close here, but it is not a guarantee. Chebyshev is conservative here, but remains valid without the exponential shape assumption. Figure 4.4 compares all three across sample sizes rather than only at one hundred.

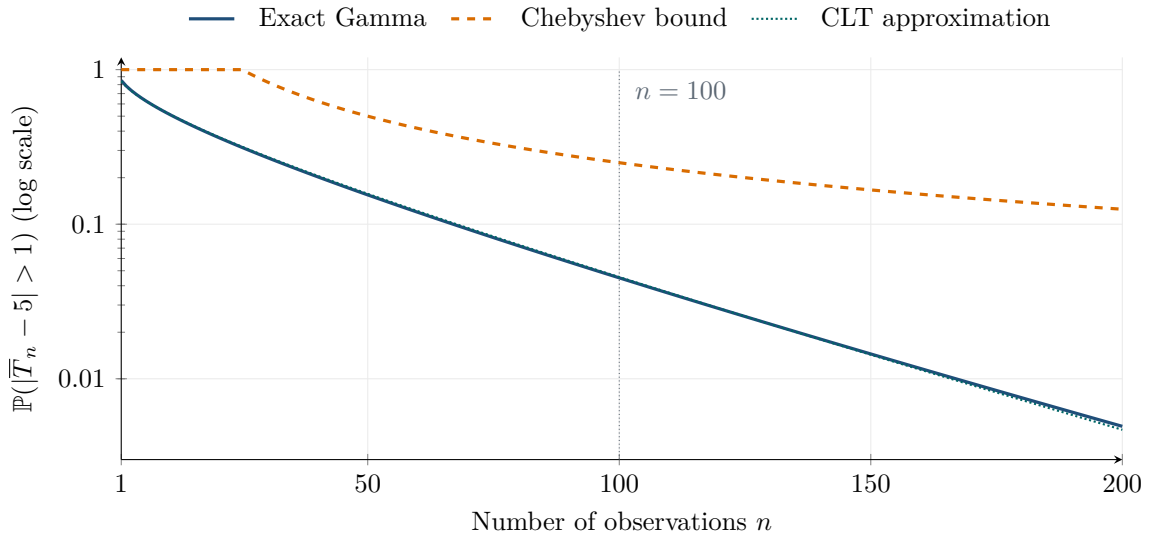


Figure 4.4: Probability that the average wait misses five minutes by more than one minute. The exact Gamma probability and CLT approximation are close at moderate sample sizes in this example; the Chebyshev curve is an upper bound, capped at one. The vertical axis is logarithmic so small probabilities remain distinguishable.

There is no universal sample size, such as thirty, at which the Gaussian approximation becomes reliable for every distribution and every probability. Strong skewness, rare extreme values, and far-tail questions can require much larger samples. The theorem states convergence, not a numerical error guarantee at a chosen finite n. In particular, approximation of an extremely small probability may have poor relative accuracy even when the CDF approximation is good in absolute terms.

4.5 The limits of averaging

4.5.1 A shared disturbance does not average away

The sensor example in Section 3.6 showed that correlated errors resist averaging. To see the consequence as sample size grows, suppose all observations have variance σ^2 and each distinct pair has correlation $\rho \in [0, 1]$. The covariance formula (3.10) gives

$$\begin{aligned} \operatorname{Var}(\bar{X}_n) &= \frac{n\sigma^2 + n(n-1)\rho\sigma^2}{n^2} \\ &= \sigma^2 \left(\rho + \frac{1-\rho}{n} \right). \end{aligned} \tag{4.10}$$

There are n diagonal terms and $n(n-1)$ off-diagonal terms. If $\rho > 0$, the variance approaches $\rho\sigma^2$, not zero.

A model that realizes this behavior is

$$X_i = \mu + B + \varepsilon_i,$$

where B is a centered calibration error shared by the whole dataset. A fresh B is drawn for a new independently calibrated dataset, but it stays fixed within that dataset's n readings. The ε_i are centered independent reading errors, independent of B. Set $\mathrm{Var}(B) = \rho\sigma^2$ and $\mathrm{Var}(\varepsilon_i) = (1-\rho)\sigma^2$. Then

$$\bar{X}_n = \mu + B + \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i.$$

The independent reading errors average toward zero, leaving the shared error. Repeating readings with the same calibration cannot resolve it. Independent calibrations would provide different information.

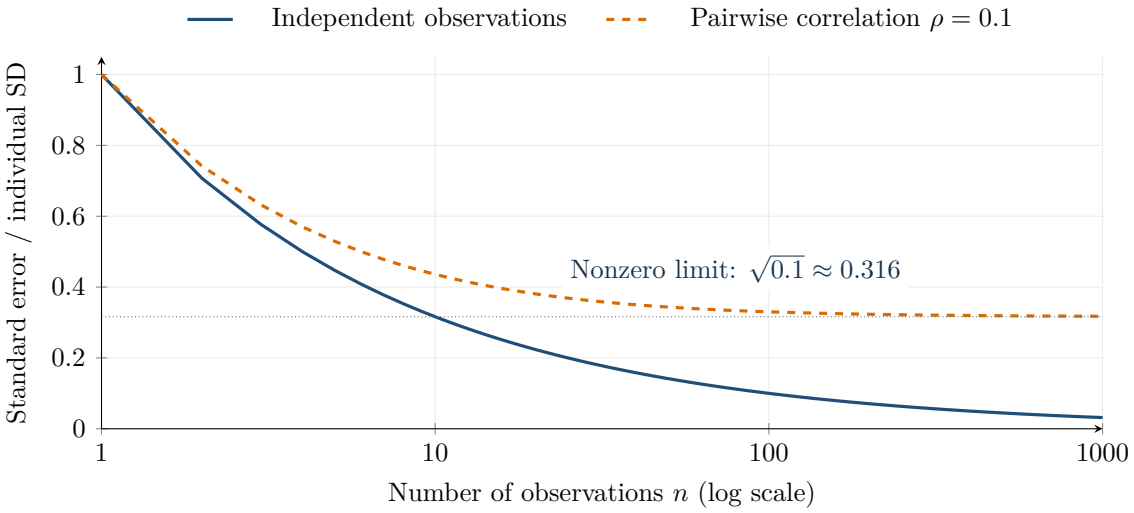


Figure 4.5: Standard error relative to the single-observation standard deviation. Under independence it decreases as $1/\sqrt{n}$. A shared component giving pairwise correlation 0.1 leaves a limiting value $\sqrt{0.1}$. Both curves use the same marginal variance; only the dependence changes.

For $n = 100$ and $\rho = 0.1$, the variance is $0.109\sigma^2$, versus $0.01\sigma^2$ under independence. Equating this variance to σ^2/n_{eff} gives

$$n_{\text{eff}} = \frac{n}{1 + (n-1)\rho} \approx 9.17.$$

In this specific equal-correlation model, one hundred readings have the same mean variance as roughly nine independent readings. This *effective sample size* is a variance comparison, not a claim that the entire datasets have identical distributions. Other forms of dependence can still permit an LLN or a CLT, but require their own assumptions and variance calculation.

4.5.2 Finite mean and finite variance are different requirements

The Student distributions in Section 2.10 already supply the examples we need. For i.i.d. observations with distribution $t_{3/2}$, the absolute first moment is finite but the variance is infinite. The general i.i.d. law of large numbers therefore applies: the sample mean converges in probability to zero. The finite-variance Chebyshev argument and the classical CLT above do not apply. There is no finite σ to insert into $\sigma/\sqrt{n}$.

Standard Cauchy observations (t_1) have an exact averaging property:

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} t_1 \quad \Longrightarrow \quad \bar{X}_n \sim t_1 \quad \text{for every } n. \tag{4.11}$$

Here a sample average is a finite arithmetic calculation for any realized finite dataset, even though its population expectation does not exist. Averaging independent Cauchy observations leaves their distribution unchanged. In particular,

$$\mathbb{P}(|\bar{X}_n| > 1) = \frac{1}{2}$$

for every n, rather than tending to zero. Zero is the distribution's center of symmetry, not its mean. Figure 4.6 contrasts this with Gaussian averaging.

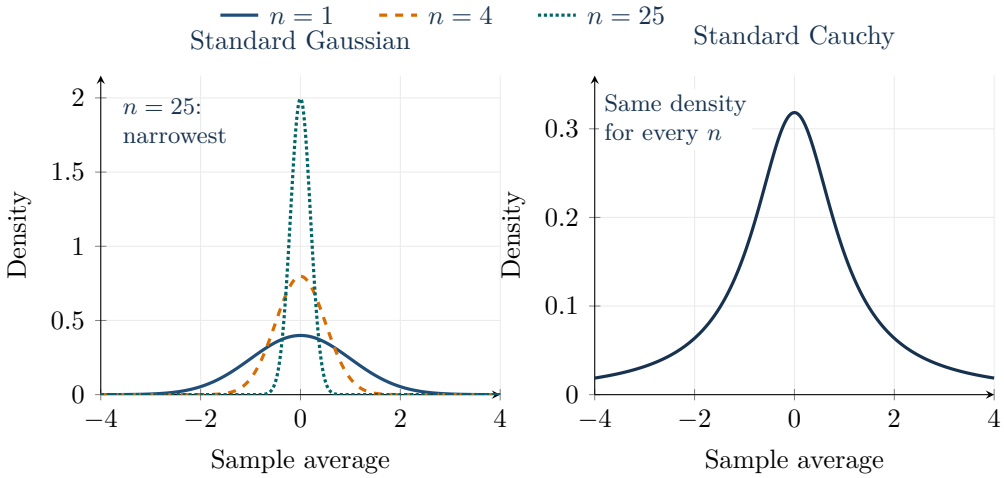


Figure 4.6: Exact densities of sample means from two centered symmetric populations. Gaussian means concentrate as n grows. For standard Cauchy observations the density is unchanged for every sample size, so the three Cauchy curves coincide. The panels have different vertical scales. Symmetry alone does not supply the moment assumptions needed for averaging to become reliable.

The conclusions are distinct: infinite variance need not destroy the LLN, and independence alone does not guarantee that a sample mean concentrates. Check the assumptions for the particular theorem being used, not just whether the data look roughly symmetric.

4.6 Monte Carlo and averages in machine learning

4.6.1 Replace an integral by independent evaluations

Suppose the quantity we need is $I = \mathbb{E}[g(X)]$. When evaluating the corresponding integral is difficult but generating observations is feasible, we can estimate it by averaging simulated values.

Definition 4.11: Monte Carlo estimation

Draw $X_1, \ldots, X_n$ independently from the target distribution and compute

$$\widehat{I}_n = \frac{1}{n} \sum_{i=1}^{n} g(X_i). \tag{4.12}$$

The earlier theorems now apply to $Y_i = g(X_i)$. If $\mathbb{E}[|g(X)|] < \infty$, the LLN gives convergence to I. If also $0 < \tau^2 = \mathrm{Var}(g(X)) < \infty$, then

$$\mathrm{SE}(\widehat{I}_n) = \frac{\tau}{\sqrt{n}}, \qquad \frac{\sqrt{n}(\widehat{I}_n - I)}{\tau} \xrightarrow{d} \mathcal{N}(0,1).$$

The relevant variance is that of the *evaluated quantity* $g(X)$, not necessarily that of X. Even a heavy-tailed input can give a bounded output, such as an event indicator, whose variance is finite.

Use the response score from Section 2.2 as a case with a known answer. For $T \sim \mathrm{Exp}(0.2)$ and $g(t) = e^{-t}$, with time expressed in minutes, the earlier moment calculation gives

$$I = \frac{1}{6}, \qquad \mathbb{E}[g(T)^2] = \frac{1}{11}, \qquad \tau^2 = \frac{1}{11} - \frac{1}{36} = \frac{25}{396}.$$

The coefficient of t in the exponent is one per minute, making the exponent dimensionless. We can compare simulated estimates against $1/6$ to check the numerical method before using it for an expectation with no closed-form answer.

4.6.2 Precision has a computational cost

Because $\mathbb{E}[\widehat{I}_n] = I$, its root mean squared error over independent datasets is

$$\mathrm{RMSE}(\widehat{I}_n) = \sqrt{\mathbb{E}[(\widehat{I}_n - I)^2]} = \frac{\tau}{\sqrt{n}}. \tag{4.13}$$

For the score, $\tau \approx 0.25126$; thus $n = 1000$ gives RMSE about 0.00795. Multiplying the number of evaluations by four halves this error scale. Reducing it by a factor of ten requires one hundred times as many evaluations, if the variance stays the same.

For a target RMSE of 0.005, a sufficient integer sample size in this example is

$$n \geq \left\lceil \frac{25/396}{0.005^2} \right\rceil = 2526,$$

where $\lceil \cdot \rceil$ rounds upward. An RMSE target is not a bound on the error in every run, nor does it specify a failure probability. The two are different accuracy requirements.

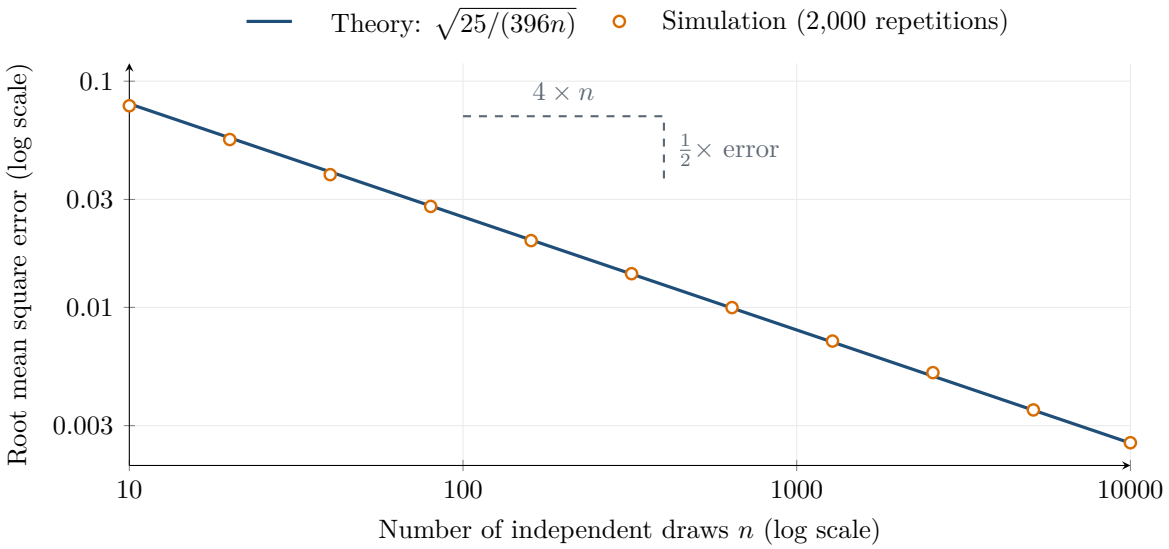


Figure 4.7: Monte Carlo error for the response score e^{-T}. The theoretical curve is the exact RMSE $\sqrt{25/396}/\sqrt{n}$. The simulated points estimate RMSE from repeated independent runs; they are not the absolute error of one running estimate. Both axes are logarithmic, so the $n^{-1/2}$ rate is a straight line.

When τ is unknown, compute the sample standard deviation S_Y of the values $g(X_i)$ and report $S_Y/\sqrt{n}$ as an estimated standard error. Increasing n does not fix a wrong sampling distribution or a faulty implementation of g.

4.6.3 A test-set average estimates a population quantity

Fix a prediction rule f before examining the evaluation data. Let (X_i, Y_i) be i.i.d. test observations, independent of any data used to choose f. For a per-observation loss $L_i = \ell(f(X_i), Y_i)$, the population quantity and its empirical average are

$$R(f) = \mathbb{E}[\ell(f(X), Y)], \qquad \widehat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(X_i), Y_i).$$

This is the same Monte Carlo construction. Conditional on the fixed model, the variation comes from which evaluation examples happen to be sampled. With finite loss variance,

$$\mathrm{SE}(\widehat{R}_n(f)) = \frac{\sqrt{\mathrm{Var}(\ell(f(X), Y))}}{\sqrt{n}}.$$

The squared-error interpretation from Section 3.4 supplies one choice of loss; classification error indicators supply another. In either case, the reported average fluctuates across test sets.

For a fixed classifier with population accuracy 0.8, four hundred independent test examples give standard error

$$\sqrt{\frac{0.8(0.2)}{400}} = 0.02.$$

This describes test-set sampling variation, not variation caused by retraining the model. Selecting the model with the best score on that same test set breaks the fixed-model premise.

More observations from an unrepresentative test population also do not establish performance on the intended population.

4.6.4 What increasing the batch size changes

At a fixed model, average the losses of b independent examples. If each has variance τ_L^2, the batch-average variance is τ_L^2/b. Thus a fourfold batch-size increase halves the standard deviation of that average. This is an averaging statement, not a claim that training becomes four times better or faster.

For a fixed finite dataset, independent sampling of indices *with replacement* gives the same rule conditional on that dataset, using its empirical loss variance. Sampling without replacement gives a different variance, and correlated examples need the covariance terms from (4.10) or the general quadratic-form rule. The number of rows in a batch is not enough to determine its information content.

Chapter 5

Statistical Estimation

5.1 An unknown parameter in a probability model

Suppose five independent waits between requests are

$$t_1, \ldots, t_5 = 1, 2, 3, 4, 5 \text{ minutes.}$$

Their average is three minutes. In Chapter 4, we specified an exponential rate and studied how sample averages fluctuate. Now we keep the exponential model but treat its rate $\lambda > 0$ as unknown. Which rate should we report from these observations?

Definition 5.1: Statistical model

A *statistical model* is a family of possible observation laws, written $\{P_\theta : \theta \in \Theta\}$. The parameter θ can be a scalar or a vector, and Θ specifies its allowed values.

For exponential waits, $\theta = \lambda$ and $\Theta = (0, \infty)$. For a Gaussian, the unknown parameter might be $\theta = (\mu, \sigma^2)$, with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. Choosing a family is an assumption about the data-generating process; estimating its parameters does not establish that the family is correct.

Definition 5.2: Estimator and estimate

An *estimator* is a rule $\widehat{\theta} = T(X_1, \ldots, X_n)$ that uses the observations to estimate a parameter. It is random before we collect the data. Its realized value $T(x_1, \ldots, x_n)$ is an *estimate*. The rule cannot use the unknown true parameter.

We retain the sample notation from Section 4.1. For the waits, $1/\bar{T}_n$ is a possible estimator, whereas $1/3$ requests per minute is its value on this dataset. A point estimate supplies a single parameter value. Quantifying its uncertainty is a further question.

The observations must also distinguish the parameter values of interest. A model is *identifiable* when $P_\theta = P_{\theta'}$ implies $\theta = \theta'$. For example, if $X \sim \mathcal{N}(a^2, 1)$ and $a \in \mathbb{R}$, the values a and $-a$ give exactly the same observation law. No amount of such data identifies the sign. Restricting $a \geq 0$, or estimating a^2, removes this particular ambiguity.

5.2 Method of moments

The model links its parameters to population summaries that we already know how to estimate by averaging. The method of moments solves those links in reverse. For exponential waits, the known relation $\mathbb{E}_\lambda[T] = 1/\lambda$ says that a large rate corresponds to a short average wait.

Replacing the population mean by the observed mean gives

$$\frac{1}{\widehat{\lambda}_{\mathrm{MM}}} = \bar{t}_n, \qquad \widehat{\lambda}_{\mathrm{MM}} = \frac{1}{\bar{t}_n}.$$

The five waits therefore give $\widehat{\lambda}_{\mathrm{MM}} = 1/3$ requests per minute. The units also follow from inverting a mean measured in minutes.

Definition 5.3: Method of moments

For a model with k unknown parameters, choose k population moments $m_j(\theta) = \mathbb{E}_\theta[X^j]$ and their empirical counterparts $\widehat{m}_j = n^{-1}\sum_i X_i^j$. A method-of-moments estimator solves

$$m_j(\widehat{\theta}_{\mathrm{MM}}) = \widehat{m}_j, \qquad j = 1, \ldots, k,$$

with a solution in Θ. Other chosen functions of X can replace the powers when their expectations identify the parameter.

For Bernoulli data the mean is p, so matching means gives $\widehat{p}_{\mathrm{MM}} = \bar{X}_n$. For Poisson data the mean is λ, so $\widehat{\lambda}_{\mathrm{MM}} = \bar{X}_n$. The population moments are given in Chapter 2.

5.2.1 Two moments for a Gamma model

Suppose instead that independent positive observations follow a Gamma distribution with shape $\alpha > 0$ and *rate* $\beta > 0$. The earlier formulas are

$$\mathbb{E}[X] = \frac{\alpha}{\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\beta^2}.$$

Let $\bar{x} = n^{-1}\sum_i x_i$ and $v_n = n^{-1}\sum_i (x_i - \bar{x})^2 = \widehat{m}_2 - \widehat{m}_1^2$. Matching the first two raw moments is equivalent to solving

$$\bar{x} = \frac{\widehat{\alpha}}{\widehat{\beta}}, \qquad v_n = \frac{\widehat{\alpha}}{\widehat{\beta}^2}.$$

Dividing the second equation by the first gives

$$\widehat{\beta}_{\mathrm{MM}} = \frac{\bar{x}}{v_n}, \qquad \widehat{\alpha}_{\mathrm{MM}} = \frac{\bar{x}^2}{v_n}. \tag{5.1}$$

For the values $1, 2, 3, 4, 5$, $\bar{x} = 3$ and $v_n = 2$, giving $\widehat{\alpha} = 4.5$ and $\widehat{\beta} = 1.5$ per minute. The Gamma fit can match both the mean and spread. An exponential fit has only one free parameter and ties the variance to the square of the mean.

The denominator here is n because we are matching empirical raw moments. Replacing v_n by the variance with denominator $n - 1$ would give a different rule. If every observation is identical, $v_n = 0$ and these equations give no finite positive Gamma parameters.

5.2.2 When matching moments is justified

The required population moments must exist, and the chosen moments must distinguish the parameters. Even then, a finite dataset can give multiple solutions, no admissible solution, or an unstable solution. For example, dividing by a very small empirical variance can produce very large estimates.

The LLN from Section 4.3 explains why the method often works: empirical moments approach their population values. If the inverse map from moments to parameters is well-defined and continuous near the true moments, the estimates approach the true parameters. This argument needs those conditions; solving equations alone does not guarantee accurate estimation.

5.3 Likelihood and maximum likelihood

Matching moments uses selected summaries. Likelihood uses the joint probability model of the observations. We now hold the recorded data fixed and vary the candidate parameter.

Definition 5.4: Likelihood and maximum likelihood estimator

For i.i.d. observations with PMF or density f_θ, the likelihood is

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^{n} f_\theta(x_i).$$

A maximum likelihood estimator (MLE) is a parameter value attaining the largest likelihood within the allowed parameter space:

$$\widehat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta \in \Theta} L(\theta; x).$$

The maximum need not be unique, and it need not exist.

For discrete observations, the likelihood is the probability of the recorded sample, viewed as a function of θ. For continuous observations it is a joint *density* value. Comparing densities compares probabilities of small neighborhoods of the recorded observations to first order, where the neighborhood sizes do not depend on θ. Likelihood is not a probability distribution over parameter values, and it need not integrate to one as a function of θ.

If observations are dependent, we must use their actual joint law. Multiplying marginal densities in that case generally gives the wrong likelihood.

5.3.1 Log-likelihood and the exponential rate

The logarithm is strictly increasing, so maximizing a positive likelihood is equivalent to maximizing its log. Under independence,

$$\ell(\theta; x) = \log L(\theta; x) = \sum_{i=1}^{n} \log f_\theta(x_i).$$

Adding logarithms is also numerically safer than multiplying many small densities. A factor independent of θ can be dropped when locating the maximum; a factor that depends on θ cannot.

For positive exponential waits, let $s = \sum_i t_i > 0$. Then

$$L(\lambda) = \lambda^n e^{-\lambda s}, \qquad \ell(\lambda) = n \log \lambda - \lambda s.$$

Differentiation gives

$$\ell'(\lambda) = \frac{n}{\lambda} - s, \qquad \ell''(\lambda) = -\frac{n}{\lambda^2} < 0.$$

Thus the unique maximum is

$$\widehat{\lambda}_{\mathrm{ML}} = \frac{n}{\sum_i t_i} = \frac{1}{\bar{t}_n}. \tag{5.2}$$

Here maximum likelihood and method of moments agree. For our five waits, both report 1/3 per minute. Agreement in this model does not mean the two methods are equivalent in general.

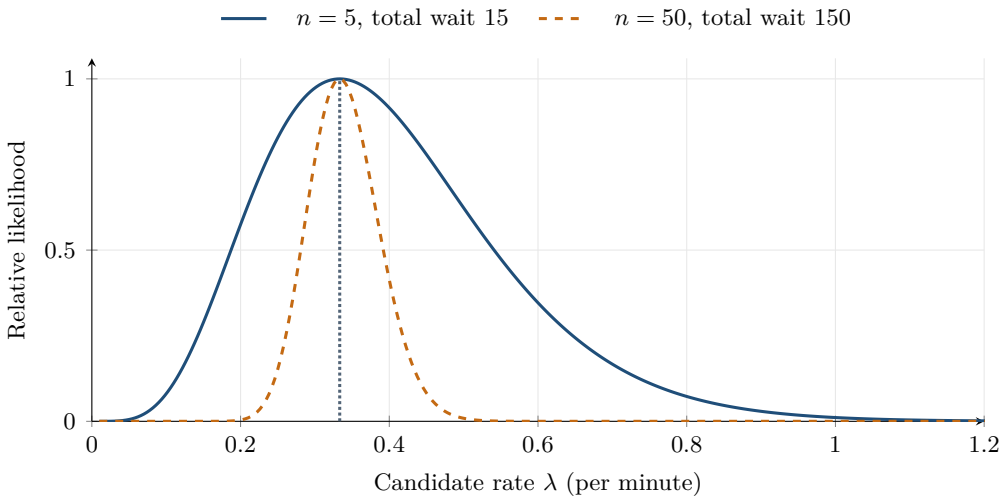


Figure 5.1: Relative likelihood $L(\lambda)/L(\widehat{\lambda})$ for two illustrative datasets with the same average wait of three minutes: $n = 5, s = 15$, and $n = 50, s = 150$. Both maximize at 1/3, but the larger sample distinguishes nearby rates more sharply. These curves are likelihood ratios, not densities over the rate.

5.3.2 The support can determine the maximum

Consider a different model, $X_i \sim \mathrm{Uniform}(0, \theta)$ with $\theta > 0$, using density $f_\theta(x) = \theta^{-1}\mathbf{1}_{\{0 \le x \le \theta\}}$. For a positive sample maximum $x_{(n)} = \max_i x_i$, the joint likelihood is

$$L(\theta) = \theta^{-n}\mathbf{1}_{\{\theta \ge x_{(n)}\}}.$$

Any upper endpoint below an observed value has likelihood zero. On the admissible region, the likelihood decreases with θ, so

$$\widehat{\theta}_{\mathrm{ML}} = x_{(n)}, \qquad \widehat{\theta}_{\mathrm{MM}} = 2\bar{x}_n,$$

where the moment estimate follows from $\mathbb{E}[X] = \theta/2$. For $x = (0.2, 0.4, 0.5, 0.7, 0.9)$, the two estimates are 0.9 and 1.08. The maximum occurs at a support boundary. Solving $\ell'(\theta) = 0$ would miss it entirely.

More generally, a stationary point need not be a global maximum. Parameter constraints, boundaries, and unbounded likelihoods must be checked. For instance, fitting a Gaussian with free variance to identical observations lets the likelihood increase without bound as the variance approaches zero.

5.4 Fitting a Gaussian mean and variance

Let X_i be i.i.d. $\mathcal{N}(\mu, v)$, writing $v = \sigma^2 > 0$ to keep variance distinct from standard deviation. The log-likelihood is

$$\ell(\mu, v) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log v - \frac{1}{2v}\sum_{i=1}^{n}(x_i - \mu)^2.$$

For each fixed v, maximizing over μ minimizes the squared deviations. The identity

$$\sum_i (x_i - \mu)^2 = \sum_i (x_i - \bar{x})^2 + n(\mu - \bar{x})^2$$

shows directly that $\widehat{\mu}_{\mathrm{ML}} = \bar{x}$. With $Q = \sum_i (x_i - \bar{x})^2 > 0$, differentiate with respect to v:

$$\frac{\partial \ell(\bar{x}, v)}{\partial v} = -\frac{n}{2v} + \frac{Q}{2v^2} = \frac{Q - nv}{2v^2}.$$

This derivative is positive for $v < Q/n$ and negative for $v > Q/n$.

Result 5.5: Gaussian maximum likelihood estimates

For a nonconstant i.i.d. Gaussian sample,

$$\widehat{\mu}_{\mathrm{ML}} = \bar{X}_n, \qquad \widehat{v}_{\mathrm{ML}} = \frac{1}{n}\sum_i (X_i - \bar{X}_n)^2.$$

If all observations coincide, there is no finite maximizer with $v > 0$.

The method of moments gives these same formulas by matching $\mathbb{E}[X] = \mu$ and $\mathbb{E}[X^2] = \mu^2 + v$. The mean and variance formulas still make sense for non-Gaussian data, but their Gaussian maximum-likelihood interpretation then depends on using a Gaussian model.

5.5 Bias, variance, and consistency

To evaluate a rule, imagine repeating the entire sampling experiment with the true parameter θ fixed. The subscript in $\mathbb{E}_\theta$ or Var_θ specifies that sampling distribution.

Definition 5.6: Bias and mean squared error

For a scalar estimator with finite second moment,

$$\mathrm{Bias}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[\widehat{\theta}] - \theta, \qquad \mathrm{MSE}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[(\widehat{\theta} - \theta)^2].$$

An estimator is *unbiased* if its bias is zero for every allowed parameter value. Expanding about $\mathbb{E}_\theta[\widehat{\theta}]$ gives

$$\mathrm{MSE}_\theta(\widehat{\theta}) = \mathrm{Var}_\theta(\widehat{\theta}) + \mathrm{Bias}_\theta(\widehat{\theta})^2.$$

The cross term vanishes because the centered estimator has expectation zero.

Bias is an average error across repeated datasets; it is not the error in one realized estimate. An unbiased estimator can fluctuate substantially. Under squared-error comparison, reducing variance can compensate for some bias.

5.5.1 Why the sample variance uses $n-1$

This answers the question left open in Section 4.1. For i.i.d. observations with mean μ and finite variance σ^2,

$$\sum_i (X_i - \bar{X}_n)^2 = \sum_i (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2.$$

Taking expectations and using $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ gives

$$\mathbb{E}\left[\sum_i (X_i - \bar{X}_n)^2\right] = n\sigma^2 - n\frac{\sigma^2}{n} = (n-1)\sigma^2.$$

Therefore

$$\mathbb{E}[\widehat{v}_{\mathrm{ML}}] = \frac{n-1}{n}\sigma^2, \qquad \mathbb{E}[S_n^2] = \sigma^2, \quad S_n^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X}_n)^2. \tag{5.3}$$

Centering at the fitted mean removes part of the observed variation. The $n-1$ correction makes the variance estimator unbiased for $n \geq 2$. This expectation calculation needs no Gaussian assumption. Under a Gaussian model the denominator n still maximizes the likelihood; unbiasedness and maximum likelihood are different criteria.

5.5.2 A biased estimate can have smaller squared error

For independent Uniform$(0, \theta)$ observations, the moment estimator $2\bar{X}_n$ is unbiased, with MSE $\theta^2/(3n)$. For $M = \max_i X_i$, $\mathbb{P}_\theta(M \leq m) = (m/\theta)^n$ on $[0, \theta]$. Its density is nm^{n-1}/θ^n; integrating m and m^2 against it gives

$$\mathbb{E}_\theta[M] = \frac{n}{n+1}\theta, \qquad \mathbb{E}_\theta[M^2] = \frac{n}{n+2}\theta^2.$$

Consequently its bias is $-\theta/(n+1)$ and

$$\mathrm{MSE}_\theta(M) = \mathbb{E}_\theta[M^2] - 2\theta\mathbb{E}_\theta[M] + \theta^2 = \frac{2\theta^2}{(n+1)(n+2)}.$$

At $n = 5, \theta = 1$, these MSEs are $1/15 \approx 0.0667$ for method of moments and $1/21 \approx 0.0476$ for the biased MLE. This comparison concerns these two rules under this model and squared error; it does not make the MLE universally best.

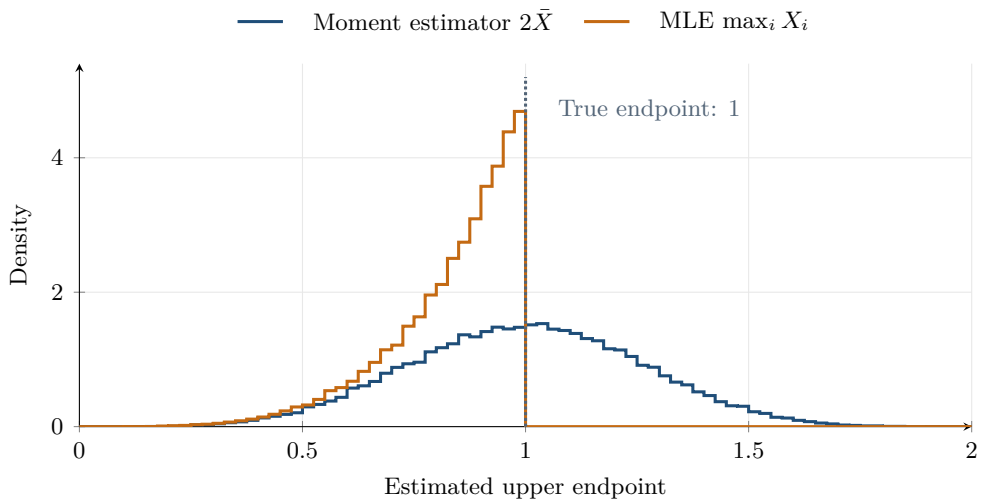


Figure 5.2: Estimates from 50,000 independent datasets, each containing five Uniform$(0, 1)$ observations. The moment estimator spreads on both sides of the true endpoint. The maximum always underestimates it, but its smaller spread yields the smaller MSE in this example. Curves show density-normalized histogram steps on common bins.

5.5.3 Consistency concerns growing sample size

Definition 5.7: Consistency

A sequence of estimators is *consistent* for θ if for every $\varepsilon > 0$,

$$\mathbb{P}_\theta(|\widehat{\theta}_n - \theta| > \varepsilon) \longrightarrow 0 \quad \text{as } n \to \infty.$$

This is convergence in probability of the estimation error to zero.

For exponential waits, the LLN gives $\bar{T}_n \to 1/\lambda$ in probability. Taking reciprocals is continuous near this positive limit, so $\widehat{\lambda} = n/\sum_i T_i \to \lambda$. This proves consistency without implying unbiasedness. In fact, for $n > 1$, the Gamma law of $\sum_i T_i$ gives

$$\mathbb{E}_\lambda[\widehat{\lambda}] = \frac{n}{n-1}\lambda.$$

The rate estimate has positive finite-sample bias, which tends to zero. For $n = 1$ its expectation is infinite, although the estimate itself is finite almost surely. Conversely, using just X_1 as an estimate of the population mean is unbiased but generally inconsistent: its spread does not decrease when more observations become available.

5.6 Likelihood gives a training loss

So far one distribution described every observation. For prediction, specify a distribution of the response Y_i conditional on its observed features x_i. Treat the feature values as fixed and assume responses are conditionally independent. The conditional likelihood is $\prod_i p_\theta(y_i \mid x_i)$. Estimation chooses θ to make this observed response sample have large conditional likelihood.

5.6.1 Gaussian regression and squared error

Let $Y_i \mid x_i \sim \mathcal{N}(f_\theta(x_i), \sigma^2)$, with the same known $\sigma^2 > 0$ for every response. Then

$$-\ell(\theta) = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_i \big(y_i - f_\theta(x_i)\big)^2. \tag{5.4}$$

The first term and the positive scale factor do not change which θ minimizes the expression. Thus maximizing this likelihood is equivalent to minimizing mean squared error. This explains a familiar loss through an explicit noise model. Chapter 3 showed why a conditional mean predicts under squared error; here we estimate the parameters of a chosen conditional-mean model from data.

If a common variance v is also unknown, keep the term $\frac{n}{2}\log v$. For any fixed θ with positive residual sum of squares, its fitted value is $\widehat{v}(\theta) = n^{-1}\sum_i (y_i - f_\theta(x_i))^2$. When the model fits every response exactly, this joint likelihood is unbounded as $v \downarrow 0$. A finite solution then requires an additional constraint or a different model.

5.6.2 A predicted variance changes the objective

Suppose the model predicts both a mean $\mu_\theta(x_i)$ and a positive variance $v_\theta(x_i)$. Up to the constant $\frac{n}{2}\log(2\pi)$,

$$-\ell(\theta) = \frac{1}{2}\sum_i \left[\log v_\theta(x_i) + \frac{(y_i - \mu_\theta(x_i))^2}{v_\theta(x_i)}\right]. \tag{5.5}$$

The inverse variance weights the squared residual, but predicting a large variance also increases the logarithmic term. Dropping that term would let the model hide every residual by making its variance arbitrarily large. A parameterization such as $v_\theta(x) = e^{s_\theta(x)}$ enforces positivity. It does not by itself prevent variance collapse at a perfectly fitted observation.

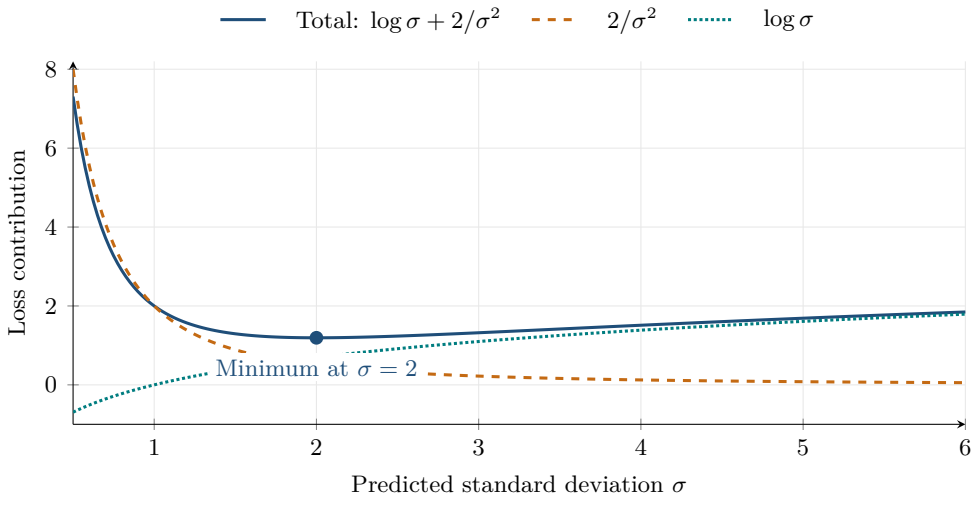


Figure 5.3: For one fixed residual $y - \mu = 2$, Gaussian negative log-likelihood, excluding its constant, is $\log\sigma + 2/\sigma^2$. Its minimum occurs at $\sigma = 2$. The logarithmic term prevents the scale from growing without cost. For residual zero, in contrast, the loss tends to $-\infty$ as $\sigma \downarrow 0$.

5.6.3 Bernoulli responses and binary log loss

For $Y_i \in \{0,1\}$, let the model predict $p_i = p_\theta(x_i) \in (0,1)$. The conditional PMF is $p_i^{y_i}(1-p_i)^{1-y_i}$, so maximizing likelihood minimizes

$$-\ell(\theta) = -\sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right]. \tag{5.6}$$

For an observed positive response, assigning probability 0.9 contributes $-\log 0.9 \approx 0.105$, whereas probability 0.1 contributes $-\log 0.1 \approx 2.303$. Confident wrong predictions incur a large cost. The information-theoretic interpretation of this loss belongs to the later chapter on entropy and cross-entropy.

The statistical model determines the objective. An optimization algorithm searches for parameters that minimize it; a numerical stationary point need not be a global optimum. Good training likelihood also does not guarantee good prediction on new data. Evaluation requires the independent-data reasoning from Section 4.6.

5.7 MAP estimation and prior information

Maximum likelihood compares how well candidate parameters account for the observations. A Bayesian model also specifies a prior distribution on the parameter before using the current data. A prior density $\pi(\theta)$ describes that parameter uncertainty. Conditional on a parameter value, the same observation model supplies the likelihood.

The conditioning rules from Chapters 1 and 3 give

$$\pi(\theta \mid x) = \frac{L(\theta; x)\pi(\theta)}{\int_\Theta L(u; x)\pi(u)\,du},$$

provided the denominator is positive and finite. For discrete parameters, replace the integral by a sum. The denominator is constant in θ, so it does not affect the location of a posterior maximum.

Definition 5.8: Maximum a posteriori estimate

A MAP estimate maximizes the posterior density (or posterior mass for a discrete parameter):

$$\widehat{\theta}_{\text{MAP}} \in \arg\max_{\theta\in\Theta}\{\ell(\theta; x) + \log \pi(\theta)\}.$$

Equivalently, it minimizes negative log-likelihood plus negative log-prior. This is a point summary of the posterior, not the entire posterior distribution.

5.7.1 A Bernoulli probability with a Beta prior

Take k successes in n independent Bernoulli observations. The likelihood is proportional to $p^k(1-p)^{n-k}$; its MLE is k/n, including boundary values if $p \in [0,1]$. To express prior information about a probability, introduce the Beta family:

$$\pi(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \quad 0 < p < 1, \quad a, b > 0, \qquad B(a,b) = \int_0^1 u^{a-1}(1-u)^{b-1}\,du.$$

Here a, b are specified prior parameters. Multiplying prior and likelihood adds their powers, yielding

$$p \mid x \sim \text{Beta}(a+k, b+n-k).$$

For $a + k > 1$ and $b + n - k > 1$, differentiation of the log posterior gives

$$\widehat{p}_{\mathrm{MAP}} = \frac{k + a - 1}{n + a + b - 2}. \tag{5.7}$$

The conditions ensure an interior mode. Outside them, check the boundaries instead of applying this formula blindly.

With a Beta$(2, 2)$ prior and seven successes in ten trials, the posterior is Beta$(9, 5)$, and

$$\widehat{p}_{\mathrm{ML}} = \frac{7}{10} = 0.7, \qquad \widehat{p}_{\mathrm{MAP}} = \frac{8}{12} = \frac{2}{3}.$$

The symmetric prior favors probabilities closer to $1/2$, moving this estimate slightly downward. A different prior can move it differently.

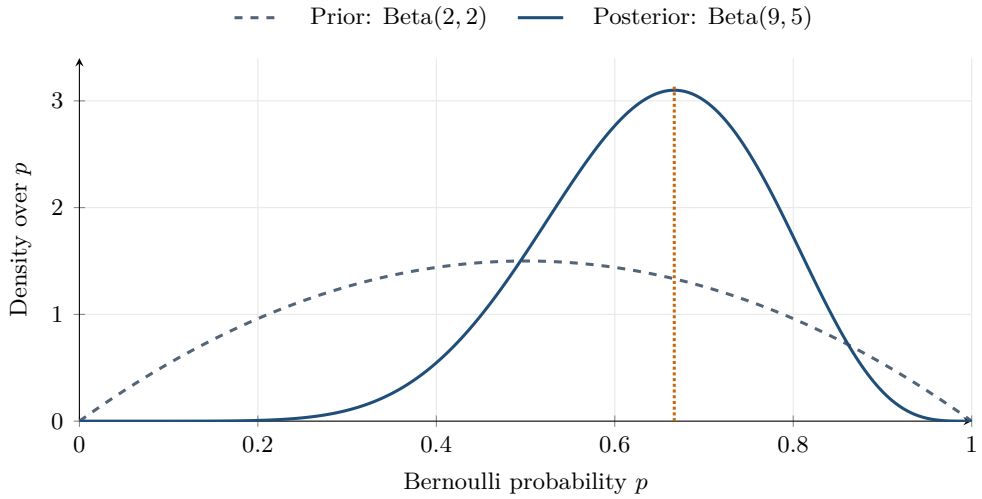


Figure 5.4: The Beta$(2, 2)$ prior and Beta$(9, 5)$ posterior after seven successes in ten trials. Both are normalized densities over p. The posterior mode is $2/3$; it summarizes the distribution without describing its whole spread.

The posterior mean is another point summary. For a Beta(A, B) law, inserting one extra factor of p into the normalizing integral gives

$$\mathbb{E}[p] = \frac{B(A + 1, B)}{B(A, B)} = \frac{A}{A + B}.$$

Here it is $9/14 \approx 0.643$, different from the MAP. The conditional-mean result in Section 3.4 makes the posterior mean the minimizer of posterior expected squared error. The mode answers a different question. For continuous parameters, MAP also depends on the chosen parameterization because a transformed density includes a Jacobian factor.

5.7.2 A Gaussian prior pulls an estimated mean toward its center

Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with known $\sigma^2 > 0$, and use prior $\mu \sim \mathcal{N}(m_0, \tau^2)$, $\tau^2 > 0$. Here σ^2 describes variation in observations, whereas τ^2 describes prior uncertainty about their mean. Terms depending on μ in negative log posterior are

$$\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 + \frac{1}{2\tau^2} (\mu - m_0)^2.$$

Setting its derivative to zero gives

$$\widehat{\mu}_{\mathrm{MAP}} = \frac{(n/\sigma^2)\bar{x} + (1/\tau^2)m_0}{n/\sigma^2 + 1/\tau^2}. \tag{5.8}$$

Completing the square shows that the posterior is Gaussian with this mean and variance $(n/\sigma^2 + 1/\tau^2)^{-1}$. Its mean and mode coincide in this special case. The weights are precisions, meaning inverse variances: the variance of $\bar{X}_n$ is σ^2/n, and the prior variance is τ^2. As n grows with the prior fixed, the data receive more relative weight.

For $n = 4, \sigma^2 = 4, \bar{x} = 1.6, m_0 = 0, \tau^2 = 1$, both precisions are one. The MLE is 1.6, the MAP is 0.8, and the posterior variance is $1/2$.

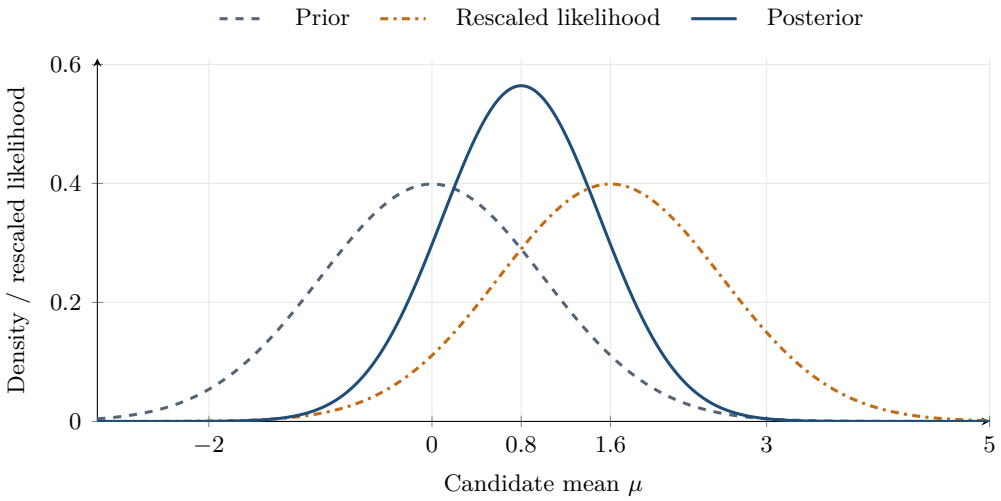


Figure 5.5: Prior $\mathcal{N}(0, 1)$, likelihood as a function of μ rescaled to integrate to one, and posterior $\mathcal{N}(0.8, 0.5)$ in the stated example. The rescaled likelihood has shape $\mathcal{N}(1.6, 1)$; its normalization is only for the plot and does not turn it into a prior or posterior.

5.7.3 Gaussian priors and quadratic regularization

In Gaussian regression with known observation variance σ^2, put a prior $w \sim \mathcal{N}(0, \tau^2 I)$ on the model coefficients. MAP minimizes, up to constants,

$$\frac{1}{2\sigma^2} \sum_i (y_i - f_w(x_i))^2 + \frac{1}{2\tau^2} \|w\|^2.$$

Multiplying by $2\sigma^2$ gives the equivalent objective

$$\sum_i (y_i - f_w(x_i))^2 + \alpha \|w\|^2, \qquad \alpha = \frac{\sigma^2}{\tau^2}. \tag{5.9}$$

For a linear model this is ridge regression, whose linear-algebra solution comes later in the course. If the data term is an *average* squared error, the same prior gives coefficient α/n, not α. The normalization of the loss matters when interpreting a penalty. An unpenalized intercept would require a different prior choice for that coefficient. More generally, MAP imposes the penalty $-\log \pi(w)$; a quadratic penalty specifically corresponds to a Gaussian prior.

Chapter 6

Sampling Distributions, Confidence Intervals, and Bootstrap

6.1 What uncertainty remains after fitting

The five waiting times from Chapter 5,

$$1,\ 2,\ 3,\ 4,\ 5 \text{ minutes},$$

gave an exponential rate estimate of $\widehat{\lambda} = 1/3$ requests per minute. This is an estimate from just five observations. A second collection of five waits would usually give a different rate. How much variation should we expect, and how can we report it alongside the estimate?

Section 4.1 already introduced sampling distributions and standard errors. We now apply that reasoning to the estimators from Chapter 5. To study a rule $\widehat{\theta} = T(X_1, \ldots, X_n)$, imagine keeping the population parameter θ fixed, drawing a new dataset, and applying the *same rule* each time. A histogram of the resulting estimates describes how the estimation procedure varies across datasets.

Three distributions answer different questions. The observation law describes a new wait or measurement. A sampling distribution describes the estimate obtained from a new dataset. A posterior, as in Section 5.7, describes uncertainty about a parameter conditional on the recorded data and a specified prior. The calculations below use sampling distributions to build intervals with a stated frequency of covering the true parameter.

For a population mean, the earlier formulas give

$$\mathrm{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}, \qquad \widehat{\mathrm{SE}}(\bar{X}_n) = \frac{S_n}{\sqrt{n}}.$$

The second expression estimates the first when σ is unknown. To turn an error scale into an interval, we also need to know, exactly or approximately, how often errors exceed a multiple of that scale.

6.2 Confidence intervals and repeated sampling

Suppose a sensor measures an unknown constant level μ. Each dataset produces a mean and an interval around it. Across repeated experiments, both endpoints move. A useful interval rule should contain μ in a specified proportion of those experiments.

Definition 6.1: Confidence interval and coverage

Let $0 < \alpha < 1$. A confidence interval procedure reports $C(X) = [L(X), U(X)]$, where both endpoints are functions of the sample. Its coverage at parameter θ is

$$\mathbb{P}_\theta\{L(X) \le \theta \le U(X)\}.$$

It has confidence level at least $1-\alpha$ if this probability is at least $1-\alpha$ for every parameter allowed by the model. An exact central construction may have equality. An approximate 95% interval targets coverage 0.95 using an approximation.

The probability concerns the random sample and the interval it produces. After we observe a particular dataset, its numerical endpoints are fixed. The frequentist model also treats θ as fixed, so its coverage statement does not supply a posterior probability for this one interval. For example, saying that a procedure covers the true sensor level in 95% of repeated experiments explains the guarantee that we used when constructing the reported interval.

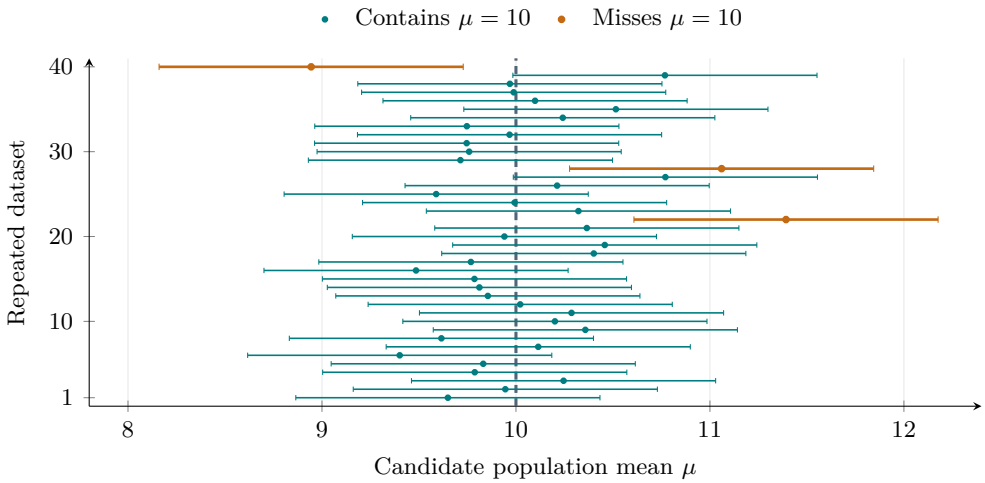


Figure 6.1: Repeated 95% Gaussian mean intervals. Each horizontal segment comes from a different dataset, with $n = 25$, true mean $\mu = 10$, and known observation standard deviation $\sigma = 2$. The dashed line marks the same true mean in every experiment. Orange intervals miss it. A finite collection need not contain exactly 95% successful intervals.

6.2.1 A Gaussian mean with known variance

For independent Gaussian observations with mean μ and known variance $\sigma^2 > 0$, the exact Gaussian sampling law from Section 4.4 gives

$$Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1).$$

The numerator is the estimation error. Dividing by its standard error puts it on a scale whose distribution is the same for every value of the unknown μ.

Definition 6.2: Quantile and pivot

For a random variable with CDF F, its p-quantile is $q_p = \inf\{x : F(x) \geq p\}$, for $0 < p < 1$. Write z_p for the standard Gaussian quantile. A *pivot* is a function of the observations and the unknown parameter whose sampling distribution does not depend on unknown parameters. The variable Z above is a pivot.

The standard Gaussian places $1-\alpha$ of its probability between $-z_{1-\alpha/2}$ and $z_{1-\alpha/2}$. Therefore

$$\mathbb{P}_\mu\left\{-z_{1-\alpha/2} \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z_{1-\alpha/2}\right\} = 1 - \alpha.$$

Rearranging the inequalities leaves the unknown μ between two quantities that we can compute from the data.

Result 6.3: Gaussian mean interval with known variance

For an i.i.d. Gaussian sample with known $\sigma^2 > 0$, the interval

$$\left[\bar{X}_n - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X}_n + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

has exact coverage $1 - \alpha$. For 95% confidence, $z_{0.975} \approx 1.96$.

For example, if $n = 25$, $\bar{x} = 10.4$, and a known $\sigma = 2$, the standard error is 0.4. The interval is

$$10.4 \pm 1.96(0.4) = [9.616,\ 11.184].$$

Its endpoints and μ share the measurement units. The number 1.96 is dimensionless.

6.2.2 Interval width and the sampling design

The margin of error in this construction is $m = z_{1-\alpha/2}\sigma/\sqrt{n}$. More observation noise increases the margin. More independent observations reduce it. Raising the confidence level increases the quantile and widens the interval. For example, the Gaussian multipliers are approximately 1.645 at 90%, 1.960 at 95%, and 2.576 at 99%.

For known σ, achieving a chosen margin $m > 0$ requires

$$n \geq \left(\frac{z_{1-\alpha/2}\sigma}{m}\right)^2,$$

rounded upward to an integer. Thus halving the margin requires four times as many independent observations. This calculation inherits the sampling assumptions. Correlated measurements do not necessarily provide that improvement, as Section 4.5 showed.

6.2.3 What the interval refers to

The interval estimates a population parameter, here the sensor's mean. It need not contain 95% of individual readings. Even if we measure the mean very accurately, a new reading still has observation noise. Section 6.3 will make this distinction explicit.

A Bayesian credible interval answers a different probability question: it contains a specified amount of the posterior probability after conditioning on the observations and a prior. The

posterior calculation in Section 5.7 supplies that distribution. A confidence interval derives its guarantee from repetition of the sampling procedure. The two approaches can sometimes produce the same numerical interval, but they justify it differently.

Coverage also depends on choosing the procedure as specified. Reporting only unusually narrow intervals, repeatedly inspecting fresh data until an interval gives a desired conclusion, or selecting a model after seeing the same test results changes the experiment. The original fixed-sample coverage statement alone does not justify such a procedure.

6.3 Mean intervals with unknown variance

In most applications σ is unknown. We estimate it by S_n, using the sample variance from Section 4.1 with denominator $n-1$. For small samples, this replacement adds appreciable uncertainty: some datasets happen to have a small observed spread and hence an overly optimistic estimated standard error. A wider reference distribution accounts for that fluctuation under a Gaussian model.

6.3.1 Why Student's distribution appears

We already met Student's t-distribution as a Gaussian with a random precision in Section 2.10. Here its random denominator comes from estimating the observation variance.

Definition 6.4: Chi-square distribution

If $Z_1, \dots, Z_\nu$ are independent standard Gaussians, then

$$U = \sum_{j=1}^{\nu} Z_j^2 \sim \chi^2_\nu.$$

The integer ν counts the independent squared Gaussian components and is called the number of degrees of freedom. In the rate convention of Section 2.5, this law is $\text{Gamma}(\nu/2, 1/2)$.

For an i.i.d. Gaussian sample, the fitted deviations $X_i - \bar{X}_n$ sum to zero. Estimating one mean leaves $n-1$ independent directions of Gaussian variation. More precisely, a Gaussian sampling result states that

$$U = \frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad Z = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim \mathcal{N}(0,1),$$

and U and Z are independent. The independence uses the Gaussian assumption. It follows by separating the mean direction from the orthogonal residual directions in a Gaussian vector, using the zero-covariance independence property from Section 3.7.

Consequently, for $\nu = n-1$,

$$\frac{Z}{\sqrt{U/\nu}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n}.$$

Conditional on U, the left-hand side is Gaussian with precision U/ν. This precision has law $\text{Gamma}(\nu/2, \nu/2)$, exactly the mixture construction that previously produced t_ν.

Theorem 6.5: Student pivot for a Gaussian mean

For $n \geq 2$ i.i.d. observations from $\mathcal{N}(\mu, \sigma^2)$, with $\sigma^2 > 0$,

$$\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t_{n-1}.$$

If $t_{\nu,p}$ is the p-quantile of t_ν, then

$$\left[\bar{X}_n - t_{n-1,1-\alpha/2}\frac{S_n}{\sqrt{n}}, \quad \bar{X}_n + t_{n-1,1-\alpha/2}\frac{S_n}{\sqrt{n}}\right]$$

has exact coverage $1 - \alpha$.

The derivation repeats the Gaussian pivot inversion, with a Student quantile in place of a Gaussian quantile. At low degrees of freedom, Student's heavier tails require a larger multiplier. As the degrees of freedom grow, the multiplier approaches the Gaussian one.

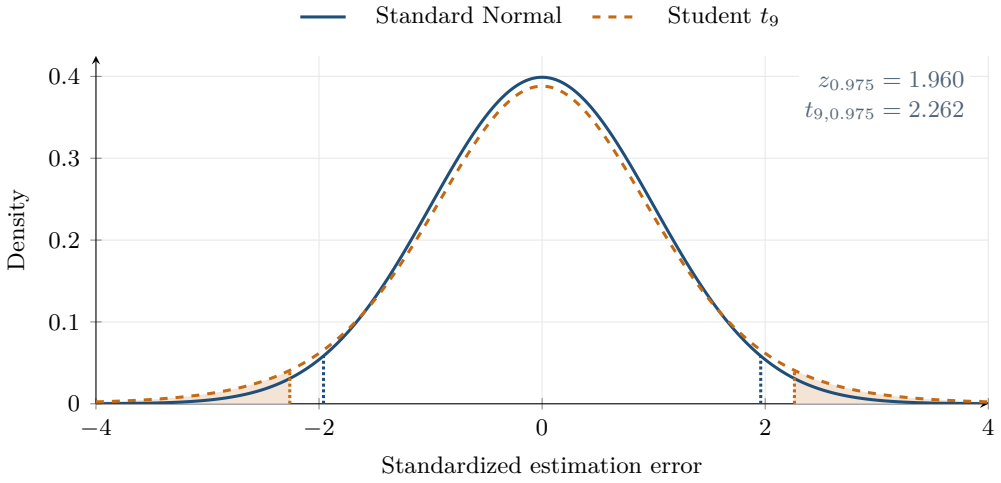


Figure 6.2: A standard Gaussian and Student's t_9 sampling reference densities. The Student reference has heavier tails. Its 97.5% quantile is approximately 2.262, compared with 1.960 for the Gaussian. Both intervals allocate 2.5% to each tail of their own reference distribution.

For ten Gaussian measurements with observed mean $\bar{x} = 10.2$ and sample standard deviation $s = 1.5$,

$$\widehat{\mathrm{SE}} = \frac{1.5}{\sqrt{10}} \approx 0.4743, \qquad t_{9,0.975} \approx 2.262.$$

The 95% interval is

$$10.2 \pm 2.262(0.4743) \approx [9.127,\ 11.273].$$

Using 1.96 with this estimated standard error would give the narrower interval $[9.270, 11.130]$. Under the Gaussian model at this sample size, that narrower procedure falls short of 95% coverage.

6.3.2 A large-sample interval for non-Gaussian observations

Gaussian sampling gives an exact Student law. For general i.i.d. observations with finite mean and finite, positive variance, use the CLT from Section 4.4. The LLN applied to X_i and

X_i^2 also makes S_n approach σ. Replacing σ by this consistent estimate preserves the limiting law:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \xrightarrow{d} \mathcal{N}(0,1).$$

This gives an approximate interval $\bar{x} \pm z_{1-\alpha/2} s/\sqrt{n}$. Using a Student multiplier is also common and has the same large-sample limit, but does not make the interval exact for non-Gaussian data.

The accuracy depends on the observation law and sample size. Strong skewness and rare extreme values can make convergence slow. There is no universal sample size at which every population has a reliable Gaussian approximation. Infinite variance and dependence require separate analysis rather than automatic use of this formula.

6.3.3 A mean interval and a prediction interval

Under the same i.i.d. Gaussian model, consider a future independent reading X_{new}. Its difference from the current sample mean has variance

$$\operatorname{Var}(X_{\text{new}} - \bar{X}_n) = \sigma^2 + \frac{\sigma^2}{n}.$$

The observation noise remains even when we know the mean accurately. The corresponding exact $1-\alpha$ prediction interval is

$$\bar{x} \pm t_{n-1,1-\alpha/2}\, s\sqrt{1 + \frac{1}{n}}.$$

Its coverage concerns the joint experiment of the training sample and a new reading. The mean interval uses $s/\sqrt{n}$ instead. As n grows, the mean interval shrinks, while the prediction interval retains the scale of individual observation noise.

6.3.4 Another exact pivot: an exponential rate

Return to the five waits at the beginning. The Gamma sum result gives, for the random total $W = \sum_i T_i$,

$$W \sim \text{Gamma}(n, \lambda), \qquad 2\lambda W \sim \text{Gamma}(n, 1/2) = \chi^2_{2n}.$$

Let $c_{\nu,p}$ denote a χ^2_ν quantile. The event $c_{2n,\alpha/2} \le 2\lambda W \le c_{2n,1-\alpha/2}$ has probability $1-\alpha$. Since $W > 0$ almost surely, dividing and inserting the observed total w gives the exact rate interval

$$\left[\frac{c_{2n,\alpha/2}}{2w}, \quad \frac{c_{2n,1-\alpha/2}}{2w}\right].$$

For $n = 5$ and observed total wait $w = 15$ minutes, $c_{10,0.025} \approx 3.247$ and $c_{10,0.975} \approx 20.483$. The 95% interval is therefore

$$[0.1082,\ 0.6828] \text{ requests per minute.}$$

Five observations leave considerable uncertainty around the point estimate $1/3$. The chi-square quantiles give unequal margins below and above this estimate: approximately 0.2251 and 0.3495. The exact coverage guarantee relies on independent exponential waits.

6.4 Proportion intervals and test accuracy

For a fixed classifier evaluated on independently sampled examples, let $X_i = 1$ when it predicts correctly and $X_i = 0$ otherwise. The population accuracy is p, and the observed accuracy is $\widehat{p} = k/n$, where k is the number correct. The Bernoulli calculation gives standard error $\sqrt{p(1-p)/n}$. Substituting $\widehat{p}$ gives the familiar Gaussian approximation.

Result 6.6: Approximate Wald interval for a proportion

For i.i.d. Bernoulli observations, the normal approximation gives

$$\widehat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

This approximation can be poor when successes or failures are rare. It is not an exact interval, and its endpoints can fall outside $[0, 1]$.

For $k = 84$ correct predictions out of $n = 100$,

$$\widehat{p} = 0.84, \qquad \widehat{\mathrm{SE}} \approx 0.0367, \qquad C_{\mathrm{Wald}} \approx [0.768,\ 0.912].$$

This quantifies uncertainty from sampling the test examples for the already fitted classifier. It does not measure how much accuracy changes when we retrain the model on a new training set.

6.4.1 Why zero observed failures do not imply certainty

If all 20 predictions are correct, then $\widehat{p} = 1$, and the plug-in standard error is zero. The Wald formula returns $[1, 1]$. Twenty correct predictions provide evidence of high accuracy, but cannot establish perfect population accuracy. Similarly, zero observed successes do not establish $p = 0$. Clipping endpoints to $[0, 1]$ does not repair the underlying coverage problem.

A useful improvement keeps the candidate population probability inside its own variance when inverting the approximate Gaussian event:

$$\frac{|\widehat{p} - p|}{\sqrt{p(1-p)/n}} \le z, \qquad z = z_{1-\alpha/2}.$$

Squaring and rearranging gives

$$(n + z^2)p^2 - (2n\widehat{p} + z^2)p + n\widehat{p}^2 \le 0.$$

The two roots supply an interval.

Result 6.7: Wilson interval for a proportion

Set $z = z_{1-\alpha/2}$. The center and half-width are

$$a = \frac{\widehat{p} + z^2/(2n)}{1 + z^2/n}, \qquad h = \frac{z}{1 + z^2/n}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n} + \frac{z^2}{4n^2}}.$$

The interval is $[a - h, a + h]$. It stays inside $[0, 1]$ and usually has much better coverage than the Wald interval near the boundaries. Its $1-\alpha$ confidence level is still approximate.

For 84/100, the 95% Wilson interval is approximately $[0.756, 0.899]$. For 0/20, it is $[0, 0.161]$, and for 20/20, it is $[0.839, 1]$. In each boundary case the unobserved outcome remains possible. Exact binomial intervals can provide a guaranteed coverage level, generally with conservative coverage because the count is discrete.

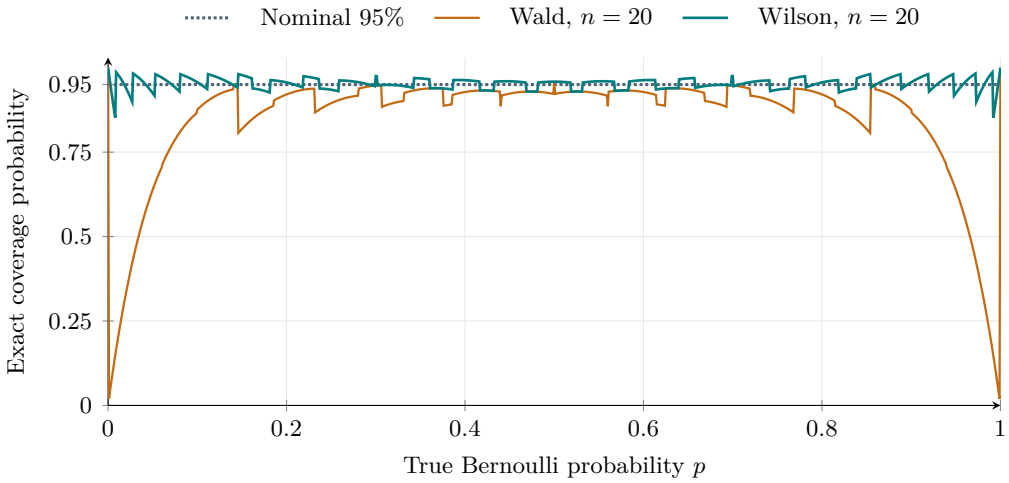


Figure 6.3: Actual coverage of nominal 95% Wald and Wilson procedures for $n = 20$. At each candidate p, coverage is the exact binomial sum over counts whose interval contains p. The jagged curves reflect the discrete count. Wilson improves the boundary behavior, but its coverage is not identically 0.95.

6.5 Bootstrap and the empirical distribution

For a mean under a Gaussian model, we could derive a sampling law. For a median, a quantile, or an F1 score, the sampling distribution can be harder to calculate. We would like to repeat the experiment many times, but usually have only one observed dataset. The bootstrap uses this dataset to construct an approximation to the population, then repeats sampling from that approximation.

6.5.1 Resampling observations with replacement

Definition 6.8: Empirical distribution and bootstrap sample

Given observations $x_1, \dots, x_n$, the empirical distribution $\widehat{F}_n$ puts probability $1/n$ on each recorded observation. Repeated values contribute their combined probability. A nonparametric bootstrap sample draws $X_1^*, \dots, X_n^*$ independently from $\widehat{F}_n$, conditional on the observed dataset. Equivalently, draw indices

$$I_1, \dots, I_n \text{ independently and uniformly from } \{1, \dots, n\}, \qquad X_j^* = x_{I_j}.$$

Sampling is with replacement, so an observation can appear several times or be absent from a resample.

For the waits $(1, 2, 3, 4, 5)$, the index sample $(1, 1, 2, 4, 5)$ produces $(1, 1, 2, 4, 5)$, whose mean is 2.6. Another draw might produce $(2, 3, 3, 3, 4)$, whose mean is 3. Both resamples have

the original sample size of five. Sampling all observations without replacement would merely permute them and leave the mean unchanged.

For any statistic T, recompute $\widehat{\theta}^* = T(X^*)$ on each resample. If the original estimate required fitting a model, recomputing it requires fitting that model again. The resamples give variation in the *estimation rule*, rather than new independent information about the population.

6.5.2 Which distribution the bootstrap approximates

An asterisk indicates randomness in resampling, conditional on the recorded data. For a population parameter θ estimated by $\widehat{\theta}$, the useful comparison is

$$\text{bootstrap errors } \widehat{\theta}^* - \widehat{\theta} \quad \text{versus sampling errors } \widehat{\theta} - \theta.$$

The empirical population plays the role of the unknown population. For a mean, its center is $\bar{x}$, so bootstrap errors about $\bar{x}$ take the place of sampling errors about μ. Similar reasoning applies to medians and other summaries when the empirical population captures the features relevant to the statistic. That approximation requires conditions and can fail, as Section 6.7 explains.

For a mean we can check the calculation directly. Conditional on the data,

$$\mathbb{E}^*[X_j^*] = \bar{x}, \qquad \mathrm{Var}^*(X_j^*) = \frac{1}{n}\sum_i (x_i - \bar{x})^2.$$

Independence within a bootstrap sample therefore gives

$$\mathbb{E}^*[\bar{X}_n^*] = \bar{x}, \qquad \mathrm{Var}^*(\bar{X}_n^*) = \frac{1}{n^2}\sum_i (x_i - \bar{x})^2 = \frac{n-1}{n}\frac{s^2}{n}. \tag{6.1}$$

For $(1, 2, 3, 4, 5)$, $\bar{x} = 3$ and $\sum_i (x_i - \bar{x})^2 = 10$. Thus the bootstrap mean has variance 0.4 and standard error $\sqrt{0.4} \approx 0.632$. The usual estimate $s/\sqrt{5} = \sqrt{0.5} \approx 0.707$ differs by the factor $\sqrt{(n-1)/n}$. The difference is appreciable at $n = 5$ and tends to zero as n grows. A bootstrap approximation does not guarantee an exact finite-sample sampling distribution.

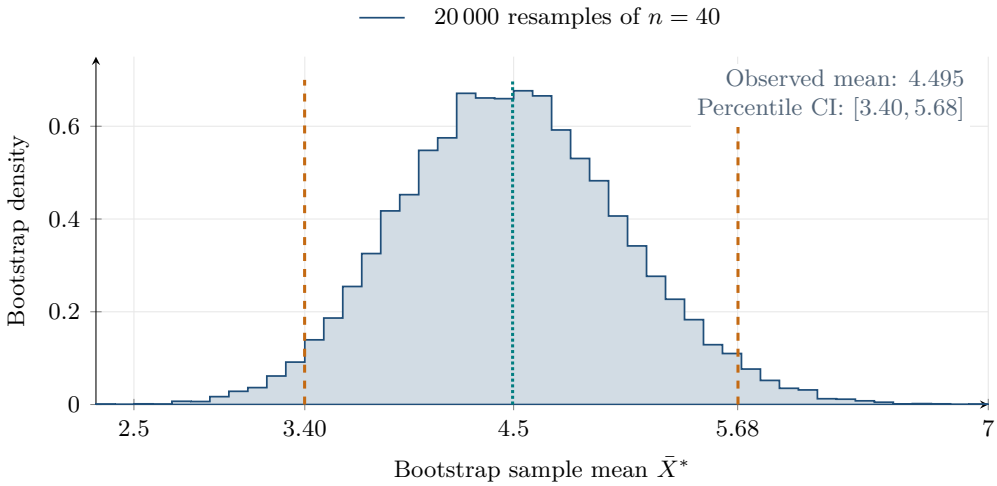


Figure 6.4: Bootstrap means from one simulated dataset of 40 exponential waits. Its observed mean is 4.495 minutes. Each of 20,000 resamples contains 40 observations drawn with replacement from this same dataset. The dotted line marks the observed mean, and the dashed lines mark the 2.5% and 97.5% bootstrap quantiles. This is a conditional bootstrap distribution, rather than the true population sampling distribution.

6.5.3 Bootstrap standard error

Choose a number of resamples B. For each $b = 1, \ldots, B$, draw n indices with replacement and recompute $\widehat{\theta}^{*(b)}$. Common choices such as $B = 10{,}000$ give many replicates for estimating spread and central quantiles. Record the random seed when reproducing a numerical illustration.

Definition 6.9: Bootstrap standard error

Let $\overline{\theta}^* = B^{-1} \sum_{b=1}^{B} \widehat{\theta}^{*(b)}$, with $B \geq 2$. The bootstrap standard error estimate is

$$\widehat{\mathrm{SE}}_{\mathrm{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} (\widehat{\theta}^{*(b)} - \overline{\theta}^*)^2}.$$

It estimates the conditional standard deviation of the bootstrap statistic, which in turn approximates the statistic's sampling standard deviation when bootstrap assumptions hold.

The divisor is $B-1$ because we estimate spread across B replicates about their replicate mean. The statistic inside each replicate still uses a dataset of size n. These are different sample sizes with different roles. Increasing B reduces simulation noise in the estimated bootstrap distribution. Increasing n provides more information about the population.

6.5.4 Percentile intervals and centered-error intervals

Definition 6.10: Percentile bootstrap interval

Let q_p^* denote the empirical p-quantile of the B bootstrap estimates. The percentile interval is

$$[q_{\alpha/2}^*,\ q_{1-\alpha/2}^*].$$

For a nominal 95% interval, retain the central 95% of the bootstrap estimates. Its coverage is approximate.

This interval can be asymmetric, following the shape of the resampled statistic. It is easy to compute, but the fact that 95% of bootstrap estimates lie between its endpoints does not by itself prove 95% coverage of the population parameter. Bias and a poor bootstrap approximation can distort coverage.

There is another construction that follows the centered-error comparison explicitly. If a^*, b^* are the lower and upper quantiles of $\widehat{\theta}^* - \widehat{\theta}$, approximate the sampling event by

$$a^* \leq \widehat{\theta} - \theta \leq b^*.$$

Inverting gives $[\widehat{\theta} - b^*, \widehat{\theta} - a^*]$, or

$$[2\widehat{\theta} - q_{1-\alpha/2}^*, \quad 2\widehat{\theta} - q_{\alpha/2}^*].$$

This is the *basic bootstrap interval*. It differs from the percentile interval unless the relevant quantiles are symmetric about $\widehat{\theta}$. Neither construction is automatically accurate for every statistic. A symmetric Gaussian approximation using $\widehat{\theta} \pm z\widehat{\mathrm{SE}}_{\mathrm{boot}}$ is another option when its shape and bias assumptions are appropriate.

For the simulated dataset in Figure 6.4, $\bar{x} \approx 4.495$ minutes and $s \approx 3.747$ minutes. The 20,000 bootstrap means give a standard error estimate of 0.586 minutes and a percentile interval $[3.399, 5.682]$ minutes. We generated the original data from an exponential population with mean five, so we can check that this particular interval covers it. Checking one successful interval does not establish the procedure's coverage. That would require repeating the whole experiment with many fresh datasets and constructing a new bootstrap interval for each one.

6.6 Comparing two models on the same examples

Suppose two fixed classifiers predict the same 200 independent test examples. Their outcomes are

	B correct	B wrong
A correct	150	30
A wrong	10	10

Model A has accuracy $180/200 = 0.90$, and B has accuracy $160/200 = 0.80$. The observed difference is 0.10. Most examples are easy or difficult for both models, so treating the two accuracy estimates as independent discards useful information.

Let A_i, B_i be the two correctness indicators on example i, and write $D_i = A_i - B_i$. Then $D_i = 1$ on the 30 examples only A gets right, $D_i = -1$ on the 10 examples only B gets right, and $D_i = 0$ on the other 160. The estimated population accuracy difference is simply $\bar{D} = 0.10$. The sample mean calculation applies to these *paired differences*:

$$\widehat{\mathrm{SE}}(\bar{D}) = \sqrt{\frac{1}{n(n-1)} \sum_i (D_i - \bar{D})^2} = \sqrt{\frac{0.20 - 0.10^2}{199}} \approx 0.0309.$$

The covariance formula in Chapter 3 explains the same result: $\mathrm{Var}(A_i - B_i) = \mathrm{Var}(A_i) + \mathrm{Var}(B_i) - 2\,\mathrm{Cov}(A_i, B_i)$.

Result 6.11: Paired bootstrap for a metric difference

Keep each test example's label and both model predictions in one row. For each replicate, sample n rows with replacement, using the same indices for both models. Recompute both metrics and their difference on the resampled rows. Apply the standard-error or interval calculation to the resulting differences.

Resampling the two models independently would remove their within-example dependence and estimate a different experiment. Pairing preserves the fact that both models encounter the same resampled easy and difficult examples.

With 20,000 paired resamples of this table, the bootstrap standard error is approximately 0.0310 and the percentile 95% interval is $[0.04, 0.16]$. These values describe an estimated accuracy improvement of 4 to 16 percentage points.

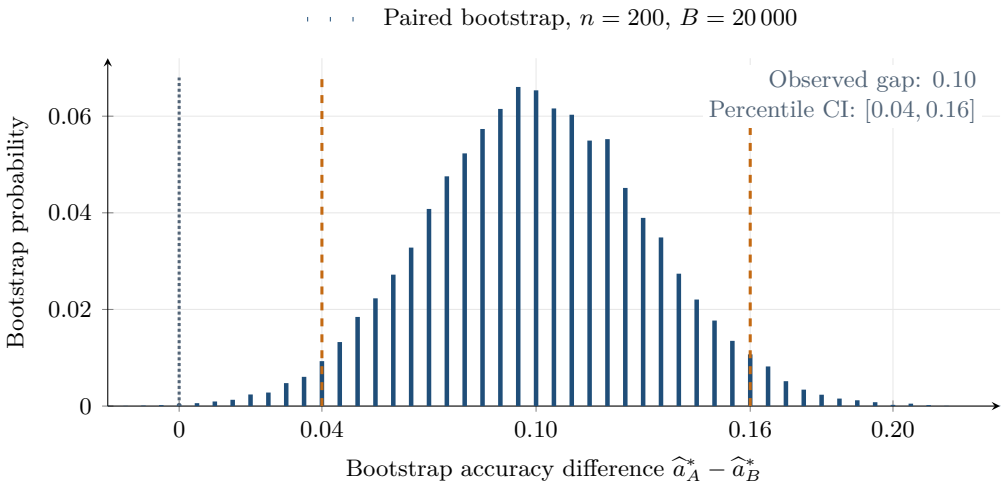


Figure 6.5: Paired bootstrap accuracy differences for the displayed 200-example table. Each replicate draws the same test rows for both classifiers. The percentile endpoints describe uncertainty from sampling test examples for these fixed fitted models.

For precision, recall, and F1, retain the label and predictions in each row, then recompute the confusion counts within each resample. For example, precision is $\mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$, recall is $\mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$, and

$$\mathrm{F1} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}.$$

These are ratios of counts, so averaging hypothetical per-example F1 scores would not reproduce the metric. With rare positive examples, some resamples may have a zero denominator. Its handling must be specified and the resulting interval may be unstable.

An interval for an accuracy difference that lies above zero is evidence of an improvement for the stated test population and sampling procedure. It does not imply an improvement for every example. If the interval includes zero, the observed experiment may be too imprecise to resolve the difference. Selecting the two models by repeatedly comparing many candidates on this same test set changes the uncertainty question.

6.7 What resampling can and cannot recover

6.7.1 Small samples and missing outcomes

The empirical distribution contains only observed values. A very small dataset may omit an important part of the population, especially a rare but influential outcome. No number of ordinary resamples can recover an event that has zero probability under the empirical distribution. For example, bootstrapping twenty observed failures as Bernoulli outcomes gives failure rate one in every resample. The resulting zero spread is the same warning we encountered with the Wald interval.

Extreme-value statistics provide a sharper counterexample. For $X_i \sim \mathrm{Uniform}(0, \theta)$, let $M = \max_i X_i$, the endpoint MLE from Section 5.3. Every ordinary bootstrap maximum satisfies $M^* \leq M$. But the true endpoint satisfies $\theta > M$ with probability one. Thus the upper endpoint of a percentile interval made from M^* is at most M, and the interval covers θ with probability zero. A sampling model can still support inference about θ, but ordinary empirical resampling fails for this endpoint problem.

6.7.2 The independent unit determines what to resample

If several measurements come from one user, independently resampling measurements loses their shared user effect. When users are independent sampling units, a cluster bootstrap resamples users and keeps their measurements together. For time series, resampling suitable consecutive blocks can retain some temporal dependence. Block length and assumptions about the process matter. Ordinary i.i.d. row resampling does not automatically apply to these settings.

The sample must also represent the target population. Repeating a biased sampling scheme or bootstrapping a nonrepresentative dataset does not remove its selection bias. The same problem arises when the deployment population differs from the test population. Heavy tails can also make bootstrap approximation unreliable, particularly for means with infinite variance.

6.7.3 Parameter uncertainty and training uncertainty

For an interval about a fitted parameter, resample the independent training units and fit the estimator again on each resample. For regression, preserve each feature vector and its response as a row. This approximates randomness in drawing those training rows under the stated sampling model. A design that holds feature values fixed requires a different resampling construction, such as resampling modeled errors, with its own assumptions.

The paired test bootstrap in Section 6.6 holds the trained models fixed. It describes test-sample uncertainty. To include variation from training data, random initialization, or hyperparameter selection, the repeated experiment must also repeat the corresponding training or selection steps. Recomputing only test metrics cannot estimate those additional sources of variation.

Chapter 7

Entropy, KL Divergence, and Fisher Information

7.1 Surprise and entropy

Suppose a population has two labels, with probabilities $\mathbb{P}(Y = 1) = 0.8$ and $\mathbb{P}(Y = 0) = 0.2$. Before seeing the next label, we expect 1. Seeing 0 is less likely and tells us more than another 1. We want a numerical measure of that difference.

Multiplying probabilities should add information. For example, two independent outcomes of probability $1/2$ have joint probability $1/4$. The function $-\log p$ has exactly this property:

$$-\log(p_1 p_2) = -\log p_1 - \log p_2.$$

It is zero for an outcome certain to occur and grows as the outcome becomes less likely.

Definition 7.1: Surprisal

If an observed outcome has probability $p > 0$, its surprisal is $-\log p$. Throughout this chapter log is the natural logarithm, so information is measured in *nats*.

For our binary example, the surprisals are

$$-\log 0.8 \approx 0.2231, \qquad -\log 0.2 \approx 1.6094.$$

The rarer label has greater surprisal. To describe the uncertainty before the label is observed, average these values using the actual label probabilities.

Definition 7.2: Entropy of a finite distribution

Let P assign probabilities $p_1, \dots, p_K$ to K labels. Its entropy is the expected surprisal:

$$H(P) = \mathbb{E}_{Y \sim P}[-\log p_Y] = -\sum_{k=1}^{K} p_k \log p_k. \tag{7.1}$$

A term with $p_k = 0$ is defined to be zero, using $\lim_{p \downarrow 0} p \log p = 0$.

Thus $H(P) = -0.8 \log 0.8 - 0.2 \log 0.2 \approx 0.5004$ nats. This is an average over repeated draws, not the surprisal of every draw. If one label has probability one, the entropy is zero: there is no uncertainty about the next label.

Write P_p for the Bernoulli distribution with success probability p. Then

$$H(P_p) = -p \log p - (1 - p) \log(1 - p).$$

For $0 < p < 1$, differentiating gives

$$\frac{d}{dp} H(P_p) = \log \frac{1-p}{p}, \qquad \frac{d^2}{dp^2} H(P_p) = -\frac{1}{p(1-p)} < 0.$$

The first derivative vanishes at $p = 1/2$, so the largest entropy is $\log 2$. A balanced binary outcome is harder to predict than an outcome that is almost always the same.

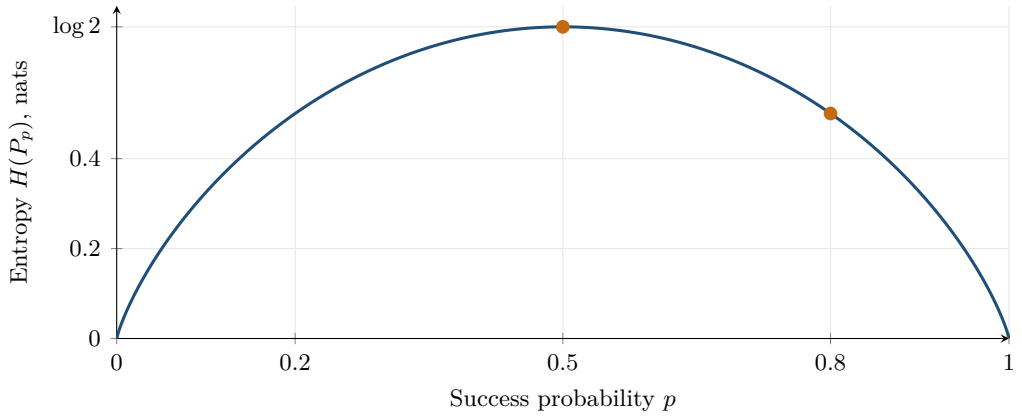


Figure 7.1: Bernoulli entropy. The maximum occurs at $p = 1/2$. The second marked point is $p = 0.8$, where $H(P_p) \approx 0.5004$. Both deterministic endpoints have entropy zero.

Changing the logarithm base changes only the unit. Dividing entropy in nats by $\log 2$ gives entropy in bits. We keep nats so that the formulas use the same logarithm as the likelihood calculations in Chapter 5.

7.2 Cross-entropy and KL divergence

Suppose the actual label distribution is $P = (p_1, \ldots, p_K)$, but a predictor reports $Q = (q_1, \ldots, q_K)$ for the same labels. When label k occurs, its log loss is $-\log q_k$. We average using P, because P describes which labels actually occur.

Definition 7.3: Cross-entropy

The cross-entropy of P relative to predictions Q is

$$\mathrm{CE}(P, Q) = -\sum_{k=1}^{K} p_k \log q_k. \tag{7.2}$$

Terms with $p_k = 0$ contribute zero. If $p_k > 0$ but $q_k = 0$, the cross-entropy is infinite: a possible outcome was declared impossible.

For the population with success probability 0.8, a predictor assigning success probability q has expected loss

$$\mathrm{CE}(P_{0.8}, P_q) = -0.8 \log q - 0.2 \log(1-q).$$

The following values compare predictions for the same population:

Predicted q	Cross-entropy	Excess over $H(P_{0.8})$
0.50	0.6931	0.1927
0.60	0.5919	0.0915
0.80	0.5004	0
0.99	0.9291	0.4287

Predicting 0.99 gives a small loss on the frequent successes but a large loss on the failures. On average it is worse than predicting 0.5. The best value is $q = 0.8$; even there, the expected loss is not zero because individual labels remain random.

Definition 7.4: Kullback–Leibler divergence

For distributions on the same finite set,

$$D_{\mathrm{KL}}(P\|Q) = \sum_{k:p_k>0} p_k \log \frac{p_k}{q_k}. \tag{7.3}$$

The value is infinite if $q_k = 0$ for any k with $p_k > 0$. Subtracting the entropy formula from the cross-entropy formula gives

$$\mathrm{CE}(P, Q) = H(P) + D_{\mathrm{KL}}(P\|Q). \tag{7.4}$$

The direction matters: the expectation in $D_{\mathrm{KL}}(P\|Q)$ is over outcomes drawn from P. Here KL measures the extra expected log loss caused by using Q in place of the actual distribution.

Result 7.5: Nonnegativity

$D_{\mathrm{KL}}(P\|Q) \geq 0$, with equality exactly when $P = Q$. Consequently, $\mathrm{CE}(P, Q) \geq H(P)$.

Proof. If the divergence is infinite, the inequality holds. For $u > 0$, the function $u - 1 - \log u$ has derivative $1 - 1/u$, so its minimum is zero at $u = 1$. Thus $-\log u \geq 1 - u$. Apply this to $u = q_k/p_k$ for every $p_k > 0$:

$$D_{\mathrm{KL}}(P\|Q) \geq \sum_{k:p_k>0} (p_k - q_k) = 1 - \sum_{k:p_k>0} q_k \geq 0.$$

The scalar inequality is an equality only at $u = 1$. Equality therefore requires $q_k = p_k$ on the support of P; normalization then leaves no mass for any other label. □

KL is not a distance. For example,

$$D_{\mathrm{KL}}(P_{0.8}\|P_{0.5}) \approx 0.1927, \qquad D_{\mathrm{KL}}(P_{0.5}\|P_{0.8}) \approx 0.2231.$$

It is not symmetric and does not generally satisfy the triangle inequality. When writing a KL penalty, we must specify its direction.

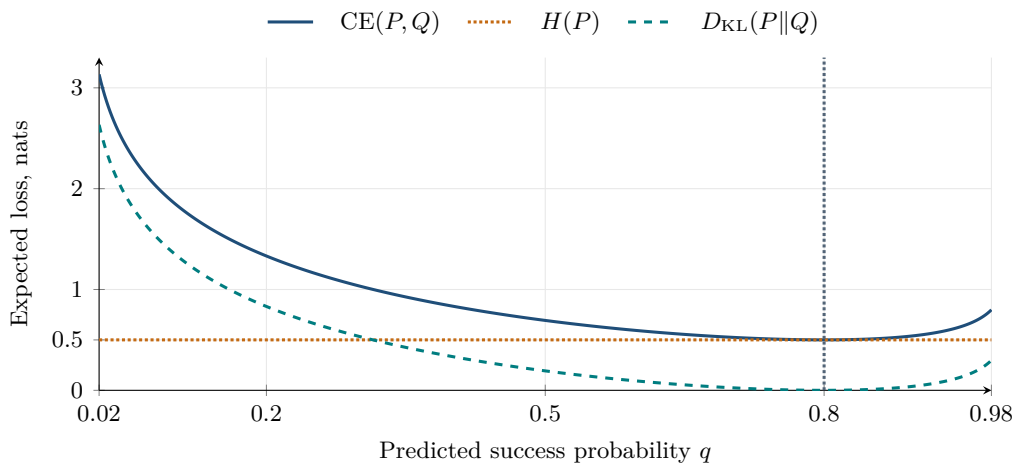


Figure 7.2: The population probability stays 0.8, while the prediction q changes. The vertical gap between cross-entropy and the constant entropy is KL. Both cross-entropy and KL are minimized at $q = 0.8$.

The same result identifies the maximum categorical entropy. If U is uniform on K labels, then

$$D_{\mathrm{KL}}(P\|U) = \log K - H(P) \geq 0.$$

Hence $0 \leq H(P) \leq \log K$, with the upper bound attained only by the uniform distribution.

7.3 Connections to learning

7.3.1 Likelihood and empirical cross-entropy

Chapter 5 derived log loss from likelihood. We can now give that loss an interpretation. Suppose $y_1, \ldots, y_n$ are independent observations taking values in K labels, and a model assigns probability $q_\theta(k)$ to label k. Let n_k be the number of occurrences of label k, and set $\widehat{p}_k = n_k/n$. Then

$$-\frac{1}{n}\log L(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\log q_\theta(y_i) = -\sum_{k=1}^{K}\widehat{p}_k \log q_\theta(k) = \mathrm{CE}(\widehat{P}, Q_\theta). \tag{7.5}$$

The empirical distribution $\widehat{P}$ is the one introduced in Section 6.5. Its entropy is fixed once the data are fixed. Therefore maximum likelihood minimizes $D_{\mathrm{KL}}(\widehat{P}\|Q_\theta)$ in this finite-label setting. This does not claim that the fitted model equals the population distribution: the empirical frequencies are noisy, and the model may impose restrictions.

For a classifier, predictions depend on input x. Write $q_\theta(k \mid x)$ for its predicted probability of label k. For an observed pair (x_i, y_i), define the *one-hot* target by $t_{ik} = 1$ if $k = y_i$, and $t_{ik} = 0$ otherwise. Its cross-entropy is

$$-\sum_{k=1}^{K} t_{ik}\log q_\theta(k \mid x_i) = -\log q_\theta(y_i \mid x_i).$$

Averaging over examples gives the multiclass negative log-likelihood. For two labels, this is the binary log loss already derived in Section 5.6.

The population expected loss also separates into entropy and KL for each input, using the true conditional label distribution $P(\cdot \mid x)$. An individual one-hot target does not mean that this true conditional distribution is deterministic.

7.3.2 Sequences and soft targets

A language model assigns a conditional distribution to the next token. For a sequence $w_1, \ldots, w_m$, the chain rule from Chapter 1 gives

$$Q_\theta(w_1, \ldots, w_m) = \prod_{t=1}^{m} q_\theta(w_t \mid w_1, \ldots, w_{t-1}).$$

At $t = 1$, the condition is empty. The average negative log-likelihood is

$$-\frac{1}{m} \sum_{t=1}^{m} \log q_\theta(w_t \mid w_1, \ldots, w_{t-1}).$$

For three observed tokens given conditional probabilities $0.5, 0.25, 0.1$, the sequence probability is 0.0125, and the average loss is $-\log(0.0125)/3 \approx 1.4607$ nats per token. The product follows from conditioning; the tokens need not be independent.

Targets can also be probability vectors. In *distillation*, a teacher model supplies probabilities P, and a student supplies Q for the same input. Minimizing $\mathrm{CE}(P, Q)$ matches all teacher probabilities, not just the most likely label. For example,

$$P = (0.7, 0.2, 0.1), \qquad Q = (0.6, 0.25, 0.15)$$

give $H(P) \approx 0.8018$, $\mathrm{CE}(P, Q) \approx 0.8245$, and $D_{\mathrm{KL}}(P\|Q) \approx 0.0227$. The teacher entropy is constant with respect to the student parameters, so minimizing this cross-entropy is equivalent to minimizing this KL.

7.3.3 KL penalties and approximate posteriors

A KL penalty can discourage a model from moving far from a reference distribution. For example, consider

$$\mathcal{L}_{\mathrm{train}}(\theta) + \beta D_{\mathrm{KL}}(Q_\theta \| Q_{\mathrm{ref}}), \qquad \beta \geq 0.$$

Here $\mathcal{L}_{\mathrm{train}}$ is the training loss, Q_θ is the model's output distribution on a specified input, and Q_{ref} is a fixed reference on the same labels. The coefficient β sets the penalty strength. For many inputs, one can average the penalty over inputs. The direction written here penalizes probabilities used by Q_θ that are small under the reference. This is an output-distribution penalty; the MAP penalty in Section 5.7 instead came from a prior on parameters.

KL also gives a way to approximate a posterior. Consider a finite unknown state Z, observed data x, and a joint model $p(x, z)$. Suppose the posterior $P(z \mid x) = p(x, z)/p(x)$ is difficult to compute. Choose a simpler candidate Q with probabilities $q(z)$, supported where $p(x, z) > 0$. Bayes' formula gives

$$D_{\mathrm{KL}}(Q\|P(\cdot \mid x)) = \sum_z q(z)\left[\log q(z) - \log p(x, z)\right] + \log p(x). \tag{7.6}$$

The evidence $p(x)$ does not depend on the choice of Q. We can therefore search for the best approximation in the chosen family by minimizing the sum in (7.6), without knowing $p(x)$. This is the basic idea of *variational inference*. It does not guarantee that the chosen family contains the true posterior.

7.3.4 Confidence is not correctness

The entropy of a model's predicted distribution measures how concentrated its own predictions are. For a prediction $q = 0.99$,

$$H(P_{0.99}) \approx 0.0560, \qquad -\log(1 - 0.99) \approx 4.6052.$$

The entropy is small, but if the observed label is zero, the log loss is large. Predictive entropy alone does not establish accuracy or calibration. This differs from the population cross-entropy, which averages prediction losses using the distribution of the actual outcomes.

7.3.5 A caution for continuous models

For densities f and g, the corresponding KL formula is

$$D_{\mathrm{KL}}(f\|g) = \int f(x) \log \frac{f(x)}{g(x)}\, dx,$$

with the same support requirement and nonnegativity. When the integrals are finite, the identity with cross-entropy remains valid if entropy is replaced by $-\int f \log f$, called *differential entropy*. This quantity can be negative and changes under a change of measurement units. A density is not a point probability.

In particular, the empirical distribution of finitely many real-valued observations places positive mass at individual sample points; a continuous model places zero probability at each such point. The KL from that atomic empirical distribution to the continuous model is infinite. Identity (7.5) therefore applies to finite labels. For continuous MLE, use the density likelihood from Chapter 5.

7.4 Score and Fisher information

Entropy describes uncertainty about an outcome. We now ask how precisely the data determine an unknown parameter. Start with the likelihood peak from Chapter 5: a broad peak leaves many nearby parameter values with similar likelihood, while a narrow peak favors a smaller range of values.

7.4.1 Observed information at a fitted maximum

Let θ be a scalar parameter, and let f_θ be the observation PMF or density. For recorded independent observations $x_1, \ldots, x_n$, write

$$\ell_n(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f_\theta(x_i), \qquad S_n(\theta) = \ell_n'(\theta).$$

The derivative S_n is the *sample score*. At a differentiable interior MLE $\widehat{\theta}$, $S_n(\widehat{\theta}) = 0$. A flat peak and a sharp peak both have zero slope at their maximum. The second derivative distinguishes them.

Definition 7.6: Observed information

For a twice differentiable log-likelihood, the observed information is

$$J_n(\theta) = -\ell_n''(\theta). \tag{7.7}$$

It measures the downward curvature for the particular recorded dataset. At an interior maximum $J_n(\widehat{\theta}) \geq 0$; away from a maximum, observed information need not be nonnegative.

Let δ be a small displacement from the fitted parameter. Since the linear term vanishes at the MLE, a quadratic approximation gives

Result 7.7: Likelihood drop near an interior MLE

If the log-likelihood is well approximated by a quadratic near $\widehat{\theta}$, with $J_n(\widehat{\theta}) > 0$, then

$$\ell_n(\widehat{\theta} + \delta) - \ell_n(\widehat{\theta}) \approx -\tfrac{1}{2} J_n(\widehat{\theta})\delta^2, \tag{7.8}$$

$$\frac{L(\widehat{\theta} + \delta)}{L(\widehat{\theta})} \approx \exp[-\tfrac{1}{2} J_n(\widehat{\theta})\delta^2]. \tag{7.9}$$

For the same displacement $\delta = 0.1$, curvature 100 gives a relative likelihood of approximately $e^{-0.5} = 0.607$, while curvature 4 gives $e^{-0.02} = 0.980$. The first likelihood loses much more of its height over the same parameter change. At distance $1/\sqrt{J_n(\widehat{\theta})}$, the approximation drops to $e^{-1/2} \approx 0.607$ of its maximum. This is a local description of the fitted peak. Boundary MLEs need separate treatment because their score need not vanish.

7.4.2 Fisher information averages over possible samples

Before collecting data, we do not know which likelihood peak we will obtain. Keep a parameter value θ fixed, repeatedly draw samples from that model, and evaluate each sample's observed information at the same θ. The average curvature is Fisher information.

Definition 7.8: Fisher information

For a regular model satisfying the conditions below, the Fisher information about θ in a sample of size n is

$$I_n(\theta) = \mathbb{E}_\theta[J_n(\theta)], \tag{7.10}$$

when this expectation exists. The expectation draws new samples under θ; the parameter at which the curvature is evaluated stays fixed.

Thus J_n describes the observed dataset, whereas I_n describes the curvature expected under the model and sampling design. For one observation this definition reads

$$I_1(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \log f_\theta(X)]. \tag{7.11}$$

The expectation in (7.10) is not taken at a new fitted maximum for each dataset. Refitting each time would be a different calculation. We can evaluate the resulting model function $I_n(\theta)$ at $\widehat{\theta}$ after obtaining an estimate.

7.4.3 The equivalent squared-score formula

To connect expected curvature to scores, assume an interior parameter value, a common support not depending on θ, positive f_θ on that support, and differentiability allowing the derivatives below to pass through the integral or sum. Assume also a finite score second moment. These conditions exclude the moving support of the uniform-endpoint model in Chapter 5.

Definition 7.9: Score of one observation

The one-observation score is

$$s_\theta(x) = \frac{\partial}{\partial\theta} \log f_\theta(x). \tag{7.12}$$

A positive score means that a small increase in θ increases the log-likelihood of this observation.

Normalization gives

$$\mathbb{E}_\theta[s_\theta(X)] = \int \frac{\partial_\theta f_\theta(x)}{f_\theta(x)} f_\theta(x)\, dx = \frac{\partial}{\partial\theta} \int f_\theta(x)\, dx = 0. \tag{7.13}$$

For a discrete model, use sums. Differentiating $\log f_\theta$ once more gives

$$\partial_\theta^2 \log f_\theta = \frac{\partial_\theta^2 f_\theta}{f_\theta} - s_\theta^2.$$

Multiply by f_θ and integrate. The first term integrates to the second derivative of 1, which is zero.

Result 7.10: Expected squared score and expected curvature

Under the stated regularity conditions,

$$I_1(\theta) = \mathbb{E}_\theta[s_\theta(X)^2] = \mathrm{Var}_\theta(s_\theta(X)). \tag{7.14}$$

The variance equality uses the zero expected score in (7.13).

Squaring prevents positive and negative scores from cancelling. For a small parameter change δ, $\log f_{\theta+\delta}(x) - \log f_\theta(x) \approx \delta s_\theta(x)$. The squared-score expression measures this sensitivity across possible observations. The recorded sample score being zero at the MLE does not make Fisher information zero: the expectation uses fresh samples at a fixed parameter, whose scores vary.

For a Gaussian observation with known σ^2,

$$s_\mu(x) = \frac{x-\mu}{\sigma^2}, \qquad I_1(\mu) = \frac{\mathbb{E}_\mu[(X-\mu)^2]}{\sigma^4} = \frac{1}{\sigma^2}.$$

A reading above μ pushes toward a larger mean, and a reading below it pushes toward a smaller mean. More observation noise gives less information about the same mean.

Result 7.11: Independent observations add information

For n i.i.d. observations,

$$S_n(\theta) = \sum_{i=1}^{n} s_\theta(X_i), \qquad I_n(\theta) = \mathbb{E}_\theta[S_n(\theta)^2] = nI_1(\theta). \tag{7.15}$$

The scores are independent and have mean zero, so their sum has variance equal to the sum of their variances. Equivalently, the sample log-likelihood is a sum, so its expected negative curvature adds. With dependent observations, the score covariance terms need not vanish.

7.4.4 When observed and expected information coincide

For independent Gaussian measurements with known variance, completing the square as in Section 5.4 gives

$$\widehat{\mu} = \bar{x}, \qquad \ell_n(\mu) = C - \frac{n}{2\sigma^2}(\mu - \bar{x})^2,$$

where C does not depend on the candidate mean μ. The curvature is the same for every possible dataset:

$$J_n(\mu) = -\ell_n''(\mu) = \frac{n}{\sigma^2} = I_n(\mu). \tag{7.16}$$

Consequently, the quadratic likelihood-drop formula is exact:

$$\frac{L(\bar{x} + \delta)}{L(\bar{x})} = \exp\left[-\frac{1}{2} I_n(\mu)\delta^2\right].$$

The equality of J_n and I_n here uses the known-variance Gaussian mean model; it is not a general identity for every model.

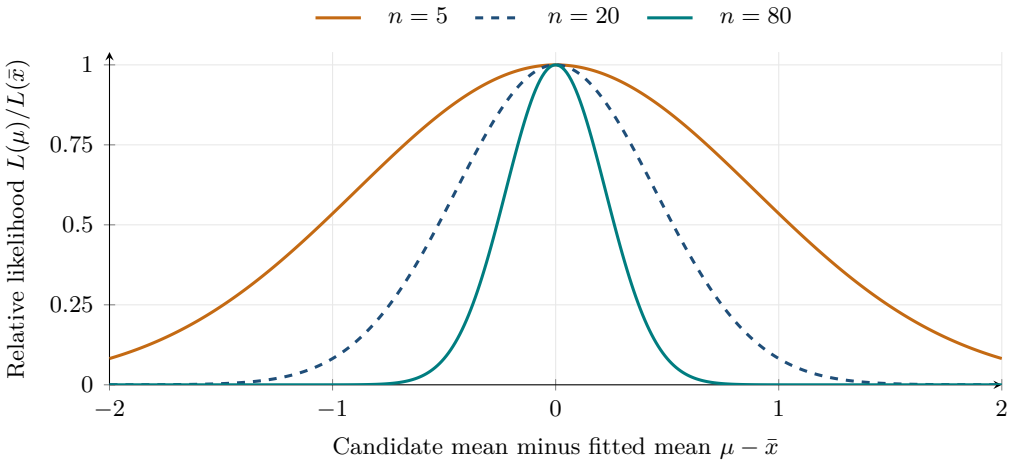


Figure 7.3: Gaussian relative likelihoods with known $\sigma = 2$, centered at each sample's fitted mean. Here $J_n = I_n = n/4$: the three curvatures are 1.25, 5, 20. The distance to relative height $e^{-1/2}$ is $1/\sqrt{I_n}$, so larger information means a narrower peak. These curves compare candidate means; their areas are not posterior probabilities.

7.4.5 Bernoulli information

Now treat the population success probability p as unknown, with $0 < p < 1$. For $Y \sim \text{Bernoulli}(p)$,

$$\log f_p(y) = y \log p + (1 - y)\log(1 - p), \qquad s_p(y) = \frac{y}{p} - \frac{1 - y}{1 - p} = \frac{y - p}{p(1 - p)}.$$

At $p = 0.8$, a success gives score 1.25, while a failure gives score -5. Their weighted average is $0.8(1.25) + 0.2(-5) = 0$. Using the Bernoulli variance already established in Chapter 2,

$$I_1(p) = \frac{\mathbb{E}_p[(Y-p)^2]}{p^2(1-p)^2} = \frac{1}{p(1-p)}, \qquad I_n(p) = \frac{n}{p(1-p)}. \tag{7.17}$$

Observed information can differ from this expectation. Let $K = \sum_{i=1}^{n} Y_i$ be the success count. Differentiating the sample log-likelihood twice gives

$$J_n(p) = \frac{K}{p^2} + \frac{n-K}{(1-p)^2}. \tag{7.18}$$

Unlike $I_n(p)$, this quantity depends on which dataset was observed. For example, evaluate two samples of size 10 at the same fixed candidate probability $p = 0.8$:

Observed successes	$J_{10}(0.8)$	$I_{10}(0.8)$
7	85.9375	62.5
9	39.0625	62.5

The likelihood curvatures differ even though the model parameter used for evaluation is the same. Averaging over all possible samples, using $\mathbb{E}_p K = np$, recovers Fisher information:

$$\mathbb{E}_p[J_n(p)] = \frac{np}{p^2} + \frac{n(1-p)}{(1-p)^2} = \frac{n}{p(1-p)} = I_n(p).$$

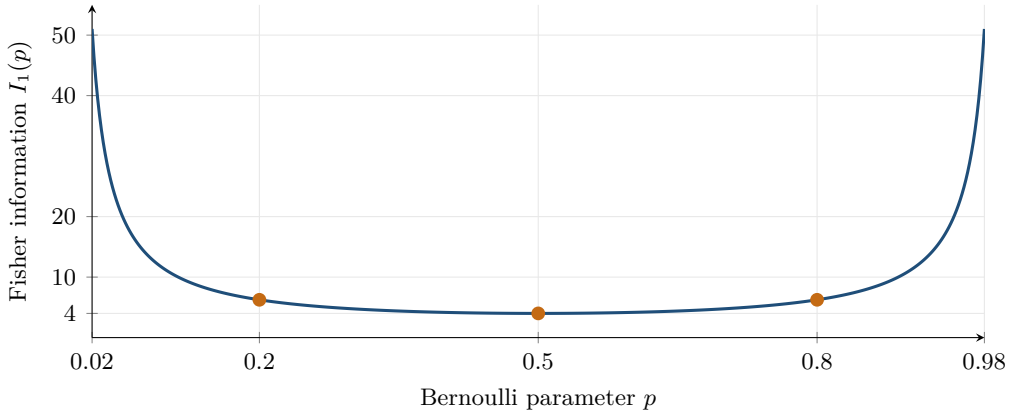


Figure 7.4: Information about the Bernoulli probability p, per observation. It is smallest at $p = 1/2$. This is not the entropy curve: entropy measures uncertainty about the next label; Fisher information measures sensitivity to a change in the parameter.

There is no contradiction between maximum entropy and minimum Fisher information at $p = 1/2$. A balanced label is most unpredictable, and the sample proportion also has its largest absolute variance $p(1-p)/n$ there. Near zero, a small absolute error in p can still be a large relative error. Moreover, Fisher information changes with the choice of parameter and its units; it is not an intrinsic score for how useful an observation is for every possible question.

7.5 The Cramér–Rao lower bound

Chapter 5 compared the bias and variance of particular estimators. Fisher information gives a lower bound for the variance of *every unbiased estimator* satisfying the regularity conditions.

Theorem 7.12: Scalar Cramér–Rao bound

Let $T = T(X_1, \dots, X_n)$ estimate θ from a regular i.i.d. model as above, with $0 < I_n(\theta) < \infty$. Suppose T has finite variance, $\mathbb{E}_\theta T = \theta$ throughout an open interval of parameter values, and differentiating its expectation under the sum or integral is valid. Then

$$\operatorname{Var}_\theta(T) \geq \frac{1}{I_n(\theta)} = \frac{1}{nI_1(\theta)}. \tag{7.19}$$

Proof. Let $g_\theta(x_1, \dots, x_n)$ be the joint sample PMF or density. Its log-derivative is the sample score $S_n(\theta)$. Differentiate the unbiasedness identity:

$$1 = \frac{\partial}{\partial \theta} \mathbb{E}_\theta T = \int T(x)\, \partial_\theta g_\theta(x)\, dx = \mathbb{E}_\theta[T S_n(\theta)].$$

The score has mean zero, so this last expectation equals $\operatorname{Cov}_\theta(T, S_n(\theta))$. The covariance bound from Section 3.3 gives

$$1 = \operatorname{Cov}_\theta(T, S_n(\theta))^2 \leq \operatorname{Var}_\theta(T) \operatorname{Var}_\theta(S_n(\theta)) = \operatorname{Var}_\theta(T) I_n(\theta).$$

Dividing proves the claim. □

Definition 7.13: Efficient unbiased estimator

An unbiased estimator is efficient at a parameter value if its variance equals the Cramér–Rao bound there. Such equality need not be attainable in every model or at every sample size.

7.5.1 Two estimators that attain the bound

For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ independently, with known σ^2, the bound and the sample-mean variance are

$$\frac{1}{I_n(\mu)} = \frac{\sigma^2}{n} = \operatorname{Var}_\mu(\bar{X}_n).$$

The sample mean is unbiased and efficient. By contrast, the estimator $T = X_1$ is also unbiased but has variance σ^2: it discards the other $n - 1$ measurements.

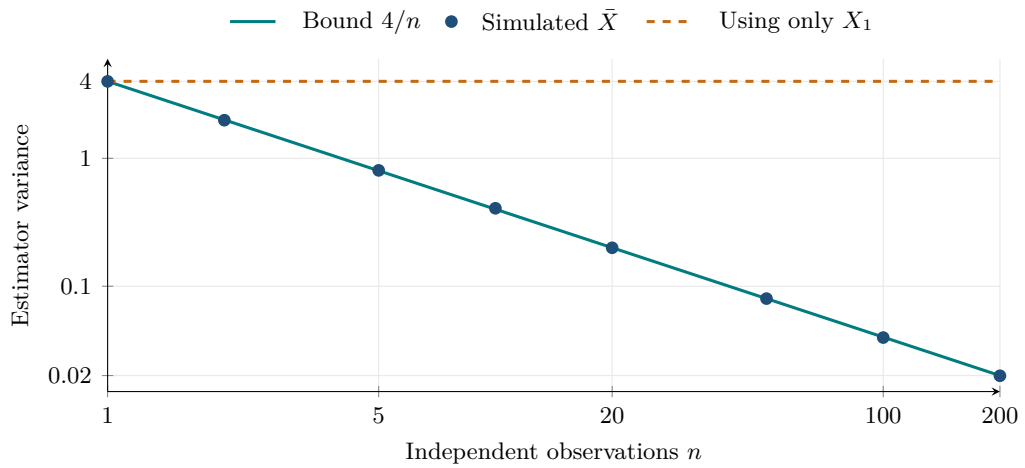


Figure 7.5: Known Gaussian variance $\sigma^2 = 4$. The line $4/n$ is both the Cramér–Rao bound and the exact variance of the sample mean. Points estimate that variance from 30,000 independent datasets at each n, using seed 20261020. Using only X_1 has variance 4, regardless of how many other observations exist. Both axes use logarithmic scales.

For i.i.d. Bernoulli labels, the sample proportion $\widehat{p} = \bar{Y}_n$ is unbiased, and

$$\mathrm{Var}_p(\widehat{p}) = \frac{p(1-p)}{n} = \frac{1}{I_n(p)}.$$

It too attains the bound for $0 < p < 1$, at every sample size. This exact variance equality does not make a Gaussian confidence approximation exact; Section 6.4 showed why small counts and boundaries remain problematic.

7.5.2 Why unbiasedness cannot be dropped

In the Gaussian mean example, consider $T = \bar{X}_n/2$. It has smaller variance,

$$\mathrm{Var}_\mu(T) = \frac{\sigma^2}{4n} < \frac{\sigma^2}{n}, \qquad \mathbb{E}_\mu T - \mu = -\frac{\mu}{2}.$$

This does not violate the bound, because the estimator is biased. Its mean squared error is

$$\mathbb{E}_\mu[(T-\mu)^2] = \frac{\sigma^2}{4n} + \frac{\mu^2}{4}.$$

Whether this is better than the sample mean depends on μ. This is the same shrinkage behavior encountered in the Gaussian MAP example in Section 5.7. Even at $\mu = 0$, the estimator is not unbiased throughout an open interval, as the theorem requires.

7.6 MLE uncertainty from information

The Cramér–Rao bound is a statement about unbiased estimators. An MLE need not be unbiased at a finite sample size. Nevertheless, regular MLEs often approach the information bound as the sample grows. The reason connects the score to the CLT from Chapter 4.

Let θ be the true parameter and $\widehat{\theta}$ an interior MLE, so $\ell_n'(\widehat{\theta}) = 0$. Near θ, a linear approximation to the score gives

$$0 \approx \ell_n'(\theta) + \ell_n''(\theta)(\widehat{\theta} - \theta).$$

The score $\ell_n'(\theta) = S_n(\theta)$ has mean zero and variance $nI_1(\theta)$. If the observed curvature is close to its expectation, $-\ell_n''(\theta) \approx nI_1(\theta)$, rearranging gives

$$\widehat{\theta} - \theta \approx \frac{S_n(\theta)}{nI_1(\theta)}. \tag{7.20}$$

The CLT makes the numerator approximately Gaussian. Dividing by $nI_1(\theta)$ gives an error scale $1/\sqrt{nI_1(\theta)}$.

Result 7.14: Asymptotic normality under explicit local conditions

Assume the regular score has finite positive information, the MLE is consistent and interior with probability tending to one, and

$$-\frac{1}{n}\ell_n''(\widetilde{\theta}_n) \xrightarrow{\mathbb{P}} I_1(\theta)$$

for any sequence of points $\widetilde{\theta}_n$ between θ and $\widehat{\theta}$. Here the convergence notation is the one from Chapter 4. Then

$$\sqrt{nI_1(\theta)}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, 1). \tag{7.21}$$

The mean value theorem gives the preceding score expansion with equality when its second derivative is evaluated at such an intermediate point. The sample-score CLT and the stated curvature convergence then give (7.21). Consistency alone is not enough; the local curvature condition is also needed.

The result justifies a Gaussian reference distribution with approximate standard error

$$\widehat{\mathrm{SE}}(\widehat{\theta}) = \frac{1}{\sqrt{nI_1(\widehat{\theta})}}.$$

It does not, by itself, prove convergence of the actual variances: that further claim needs control of the tails of the estimation errors. When the approximation and substitution are valid, the interval formula from Chapter 6 becomes

$$\widehat{\theta} \pm z_{1-\alpha/2}\frac{1}{\sqrt{nI_1(\widehat{\theta})}}. \tag{7.22}$$

The observed-information alternative uses $1/\sqrt{J_n(\widehat{\theta})}$ when the fitted curvature is positive and consistently estimates the same information.

For 80 successes in 100 independent Bernoulli observations,

$$\widehat{p} = 0.8, \qquad nI_1(\widehat{p}) = \frac{100}{0.8(0.2)} = 625, \qquad \widehat{\mathrm{SE}} = \frac{1}{\sqrt{625}} = 0.04.$$

The approximate 95% interval is

$$0.8 \pm 1.96(0.04) = [0.7216,\ 0.8784].$$

This is the Wald interval already discussed in Section 6.4. Fisher information explains its standard error; it does not fix the boundary and small-sample approximation problems.

7.7 A local connection between KL and Fisher information

KL compared two distributions; Fisher information described changing a parameter. They meet when we compare nearby members of a regular family. Fix θ, let δ be a small parameter change, and define

$$F(\delta) = D_{\mathrm{KL}}(P_\theta \| P_{\theta+\delta}) = \mathbb{E}_\theta[\log f_\theta(X) - \log f_{\theta+\delta}(X)].$$

The expectation stays under P_θ as δ changes. Under the differentiation assumptions used above,

$$F(0) = 0, \qquad F'(0) = -\mathbb{E}_\theta[s_\theta(X)] = 0, \qquad F''(0) = -\mathbb{E}_\theta[\partial_\theta^2 \log f_\theta(X)] = I_1(\theta).$$

A quadratic approximation with this value, slope, and curvature gives

Result 7.15: KL for a small parameter change

For a sufficiently smooth regular family and small δ,

$$D_{\mathrm{KL}}(P_\theta \| P_{\theta+\delta}) \approx \frac{1}{2} I_1(\theta)\delta^2. \tag{7.23}$$

This is a local approximation, not a formula for arbitrary parameter changes.

For Gaussians with the same known variance, the formula is exact:

$$\log \frac{f_\mu(x)}{f_{\mu+\delta}(x)} = \frac{-2\delta(x-\mu)+\delta^2}{2\sigma^2}, \qquad D_{\mathrm{KL}}\Big(\mathcal{N}(\mu,\sigma^2)\|\mathcal{N}(\mu+\delta,\sigma^2)\Big) = \frac{\delta^2}{2\sigma^2}.$$

The term linear in $X-\mu$ disappears on taking the expectation. Since $I_1(\mu) = 1/\sigma^2$, this equals $\frac{1}{2}I_1(\mu)\delta^2$ for every δ.

For Bernoulli probabilities near $p = 0.8$, the approximation is

$$D_{\mathrm{KL}}(P_{0.8}\|P_q) \approx \frac{(q-0.8)^2}{2(0.8)(0.2)} = 3.125(q-0.8)^2.$$

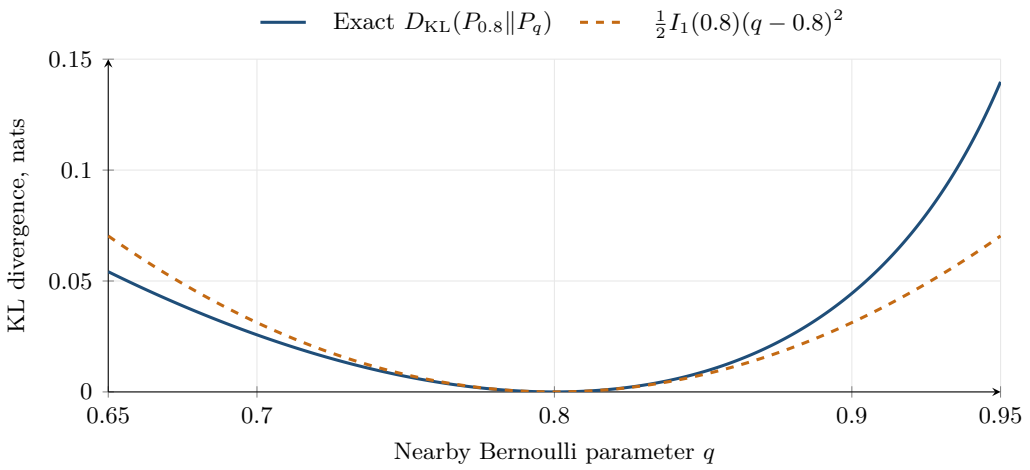


Figure 7.6: Exact Bernoulli KL and its quadratic approximation around $p = 0.8$. They agree near $q = 0.8$, but the quadratic expression misses the asymmetric growth farther away.

For independent samples, KL adds just as Fisher information does: the log ratio of joint probabilities is the sum of the observation log ratios, and taking the expectation gives n times the one-observation KL. Thus nearby parameter changes have sample divergence approximately $nI_1(\theta)\delta^2/2$. Setting $|\delta| = 1/\sqrt{nI_1(\theta)}$, the standard-error scale from the previous section, makes that approximation equal to $1/2$. The same information quantity therefore controls both the separation of nearby observation laws and the scale of regular estimation error.